

Notes on Introduction to Real Analysis

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1 Multivariable Differential Calculus

1.1 Linear Approximation

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be defined by $f(\mathbf{x}) = (f_1(\mathbf{x}), f_2(\mathbf{x}), \dots, f_m(\mathbf{x}))$, where each component $f_i : \mathbb{R}^n \rightarrow \mathbb{R}$ is a real-valued function.

Regardless of whether f is linear or non-linear, a natural question arises: can it be locally approximated by a linear function? And if so, how accurate is this approximation?

Ultimately, given a point $\mathbf{x}_0 \in \mathbb{R}^n$, our goal is to approximate f near $\mathbf{x}_0$ using its linearization:

$$f(\mathbf{x}) \approx f(\mathbf{x}_0) + Df(\mathbf{x}_0)(\mathbf{x} - \mathbf{x}_0)$$

Definition 1.1 (Differentiability)

Let $E \subseteq \mathbb{R}^n$, $f : E \rightarrow \mathbb{R}^m$ be a function, and $\mathbf{x}_0$ be an interior point of E . Let $L : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation. We say f is differentiable at $\mathbf{x}_0$ with derivative L if

$$\lim_{\mathbf{x} \rightarrow \mathbf{x}_0} \frac{\|f(\mathbf{x}) - f(\mathbf{x}_0) - L(\mathbf{x} - \mathbf{x}_0)\|}{\|\mathbf{x} - \mathbf{x}_0\|} = 0.$$

Equivalently,

$$\lim_{\mathbf{h} \rightarrow \mathbf{0}} \frac{\|f(\mathbf{x}_0 + \mathbf{h}) - f(\mathbf{x}_0) - L(\mathbf{h})\|}{\|\mathbf{h}\|} = 0.$$

Therefore, multivariable differentiability is defined such that the relative error between the function and its linear approximation vanishes as the distance approaches zero.

Theorem 1.1

Let $E \subseteq \mathbb{R}$, $f : E \rightarrow \mathbb{R}$ be a function, $L \in \mathbb{R}$, and x_0 be an interior point of E . TFAE:

1. f is differentiable at x_0 and $f'(x_0) = L$.
2. $\lim_{x \rightarrow x_0} \frac{|f(x) - f(x_0) - L(x - x_0)|}{|x - x_0|} = 0$.

Proof.

For $x \neq x_0$, write

$$\frac{f(x) - f(x_0)}{x - x_0} = L + \frac{f(x) - f(x_0) - L(x - x_0)}{x - x_0}.$$

Rearrange L to the left, take the absolute value and limit $x \rightarrow x_0$, and we can show the limit on the both sides imply each other. The second expression is exactly the special case of definition (1.1). ■

Theorem 1.2 (Uniqueness of derivatives)

Let $\mathbf{x}_0 \in E$ be an interior point of E . Suppose $L_1, L_2 : \mathbb{R}^n \rightarrow \mathbb{R}^m$ are linear transformations such that f is differentiable at $\mathbf{x}_0$ with derivative L_1 and also differentiable at $\mathbf{x}_0$ with derivative L_2 . Then $L_1 = L_2$.

Proof.

For $\mathbf{h} \neq \mathbf{0}$,

$$\frac{\|L_1(\mathbf{h}) - L_2(\mathbf{h})\|}{\|\mathbf{h}\|} \leq \frac{\|f(\mathbf{x}_0 + \mathbf{h}) - f(\mathbf{0}) - L_1(\mathbf{h})\|}{\|\mathbf{h}\|} + \frac{\|f(\mathbf{x}_0 + \mathbf{h}) - f(\mathbf{0}) - L_2(\mathbf{h})\|}{\|\mathbf{h}\|}.$$

By sandwich, $\lim_{\mathbf{h} \rightarrow \mathbf{0}} \frac{\|L_1(\mathbf{h}) - L_2(\mathbf{h})\|}{\|\mathbf{h}\|} = 0$.

Let $\mathbf{h} = t\mathbf{v}$, $t \neq 0$, we have

$$\lim_{\mathbf{v} \rightarrow \mathbf{0}} \frac{\|L_1(t\mathbf{v}) - L_2(t\mathbf{v})\|}{\|t\mathbf{v}\|} = \lim_{\mathbf{v} \rightarrow \mathbf{0}} \frac{|t| \|L_1(\mathbf{v}) - L_2(\mathbf{v})\|}{|t| \|\mathbf{v}\|} \quad (L_1 \text{ and } L_2 \text{ are linear.})$$

Hence, for $\mathbf{v} \neq \mathbf{0}$, $L_1(\mathbf{v}) = L_2(\mathbf{v})$, which is also true when $\mathbf{v} = \mathbf{0}$. ■

We may prove by contradiction that f is not differentiable by the above theorem.

Theorem 1.3

If f is differentiable at $\mathbf{x}_0$, then f is continuous at $\mathbf{x}_0$.

Proof.

For $\mathbf{x} \neq \mathbf{x}_0$, write

$$\|f(\mathbf{x}) - f(\mathbf{x}_0) - L(\mathbf{x} - \mathbf{x}_0)\| = \underbrace{\frac{\|f(\mathbf{x}) - f(\mathbf{x}_0) - L(\mathbf{x} - \mathbf{x}_0)\|}{\|\mathbf{x} - \mathbf{x}_0\|}}_{\rightarrow 0 \text{ by assumption}} \underbrace{\|\mathbf{x} - \mathbf{x}_0\|}_{\rightarrow 0 \text{ as } \mathbf{x} \rightarrow \mathbf{x}_0}.$$

Next, use

$$\|f(\mathbf{x}) - f(\mathbf{x}_0)\| \leq \underbrace{\|f(\mathbf{x}) - f(\mathbf{x}_0) - L(\mathbf{x} - \mathbf{x}_0)\|}_{\rightarrow 0 \text{ from the previous equality}} + \underbrace{\|L(\mathbf{x} - \mathbf{x}_0)\|}_{\rightarrow 0 \text{ as } \mathbf{x} \rightarrow \mathbf{x}_0}.$$

By Sandwich, $\lim_{\mathbf{x} \rightarrow \mathbf{x}_0} f(\mathbf{x}) = f(\mathbf{x}_0)$. ■

Definition 1.2 (Directional Derivative)

Let $E \subseteq \mathbb{R}^n$, $f : E \rightarrow \mathbb{R}^m$ be a function, and $\mathbf{x}_0$ be an interior point of E . Fix $\mathbf{v} \in \mathbb{R}^n$, if the limit

$$D_{\mathbf{v}}f(\mathbf{x}_0) = \lim_{t \rightarrow 0} \frac{f(\mathbf{x}_0 + t\mathbf{v}) - f(\mathbf{x}_0)}{t}$$

exists, we call this limit the directional derivative of f at $\mathbf{x}_0$ in the direction $\mathbf{v}$.

Definition 1.3 (Partial Derivative)

Let $E \subseteq \mathbb{R}^n$, $f : E \rightarrow \mathbb{R}^m$ be a function, and $\mathbf{x}_0$ be an interior point of E . Let $1 \leq j \leq n$, if the limit

$$\frac{\partial f}{\partial x_j}(\mathbf{x}_0) = \lim_{t \rightarrow 0} \frac{f(\mathbf{x}_0 + t\mathbf{e}_j) - f(\mathbf{x}_0)}{t}$$

exists, we call this limit the partial derivative of f with respect to the variable x_j .

So partial derivative is just a special case of directional derivative.

Theorem 1.4

If f is differentiable at $\mathbf{x}_0$, then f is differentiable in the direction $\mathbf{v}$ at $\mathbf{x}_0$, and:

$$D_{\mathbf{v}}f(\mathbf{x}_0) = f'(\mathbf{x}_0) \cdot \mathbf{v}.$$

Proof.

For $\mathbf{h} \neq 0$, let $\mathbf{h} = t\mathbf{v}$, $t \neq 0$, write

$$\frac{f(\mathbf{x}_0 + t\mathbf{v}) - f(\mathbf{x}_0)}{t} = \frac{f(\mathbf{x}_0 + t\mathbf{v}) - f(\mathbf{x}_0) - L(t\mathbf{v})}{t} + \frac{L(t\mathbf{v})}{t}.$$

Rearrange $L(\mathbf{v})$ to the left, and take the Euclidean norm and $t \rightarrow 0$, the RHS converge to 0 by differentiability ($\|\mathbf{v}\| \neq 0$). So we have

$$\lim_{t \rightarrow 0} \frac{f(\mathbf{x}_0 + t\mathbf{v}) - f(\mathbf{x}_0)}{t} = L(\mathbf{v}) = f'(\mathbf{x}_0) \cdot \mathbf{v}.$$

■

We may also disprove the differentiability of f provided its directional derivative exists. The next question is: how to write the linear map $f'(\mathbf{x}_0)$ explicitly?

Theorem 1.5

Let $E \subseteq \mathbb{R}^n$, $f : E \rightarrow \mathbb{R}^m$ be a function, and $\mathbf{x}_0$ be an interior point of E . If all the partial derivatives of f exist on E and are continuous at $\mathbf{x}_0$, then f is differentiable at $\mathbf{x}_0$, and

$$f'(\mathbf{x}_0) \cdot \mathbf{v} = \sum_{j=1}^n v_j \frac{\partial f}{\partial x_j}(\mathbf{x}_0).$$

Proof.

Step 1: Setup

Let $L(\mathbf{v}) = \sum_{j=1}^n v_j \frac{\partial f}{\partial x_j}(\mathbf{x}_0)$ be our candidate for the derivative. Since $\mathbf{x}_0$ is an interior point of E , there exists $r > 0$ such that the ball $B(\mathbf{x}_0, r) \subseteq E$.

For $\epsilon > 0$, since each partial derivative $\frac{\partial f_i}{\partial x_j}$ is continuous at $\mathbf{x}_0$, there exists $\delta_{i,j} > 0$ such that $\|\mathbf{x} - \mathbf{x}_0\| < \delta_{i,j}$ implies

$$\left| \frac{\partial f_i}{\partial x_j}(\mathbf{x}) - \frac{\partial f_i}{\partial x_j}(\mathbf{x}_0) \right| < \frac{\epsilon}{mn}$$

for all $1 \leq i \leq m$ and $1 \leq j \leq n$.

Let $\delta = \min\{r, \delta_{i,j} \mid 1 \leq i \leq m, 1 \leq j \leq n\}$. Now, fix $\mathbf{x} \in E$ such that $\|\mathbf{x} - \mathbf{x}_0\| < \delta$.

Let $\mathbf{v} = \mathbf{x} - \mathbf{x}_0 = \sum_{j=1}^n v_j \mathbf{e}_j$. Note that $|v_j| \leq \|\mathbf{x} - \mathbf{x}_0\| < \delta$. We construct a path

from $\mathbf{x}_0$ to $\mathbf{x}$ along the coordinate axes. Define the sequence of points:

$$\begin{aligned}\mathbf{x}^{(0)} &= \mathbf{x}_0 \\ \mathbf{x}^{(1)} &= \mathbf{x}_0 + v_1 \mathbf{e}_1 \\ \mathbf{x}^{(2)} &= \mathbf{x}_0 + v_1 \mathbf{e}_1 + v_2 \mathbf{e}_2 \\ &\vdots \\ \mathbf{x}^{(j)} &= \mathbf{x}^{(j-1)} + v_j \mathbf{e}_j \\ &\vdots \\ \mathbf{x}^{(n)} &= \mathbf{x}_0 + \sum_{j=1}^n v_j \mathbf{e}_j = \mathbf{x}\end{aligned}$$

We can express the difference $f(\mathbf{x}) - f(\mathbf{x}_0)$ as a telescoping sum:

$$f(\mathbf{x}) - f(\mathbf{x}_0) = \sum_{j=1}^n \left(f(\mathbf{x}^{(j)}) - f(\mathbf{x}^{(j-1)}) \right)$$

For each component i , consider $f_i(\mathbf{x}^{(j)}) - f_i(\mathbf{x}^{(j-1)})$. Let $\varphi_i^j(t) = f_i(\mathbf{x}^{(j-1)} + tv_j \mathbf{e}_j)$, then $f_i(\mathbf{x}^{(j)}) - f_i(\mathbf{x}^{(j-1)}) = \varphi_i^j(1) - \varphi_i^j(0)$. By the Mean Value Theorem, $\exists t_j \in (0, 1)$ such that $\varphi_i^j(1) - \varphi_i^j(0) = (\varphi_i^j)'(t_j)$.

Step 2: Continuity of f_i along the coordinate $\mathbf{x}_j$

Applying Lemma (1.1):

$$f_i(\mathbf{x}^{(j)}) - f_i(\mathbf{x}^{(j-1)}) = (\varphi_i^j)'(t_j) = \frac{\partial f_i}{\partial x_j}(\boldsymbol{\xi}_j) v_j,$$

where $\boldsymbol{\xi}_j = \mathbf{x}^{(j-1)} + t_j v_j \mathbf{e}_j$.

Note that

$$\begin{aligned}\|\mathbf{x}^{(j-1)} + t_j v_j \mathbf{e}_j - \mathbf{x}_0\| &= \|v_1 \mathbf{e}_1 + v_2 \mathbf{e}_2 + \cdots + v_{j-1} \mathbf{e}_{j-1} + t_j v_j \mathbf{e}_j\| \\ &\leq \left\| \sum_{i=1}^n v_i \mathbf{e}_i \right\| \quad (\because |t_j| < 1) \\ &= \|\mathbf{x} - \mathbf{x}_0\| \leq \delta,\end{aligned}$$

which implies

$$\begin{aligned}\left| f_i(\mathbf{x}^{(j)}) - f_i(\mathbf{x}^{(j-1)}) - \frac{\partial f_i}{\partial x_j}(\mathbf{x}_0) v_j \right| &= \left| \frac{\partial f_i}{\partial x_j}(\mathbf{x}^{(j-1)} + t_j v_j \mathbf{e}_j) - \frac{\partial f_i}{\partial x_j}(\mathbf{x}_0) \right| |v_j| \\ &< \frac{\epsilon}{mn} |v_j|.\end{aligned}$$

Step 3: Bound the error for each component f_i

$$\begin{aligned}
 \left| f_i(\mathbf{x}) - f_i(\mathbf{x}_0) - \sum_{j=1}^n \frac{\partial f_i}{\partial x_j}(\mathbf{x}_0) v_j \right| &= \left| \sum_{j=1}^n \left(f_i(\mathbf{x}^{(j)}) - f_i(\mathbf{x}^{(j-1)}) - \frac{\partial f_i}{\partial x_j}(\mathbf{x}_0) v_j \right) \right| \\
 &\leq \sum_{j=1}^n \left| \frac{\partial f_i}{\partial x_j}(\xi_j) - \frac{\partial f_i}{\partial x_j}(\mathbf{x}_0) \right| |v_j| \\
 &< \sum_{j=1}^n \frac{\epsilon}{mn} |v_j| \\
 &\leq \sum_{j=1}^n \frac{\epsilon}{mn} \|\mathbf{x} - \mathbf{x}_0\| = \frac{\epsilon}{m} \|\mathbf{x} - \mathbf{x}_0\|
 \end{aligned}$$

Step 4: Extend this to the vector-valued function f

$$\begin{aligned}
 \|f(\mathbf{x}) - f(\mathbf{x}_0) - L(\mathbf{x} - \mathbf{x}_0)\| &\leq \sum_{i=1}^m \left| f_i(\mathbf{x}) - f_i(\mathbf{x}_0) - \sum_{j=1}^n \frac{\partial f_i}{\partial x_j}(\mathbf{x}_0) v_j \right| \\
 &< \sum_{i=1}^m \frac{\epsilon}{m} \|\mathbf{x} - \mathbf{x}_0\| = \epsilon \|\mathbf{x} - \mathbf{x}_0\|
 \end{aligned}$$

Dividing both sides by $\|\mathbf{x} - \mathbf{x}_0\|$, we get:

$$\frac{\|f(\mathbf{x}) - f(\mathbf{x}_0) - L(\mathbf{x} - \mathbf{x}_0)\|}{\|\mathbf{x} - \mathbf{x}_0\|} < \epsilon$$

Since this holds for any $\epsilon > 0$ when $\|\mathbf{x} - \mathbf{x}_0\|$ is sufficiently small, we conclude:

$$\lim_{\mathbf{x} \rightarrow \mathbf{x}_0} \frac{\|f(\mathbf{x}) - f(\mathbf{x}_0) - L(\mathbf{x} - \mathbf{x}_0)\|}{\|\mathbf{x} - \mathbf{x}_0\|} = 0$$

This proves that f is differentiable at $\mathbf{x}_0$, and its derivative is given by L . ■

Lemma (Chain rule along a coordinate segment)

Let $E \subset \mathbb{R}^n$ and $f : E \rightarrow \mathbb{R}^m$. Fix $1 \leq i \leq m$ and $1 \leq j \leq n$. Define

$$\varphi_i^j(t) := f_i(\mathbf{z} + t v \mathbf{e}_j), \quad 0 \leq t \leq 1,$$

where $\mathbf{z} \in E$ and $v \in \mathbb{R}$. Assume that for some $t_0 \in (0, 1)$ there exists $\rho > 0$ such that

$$\mathbf{z} + (t_0 + h) v \mathbf{e}_j \in E \quad \text{for all } |h| < \rho,$$

and that the partial derivative $\frac{\partial f_i}{\partial x_j}$ exists at the point $\mathbf{p} := \mathbf{z} + t_0 v \mathbf{e}_j$. Then φ_i^j is differen-

tiabile at t_0 and

$$(\varphi_i^j)'(t_0) = \frac{\partial f_i}{\partial x_j}(\mathbf{p})v.$$

Proof.

Let

$$\begin{aligned} (\varphi_i^j)'(t_0) &= \lim_{h \rightarrow 0} \frac{\varphi_i^j(t_0 + h) - \varphi_i^j(t_0)}{h} = \lim_{h \rightarrow 0} \frac{f_i(\mathbf{z} + (t_0 + h)v\mathbf{e}_j) - f_i(\mathbf{z} + t_0v\mathbf{e}_j)}{h} \\ &= \lim_{h \rightarrow 0} \frac{f_i(\mathbf{p} + hv\mathbf{e}_j) - f_i(\mathbf{p})}{h}. \end{aligned}$$

If $v = 0$, $f_i(\mathbf{p} + hv\mathbf{e}_j) - f_i(\mathbf{p}) = 0$. Now consider $v \neq 0$.

For arbitrarily small h , $\mathbf{z} + (t_0 + h)v\mathbf{e}_j = \mathbf{p} + hv\mathbf{e}_j \in B(\mathbf{p}, \rho) \subseteq E$. Set $hv = s$, the above quotient becomes:

$$\lim_{s \rightarrow 0} \frac{f_i(\mathbf{p} + s\mathbf{e}_j) - f_i(\mathbf{p})}{s} \cdot v = \frac{\partial f_i}{\partial x_j}(\mathbf{p})v,$$

which shows that φ_i^j is differentiable at t_0 and $(\varphi_i^j)'(t_0) = \frac{\partial f_i}{\partial x_j}(\mathbf{p})v$. ■

Remark. Let $E \subseteq \mathbb{R}^n$, $f : E \rightarrow \mathbb{R}^m$ be a function, and $\mathbf{x}_0$ be an interior point of E . If f is differentiable at $\mathbf{x}_0$, by definition, there exists a linear map such that the relative error converge to zero near $\mathbf{x}_0$. Then we call this linear map (total) derivative of f at $\mathbf{x}_0$, denoted by

$$\begin{aligned} Df(\mathbf{x}_0) = f'(\mathbf{x}_0) &= \begin{pmatrix} - & \nabla f_1(\mathbf{x}_0) & - \\ - & \nabla f_2(\mathbf{x}_0) & - \\ & \vdots & \\ - & \nabla f_m(\mathbf{x}_0) & - \end{pmatrix}_{m \times n} \\ &= \begin{pmatrix} | & | & & | \\ \frac{\partial f}{\partial x_1}(\mathbf{x}_0) & \frac{\partial f}{\partial x_2}(\mathbf{x}_0) & \cdots & \frac{\partial f}{\partial x_n}(\mathbf{x}_0) \\ | & | & & | \end{pmatrix}_{m \times n}. \end{aligned}$$

Hence, we have:

1. The partial derivative is simply the directional derivative along the standard basis vector $\mathbf{e}_j$:

$$\frac{\partial f}{\partial x_j}(\mathbf{x}_0) = D_{\mathbf{e}_j}f(\mathbf{x}_0) = f'(\mathbf{x}_0)\mathbf{e}_j$$

2. If f is differentiable, then for any direction vector $\mathbf{v} = \sum_{j=1}^n v_j \mathbf{e}_j$, the directional derivative can be expanded as:

$$D_{\mathbf{v}}f(\mathbf{x}_0) = f'(\mathbf{x}_0)\mathbf{v} = f'(\mathbf{x}_0) \left(\sum_{j=1}^n v_j \mathbf{e}_j \right) = \sum_{j=1}^n v_j f'(\mathbf{x}_0) \mathbf{e}_j = \sum_{j=1}^n v_j \frac{\partial f}{\partial x_j}(\mathbf{x}_0)$$

This can be naturally represented in matrix multiplication form (where the derivative is a $m \times n$ matrix):

$$\left(\begin{array}{c|c|c|c} \frac{\partial f}{\partial x_1}(\mathbf{x}_0) & \frac{\partial f}{\partial x_2}(\mathbf{x}_0) & \cdots & \frac{\partial f}{\partial x_n}(\mathbf{x}_0) \\ \hline & & & \end{array} \right)_{m \times n} \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix}_{n \times 1}.$$

Definition 1.4 (Frobenius Norm)

Let $A = (a_{ij})_{i,j} \in M_{m \times n}(\mathbb{R})$. We can regard A as a vector in $\mathbb{R}^{mn}$, and define its Frobenius norm by

$$\|A\|_F = \sqrt{\sum_{i=1}^m \sum_{j=1}^n a_{ij}^2}.$$

Remark. The Frobenius norm can also be elegantly expressed using the trace of a matrix. For $A \in M_{m \times n}(\mathbb{R})$, consider the matrix product $AA^\top$. By the definition of matrix multiplication, the i -th diagonal entry of $AA^\top$ is given by:

$$(AA^\top)_{ii} = \sum_{j=1}^n a_{ij}(A^\top)_{ji} = \sum_{j=1}^n a_{ij}^2$$

Since the trace of a matrix is the sum of its diagonal entries, taking the trace of $AA^\top$ yields:

$$\text{tr}(AA^\top) = \sum_{i=1}^m (AA^\top)_{ii} = \sum_{i=1}^m \sum_{j=1}^n a_{ij}^2$$

Therefore, the Frobenius norm can be equivalently written as:

$$\|A\|_F = \sqrt{\text{tr}(AA^\top)}.$$

Property 1.1

Let $A, B \in M_{n \times n}(\mathbb{R})$ and $c \in \mathbb{R}$. The Frobenius norm is defined as $\|A\|_F = \sqrt{\sum_{i,j} a_{ij}^2} = \sqrt{\text{tr}(AA^\top)}$. It satisfies the following:

1. **Positive Definiteness:** $\|A\|_F \geq 0$, and $\|A\|_F = 0 \iff A = \mathbf{0}$.

2. **Absolute Homogeneity:** $\|cA\|_F = |c|\|A\|_F$.

3. **Triangle Inequality:** $\|A + B\|_F \leq \|A\|_F + \|B\|_F$.

4. **Sub-multiplicativity:** $\|AB\|_F \leq \|A\|_F \|B\|_F$.

5. **Power Inequality:** $\|A^k\|_F \leq \|A\|_F^k$ for $k \in \mathbb{N}$.

6. **Continuity:** For any column vector $\mathbf{x} \in \mathbb{R}^n$, $\|A\mathbf{x}\| \leq \|A\|_F \|\mathbf{x}\|$. This implies that the linear map $\mathbf{x} \mapsto A\mathbf{x}$ is continuous on $\mathbb{R}^n$.

Proof.

We will only prove the sub-multiplicativity, and the rest is left as an exercise. By the definition of matrix multiplication, the (i, j) -th entry of the product matrix AB is given by:

$$(AB)_{ij} = \sum_{k=1}^n a_{ik} b_{kj}$$

Applying the Cauchy-Schwarz inequality to the vectors $(a_{i1}, \dots, a_{in})$ and $(b_{1j}, \dots, b_{nj})$, we obtain an upper bound for the square of this entry:

$$|(AB)_{ij}|^2 = \left| \sum_{k=1}^n a_{ik} b_{kj} \right|^2 \leq \left(\sum_{k=1}^n a_{ik}^2 \right) \left(\sum_{k=1}^n b_{kj}^2 \right)$$

Now, we compute the squared Frobenius norm of the matrix AB by summing over all entries ($1 \leq i \leq m$ and $1 \leq j \leq p$):

$$\begin{aligned} \|AB\|_F^2 &= \sum_{i=1}^m \sum_{j=1}^p (AB)_{ij}^2 \\ &\leq \sum_{i=1}^m \sum_{j=1}^p \left(\sum_{k=1}^n a_{ik}^2 \right) \left(\sum_{k=1}^n b_{kj}^2 \right) \end{aligned}$$

Notice that the sum over i only depends on the terms a_{ik} , and the sum over j only depends on the terms b_{kj} . Therefore, we can separate and factor the summations:

$$\begin{aligned} \|AB\|_F^2 &\leq \left(\sum_{i=1}^m \sum_{k=1}^n a_{ik}^2 \right) \left(\sum_{j=1}^p \sum_{k=1}^n b_{kj}^2 \right) \\ &= \|A\|_F^2 \cdot \|B\|_F^2 \end{aligned}$$

Finally, taking the square root of both sides, we get:

$$\|AB\|_F \leq \|A\|_F \|B\|_F$$

This completes the proof. ■

Theorem 1.6 (Neumann Series)

Let $M \in \mathbb{R}^{n \times n}$. If $\|M\|_F < 1$, then $(I + M)$ is invertible, and its inverse is given by the convergent series:

$$(I + M)^{-1} = \sum_{i=0}^{\infty} (-1)^i M^i$$

Proof.

Let $S_k = \sum_{i=0}^k (-1)^i M^i$. Since $(\mathbb{R}^{n \times n}, \|\cdot\|_F)$ is a complete metric space, it suffices to show that $\{S_k\}$ is a Cauchy sequence.

For $k > m$:

$$\|S_k - S_m\|_F = \left\| \sum_{i=m+1}^k (-1)^i M^i \right\|_F \leq \sum_{i=m+1}^k \|M\|_F^i \leq \frac{\|M\|_F^{m+1}}{1 - \|M\|_F}$$

As $m \rightarrow \infty$, the right-hand side vanishes since $\|M\|_F < 1$. Thus, $\{S_k\}$ is Cauchy and converges to some matrix K .

To verify $K = (I + M)^{-1}$, observe:

$$\begin{aligned} (I + M)S_k &= (I + M) \sum_{i=0}^k (-1)^i M^i = \sum_{i=0}^k (-1)^i M^i + \sum_{i=0}^k (-1)^i M^{i+1} \\ &= \sum_{i=0}^k (-1)^i M^i + \sum_{j=1}^{k+1} (-1)^{j-1} M^j = \sum_{i=0}^k (-1)^i M^i - \sum_{j=1}^{k+1} (-1)^j M^j. \end{aligned}$$

Canceling the telescoping terms, we obtain:

$$(I + M)S_k = I - (-1)^{k+1} M^{k+1}$$

Taking the limit as $k \rightarrow \infty$ (since $\|M\|_F < 1 \implies M^{k+1} \rightarrow \mathbf{0}$):

$$(I + M)K = I + \mathbf{0} = I$$

Therefore, $K = (I + M)^{-1}$. ■

Theorem 1.7 (Chain Rule)

Let $E \subseteq \mathbb{R}^n$, $F \subseteq \mathbb{R}^m$, $f : E \rightarrow F$, $g : F \rightarrow \mathbb{R}^p$ and $\mathbf{x}_0 \in \text{int}(E)$. Suppose f is differentiable at $\mathbf{x}_0$, $f(\mathbf{x}_0)$ lies in the interior of F , and g is differentiable at $f(\mathbf{x}_0)$. Then $g \circ f$ is differentiable at $\mathbf{x}_0$, and

$$(g \circ f)'(\mathbf{x}_0) = g'(f(\mathbf{x}_0))f'(\mathbf{x}_0).$$

Proof.

Let $\mathbf{y}_0 = f(\mathbf{x}_0)$ and denote the derivatives by $A = f'(\mathbf{x}_0) : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $B =$

$$g'(f(\mathbf{x}_0)) = g'(\mathbf{y}_0) : \mathbb{R}^m \rightarrow \mathbb{R}^p.$$

Step 1: Define the error function η

Define $\eta : \mathbb{R}^m \rightarrow \mathbb{R}^p$ by:

$$\eta(\mathbf{k}) = \begin{cases} \frac{g(\mathbf{y}_0 + \mathbf{k}) - g(\mathbf{y}_0) - B\mathbf{k}}{\|\mathbf{k}\|} & \text{if } \mathbf{k} \neq \mathbf{0} \\ \mathbf{0} & \text{if } \mathbf{k} = \mathbf{0} \end{cases}$$

Because g is differentiable at $\mathbf{y}_0$, $\lim_{\mathbf{k} \rightarrow \mathbf{0}} \eta(\mathbf{k}) = \mathbf{0}$, which implies η is continuous at $\mathbf{0}$. Moreover, the relation

$$g(\mathbf{y}_0 + \mathbf{k}) - g(\mathbf{y}_0) - B\mathbf{k} = \eta(\mathbf{k})\|\mathbf{k}\|$$

is true for all $\mathbf{k} \in \mathbb{R}^m$.

Step 2: Expand the composition difference

Take $\mathbf{h}$ such that $\mathbf{x}_0 + \mathbf{h} \in E$. Let $\mathbf{k} = f(\mathbf{x}_0 + \mathbf{h}) - f(\mathbf{x}_0)$. Consider the difference:

$$\begin{aligned} (g \circ f)(\mathbf{x}_0 + \mathbf{h}) - (g \circ f)(\mathbf{x}_0) &= g(f(\mathbf{x}_0 + \mathbf{h})) - g(f(\mathbf{x}_0)) \\ &= g(\mathbf{y}_0 + \mathbf{k}) - g(\mathbf{y}_0) \\ &= B(f(\mathbf{x}_0 + \mathbf{h}) - f(\mathbf{x}_0)) \\ &\quad + \eta(f(\mathbf{x}_0 + \mathbf{h}) - f(\mathbf{x}_0))\|f(\mathbf{x}_0 + \mathbf{h}) - f(\mathbf{x}_0)\| \end{aligned}$$

Subtracting $BA\mathbf{h}$ from both sides, we get:

$$\begin{aligned} (g \circ f)(\mathbf{x}_0 + \mathbf{h}) - (g \circ f)(\mathbf{x}_0) - BA\mathbf{h} \\ = B(f(\mathbf{x}_0 + \mathbf{h}) - f(\mathbf{x}_0) - A\mathbf{h}) + \eta(f(\mathbf{x}_0 + \mathbf{h}) - f(\mathbf{x}_0))\|f(\mathbf{x}_0 + \mathbf{h}) - f(\mathbf{x}_0)\| \end{aligned}$$

Step 3: Estimate the first term

Note that:

$$\frac{\|B(f(\mathbf{x}_0 + \mathbf{h}) - f(\mathbf{x}_0) - A\mathbf{h})\|}{\|\mathbf{h}\|} \leq \|B\| \frac{\|f(\mathbf{x}_0 + \mathbf{h}) - f(\mathbf{x}_0) - A\mathbf{h}\|}{\|\mathbf{h}\|}$$

By the assumption that f is differentiable at $\mathbf{x}_0$, $\lim_{\mathbf{h} \rightarrow \mathbf{0}} \frac{\|f(\mathbf{x}_0 + \mathbf{h}) - f(\mathbf{x}_0) - A\mathbf{h}\|}{\|\mathbf{h}\|} = 0$.

By the Squeeze Theorem, the first term converges to zero.

Step 4: Estimate the second term

Since f is differentiable at $\mathbf{x}_0$, f is continuous at $\mathbf{x}_0$, meaning $\lim_{\mathbf{h} \rightarrow \mathbf{0}} (f(\mathbf{x}_0 + \mathbf{h}) - f(\mathbf{x}_0)) = \mathbf{0}$. On the other hand, η is continuous at $\mathbf{0}$ (and $\lim_{\mathbf{k} \rightarrow \mathbf{0}} \eta(\mathbf{k}) = \mathbf{0}$), it follows that $\lim_{\mathbf{h} \rightarrow \mathbf{0}} \eta(f(\mathbf{x}_0 + \mathbf{h}) - f(\mathbf{x}_0)) = \mathbf{0}$

Next, consider:

$$\begin{aligned}
\frac{\|f(\mathbf{x}_0 + \mathbf{h}) - f(\mathbf{x}_0)\|}{\|\mathbf{h}\|} &= \frac{\|f(\mathbf{x}_0 + \mathbf{h}) - f(\mathbf{x}_0) - A\mathbf{h} + A\mathbf{h}\|}{\|\mathbf{h}\|} \\
&\leq \frac{\|f(\mathbf{x}_0 + \mathbf{h}) - f(\mathbf{x}_0) - A\mathbf{h}\|}{\|\mathbf{h}\|} + \frac{\|A\mathbf{h}\|}{\|\mathbf{h}\|} \\
&\leq \underbrace{\frac{\|f(\mathbf{x}_0 + \mathbf{h}) - f(\mathbf{x}_0) - A\mathbf{h}\|}{\|\mathbf{h}\|}}_{\rightarrow 0 \text{ by differentiability}} + \underbrace{\|A\|}_{\text{fixed}}
\end{aligned}$$

By the Squeeze Theorem:

$$\lim_{\mathbf{h} \rightarrow \mathbf{0}} \frac{\|\eta(f(\mathbf{x}_0 + \mathbf{h}) - f(\mathbf{x}_0))\| \|f(\mathbf{x}_0 + \mathbf{h}) - f(\mathbf{x}_0)\|}{\|\mathbf{h}\|} = 0$$

Combining Step 3 and Step 4, we conclude:

$$\lim_{\mathbf{h} \rightarrow \mathbf{0}} \frac{\|(g \circ f)(\mathbf{x}_0 + \mathbf{h}) - (g \circ f)(\mathbf{x}_0) - BA\mathbf{h}\|}{\|\mathbf{h}\|} = 0$$

This proves that $g \circ f$ is differentiable at $\mathbf{x}_0$ with derivative $BA = g'(f(\mathbf{x}_0))f'(\mathbf{x}_0)$. ■

Definition 1.5

Let $E \subseteq \mathbf{R}^n$ be open and $f : E \rightarrow \mathbf{R}^m$. We say f is twice continuously differentiable on E , denoted by $f \in C^2(E)$.

1. f is continuously differentiable (its first order partial derivatives exist and are continuous).
2. For each $i = 1, 2, \dots, n$, the partial derivative $\frac{\partial f}{\partial x_i} : E \rightarrow \mathbf{R}^m$ is continuously differentiable on E (its second order partial derivatives exist and are continuous on E).

Theorem 1.8 (Clairaut)

Let E be an open set and $f \in C^2(E)$. Then for all $1 \leq i, j \leq n$ and all $\mathbf{x}_0 \in E$:

$$\frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{x}_0) = \frac{\partial^2 f}{\partial x_j \partial x_i}(\mathbf{x}_0).$$

Proof.

Let $f = (f_1, f_2, \dots, f_m)$ where $f_k : E \rightarrow \mathbf{R}$. Without loss of generality, we can assume $f : E \rightarrow \mathbf{R}$ and the point of interest is $\mathbf{x}_0 = \mathbf{0}$. (By some shifting, $F(\mathbf{x}) = f(\mathbf{x} + \mathbf{x}_0) \Rightarrow F(\mathbf{0}) = f(\mathbf{x}_0)$). We also assume $i \neq j$.

Since $f \in C^2(E)$ and $\mathbf{0} \in E$, given any $\epsilon > 0$, there exists $\delta_0 > 0$ such that:

$$\left| \frac{\partial^2 f}{\partial x_j \partial x_i}(\mathbf{x}) - \frac{\partial^2 f}{\partial x_j \partial x_i}(\mathbf{0}) \right| \leq \frac{\epsilon}{2} \quad \text{and} \quad \left| \frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{x}) - \frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{0}) \right| \leq \frac{\epsilon}{2}$$

if $\|\mathbf{x}\| < 2\delta_0$.

Step 1: The Second-Order Difference (DiD Approximation)

Fix $\delta < \delta_0$, and consider the rectangle. Define the second-order difference X :

$$X = f(\delta \mathbf{e}_i + \delta \mathbf{e}_j) - f(\delta \mathbf{e}_i) - f(\delta \mathbf{e}_j) + f(\mathbf{0})$$

We can group the terms in two different ways:

$$X = (f(\delta \mathbf{e}_i + \delta \mathbf{e}_j) - f(\delta \mathbf{e}_i)) - (f(\delta \mathbf{e}_j) - f(\mathbf{0})) \quad (1)$$

$$X = (f(\delta \mathbf{e}_i + \delta \mathbf{e}_j) - f(\delta \mathbf{e}_j)) - (f(\delta \mathbf{e}_i) - f(\mathbf{0})) \quad (2)$$

Step 2: Apply Mean Value Theorem for Grouping (1)

Let $\phi(t) = f(t \mathbf{e}_i + \delta \mathbf{e}_j) - f(t \mathbf{e}_i)$. Then from (1), we have:

$$X = \phi(\delta) - \phi(0)$$

By the MVT, $X = \delta \phi'(c_i)$ for some $c_i \in (0, \delta)$ (*).

Note that $\phi'(t) = \frac{\partial f}{\partial x_i}(t \mathbf{e}_i + \delta \mathbf{e}_j) - \frac{\partial f}{\partial x_i}(t \mathbf{e}_i)$. So, $\phi'(c_i) = \frac{\partial f}{\partial x_i}(c_i \mathbf{e}_i + \delta \mathbf{e}_j) - \frac{\partial f}{\partial x_i}(c_i \mathbf{e}_i)$.

Now, let $g(s) = \frac{\partial f}{\partial x_i}(c_i \mathbf{e}_i + s \mathbf{e}_j)$. Then $g'(s) = \frac{\partial^2 f}{\partial x_j \partial x_i}(c_i \mathbf{e}_i + s \mathbf{e}_j)$. We can rewrite (*) as:

$$\phi'(c_i) = g(\delta) - g(0)$$

By applying MVT again to $g(s)$, there exists $c_j \in (0, \delta)$ such that $g(\delta) - g(0) = \delta g'(c_j)$. Thus, $\phi'(c_i) = \delta \frac{\partial^2 f}{\partial x_j \partial x_i}(c_i \mathbf{e}_i + c_j \mathbf{e}_j)$ (**).

From (*) and (**), we obtain:

$$X = \delta^2 \frac{\partial^2 f}{\partial x_j \partial x_i}(c_i \mathbf{e}_i + c_j \mathbf{e}_j) \quad \text{where } 0 < c_i < \delta, 0 < c_j < \delta.$$

Step 3: Symmetric Argument for Grouping (2)

Similarly, starting from (2) and taking derivatives first with respect to x_j and then x_i , MVT gives us:

$$X = \delta^2 \frac{\partial^2 f}{\partial x_i \partial x_j}(d_i \mathbf{e}_i + d_j \mathbf{e}_j) \quad \text{where } 0 < d_i < \delta, 0 < d_j < \delta.$$

Step 4: Bounding the Error using Continuity

Notice that the norm of the points we found is strictly bounded:

$$\|c_i \mathbf{e}_i + c_j \mathbf{e}_j\| = \sqrt{c_i^2 + c_j^2} < \sqrt{\delta^2 + \delta^2} = \sqrt{2}\delta < \sqrt{2}\delta_0 < 2\delta_0$$

Because these points fall within the $2\delta_0$ radius, our initial continuity conditions apply:

$$\left| \frac{\partial^2 f}{\partial x_j \partial x_i}(c_i \mathbf{e}_i + c_j \mathbf{e}_j) - \frac{\partial^2 f}{\partial x_j \partial x_i}(\mathbf{0}) \right| \leq \frac{\epsilon}{2} \implies \left| \frac{X}{\delta^2} - \frac{\partial^2 f}{\partial x_j \partial x_i}(\mathbf{0}) \right| \leq \frac{\epsilon}{2}$$

Similarly,

$$\left| \frac{X}{\delta^2} - \frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{0}) \right| \leq \frac{\epsilon}{2}$$

By the triangle inequality:

$$\left| \frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{0}) - \frac{\partial^2 f}{\partial x_j \partial x_i}(\mathbf{0}) \right| \leq \epsilon \quad \forall \epsilon > 0$$

Hence, we conclude that:

$$\frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{0}) = \frac{\partial^2 f}{\partial x_j \partial x_i}(\mathbf{0}).$$

■

Remark (Numerical Approximation via Second-Order Difference).

From the proof above, we know that by the continuity of $\frac{\partial^2 f}{\partial x_j \partial x_i}$ and $\frac{\partial^2 f}{\partial x_i \partial x_j}$,

$$\lim_{\delta \rightarrow 0} \frac{X}{\delta^2} = \frac{\partial^2 f}{\partial x_j \partial x_i}(\mathbf{0}) \quad \text{and} \quad \lim_{\delta \rightarrow 0} \frac{X}{\delta^2} = \frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{0})$$

Hence, if $f \in C^2(E)$, for any point $\mathbf{x}_0$, we can use numerical approximation by a second-order finite difference without computing the derivative directly:

$$\lim_{\delta \rightarrow 0} \frac{f(\mathbf{x}_0 + \delta \mathbf{e}_i + \delta \mathbf{e}_j) - f(\mathbf{x}_0 + \delta \mathbf{e}_i) - f(\mathbf{x}_0 + \delta \mathbf{e}_j) + f(\mathbf{x}_0)}{\delta^2} = \frac{\partial^2 f}{\partial x_j \partial x_i}(\mathbf{x}_0) = \frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{x}_0)$$

1.2 Inverse Function Theorem

Definition 1.6 (Metric Space)

The pair (X, d) is called a metric space if a set X endowed with the function $d : X \times X \rightarrow \mathbb{R}$ such that ...

Definition 1.7 (Convergence)

Given a sequence $\{x_n\}_{n=1}^\infty$ in X . We say the sequence converges to x if $\forall \epsilon > 0 \exists N \in \mathbb{N}$ s.t. for $n \geq N$, $d(x_n, x) < \epsilon$.

Definition 1.8 (Cauchy Sequence)

A sequence $\{x_n\}_{n=1}^{\infty}$ is said to be Cauchy if $\forall \varepsilon > 0, \exists N \in \mathbb{N}$ s.t. for $m, n \geq N, d(x_n, x_m) < \varepsilon$.

Definition 1.9 (Completeness)

(X, d) is complete if every Cauchy sequence in X converges to a limit in X .

Theorem 1.9

Let E be a closed subset in X , and (X, d) is a complete metric space. Then (E, d) is also complete.

Proof.

Take any Cauchy sequence in E , which is also a Cauchy in X . By completeness of X , the sequence converges to some point in X . Since a closed set contains all limit point, the Cauchy sequence converges to a limit in E . ■

Definition 1.10 (Contraction)

Let (X, d) be a metric space and $T : X \rightarrow X$ be a map.

1. We say T is a contraction if $d(T(x), T(y)) \leq d(x, y) \forall x, y \in X$.
2. T is a strict contraction if there exists a constant $0 < c < 1$ s.t. $d(T(x), T(y)) \leq cd(x, y)$.

Remark. Under the standard setting, the term **contraction** simply refers to a **strict contraction**. Terence Tao use “contraction” for maps where $d(T(x), T(y)) \leq d(x, y)$ and explicitly write “strict contraction” when the constant is strictly less than 1.

Definition 1.11 (Fixed Point)

Let $T : X \rightarrow X$ be a map and $x^* \in X$. We say x^* is a fixed point of T if $T(x^*) = x^*$.

Remark. Suppose X is a subset of $\mathbb{R}$. The existence of a fixed point means that the graph of $y = f(x)$ intersect with $y = x$.

Theorem 1.10 (Contraction Mapping Theorem)

Let (X, d) be a metric space and $T : X \rightarrow X$ be a (strict) contraction. Then T has at most one fixed point. Moreover, if we assume X is non-empty and complete, then T has exactly one fixed point.

Proof.

Suppose x, y are distinct fixed point of T , consider $d(x, y) = d(T(x), T(y)) \leq cd(x, y)$ for some $c \in (0, 1)$. The inequality $(1 - c)d(x, y) \leq 0$ forces $x = y$.

Now we further assume X is complete. Take any point x_0 , construct the sequence $\{x_n\}_{n=0}^{\infty}$, $x_{n+1} = T(x_n) \forall n \in \mathbb{N}$. Observe that $d(x_{n+1}, x_n) \leq c^n d(x_1, x_0)$ for $n \geq 1$. Suppose $n > m \geq 1$,

$$\begin{aligned} d(x_n, x_m) &\leq d(x_n, x_{n-1}) + d(x_{n-1}, x_{n-2}) + \cdots + d(x_{m+1}, x_m) \\ &\leq c^{n-1} d(x_1, x_0) + c^{n-2} d(x_1, x_0) + \cdots + c^m d(x_1, x_0) \\ &\leq \frac{c^m}{1-c} d(x_1, x_0). \end{aligned}$$

Take $m \rightarrow \infty$, $c^m \rightarrow 0$, which shows $\{x_n\}_{n=1}^{\infty}$ is a Cauchy in X . Since X is complete, $\lim_{n \rightarrow \infty} x_n$ exists and converges to x^* . Finally,

$$\begin{aligned} d(x^*, T(x^*)) &\leq d(x^*, x_{n+1}) + d(x_{n+1}, T(x^*)) \\ &\leq d(x^*, x_{n+1}) + d(T(x_n), T(x^*)) \\ &\leq d(x^*, x_{n+1}) + c d(x_n, x^*). \end{aligned}$$

This shows $x^* = T(x^*)$. ■

Definition 1.12 (Lipschitz continuous)

Given (X, d_X) , (Y, d_Y) and a map $f : X \rightarrow Y$. We say f is *Lipschitz continuous* if there exists a constant $L \geq 0$ such that $d_Y(f(x_1), f(x_2)) \leq L d_X(x_1, x_2)$ for any $x_1, x_2 \in X$. Any such L is referred to as a *Lipschitz constant* for the function f , and f may also be referred to as *L -Lipschitz*.

Definition 1.13 (Uniform Continuity)

Let $f : X \rightarrow Y$ be a map btw (X, d_X) and (Y, d_Y) . f is *uniformly continuous* if $\forall \epsilon > 0 \exists \delta > 0$ s.t. for any $x, y \in X$ $d_X(x, y) < \delta \rightarrow d_Y(f(x), f(y)) < \epsilon$. This means δ is independent of the location of $x \in X$.

Theorem 1.11

If a function f is Lipschitz continuous, then

1. f is continuous.
2. f is uniformly continuous.

Proof.

Let $f : X \rightarrow Y$ be a L -Lipschitz continuous map. Fix $\epsilon > 0$ and a point $a \in X$, choose $\frac{\epsilon}{L}$ such that if $d(x, a) < \frac{\epsilon}{L}$, one has $d(f(x), f(a)) \leq L d(x, a) < L \frac{\epsilon}{L} = \epsilon$, which shows that f is continuous at a . The argument for 2 is analogous. ■

Corollary

A contraction $T : X \rightarrow X$ is a continuous function.

Proof.

A contraction with contraction ratio c is c -Lipschitz, by theorem (1.11), T is continuous. ■

Theorem 1.12

Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation. Then T is continuous.

Proof.

Let $\{e_1, e_2, \dots, e_n\}$ be the standard basis for $\mathbb{R}^n$. Any vector $x \in \mathbb{R}^n$ can be expressed as $x = \sum_{i=1}^n x_i e_i$. By the linearity of T and the triangle inequality, we have:

$$\|T(x)\| = \left\| T \left(\sum_{i=1}^n x_i e_i \right) \right\| = \left\| \sum_{i=1}^n x_i T(e_i) \right\| \leq \sum_{i=1}^n |x_i| \|T(e_i)\|$$

In the standard Euclidean norm, the absolute value of any component is bounded by the norm of the vector, so $|x_i| \leq \|x\|$ for all $i = 1, \dots, n$. Substituting this yields:

$$\|T(x)\| \leq \sum_{i=1}^n \|x\| \|T(e_i)\| = \|x\| \sum_{i=1}^n \|T(e_i)\|$$

Define $M = \sum_{i=1}^n \|T(e_i)\|$. Since $\{T(e_i)\}$ is a finite set of fixed vectors in $\mathbb{R}^m$, M is a finite non-negative constant independent of x . Thus, we obtain the boundedness condition $\|T(x)\| \leq M\|x\|$.

For any $x, y \in \mathbb{R}^n$, the linearity of T implies:

$$\|T(x) - T(y)\| = \|T(x - y)\| \leq M\|x - y\|$$

This demonstrates that T is M -Lipschitz continuous. By theorem (1.11), it follows that T is continuous. ■

Lemma

Let $B(\mathbf{0}, r)$ be an open ball in $\mathbb{R}^n$ and $g : B(\mathbf{0}, r) \rightarrow \mathbb{R}^n$ be a map with $g(\mathbf{0}) = \mathbf{0}$. If $\|g(\mathbf{x}) - g(\mathbf{y})\| \leq \frac{1}{2}\|\mathbf{x} - \mathbf{y}\|$ for $\mathbf{x}, \mathbf{y} \in B(\mathbf{0}, r)$, then the function $f : B(\mathbf{0}, r) \rightarrow \mathbb{R}^n$ defined by $f(\mathbf{x}) = \mathbf{x} + g(\mathbf{x})$ is one-to-one, and the image of $f(B(\mathbf{0}, r))$ contains $B(\mathbf{0}, \frac{r}{2})$.

Proof.

We first show injection. Suppose $f(\mathbf{x}) = f(\mathbf{y})$, by construction, we have $\|\mathbf{x} - \mathbf{y}\| = \|g(\mathbf{x}) - g(\mathbf{y})\| \leq \frac{1}{2}\|\mathbf{x} - \mathbf{y}\|$. The inequality $\frac{1}{2}\|\mathbf{x} - \mathbf{y}\| \leq 0$ forces $\|\mathbf{x} - \mathbf{y}\| = 0$. So f is one-to-one.

Next, we want to show $B(\mathbf{0}, \frac{r}{2}) \subseteq f(B(\mathbf{0}, r))$. It suffices to show $\forall \mathbf{y} \in B(\mathbf{0}, \frac{r}{2})$, $\exists \mathbf{x} \in B(\mathbf{0}, r)$ s.t. $f(\mathbf{x}) = \mathbf{y}$. Define $F(\mathbf{x}) = \mathbf{y} - g(\mathbf{x})$. If we can find the fixed point of F , then we equivalently find $\mathbf{y} = f(\mathbf{x})$. Fix $\mathbf{y} \in B(\mathbf{0}, \frac{r}{2})$. $\|\mathbf{y}\| < \frac{r}{2}$. $\exists \varepsilon > 0$ s.t. $\|\mathbf{y}\| \leq \frac{r-\varepsilon}{2}$. Consider the map $F : \overline{B(\mathbf{0}, r-\varepsilon)} \rightarrow \overline{B(\mathbf{0}, r-\varepsilon)}$. Take $\mathbf{x} \in \overline{B(\mathbf{0}, r-\varepsilon)}$, $\|\mathbf{x}\| \leq r-\varepsilon$. Then we have $\|F(\mathbf{x})\| \leq \|\mathbf{y}\| + \|g(\mathbf{x})\| \leq \frac{r-\varepsilon}{2} + \|g(\mathbf{x})\| \leq \frac{r-\varepsilon}{2} + \frac{r-\varepsilon}{2} = r-\varepsilon$, where $\|g(\mathbf{x})\| = \|g(\mathbf{x}) - g(\mathbf{0})\| \leq \frac{1}{2}\|\mathbf{x}\| \leq \frac{1}{2}(r-\varepsilon)$. So $F(\mathbf{x}) \in \overline{B(\mathbf{0}, r-\varepsilon)}$. Given a fixed $\mathbf{y}$, $\|F(\mathbf{x}_1) - F(\mathbf{x}_2)\| = \|\mathbf{y} - g(\mathbf{x}_1) - (\mathbf{y} - g(\mathbf{x}_2))\| \leq \|g(\mathbf{x}_2) - g(\mathbf{x}_1)\| \leq \frac{1}{2}\|\mathbf{x}_2 - \mathbf{x}_1\|$. Hence, $F : \overline{B(\mathbf{0}, r-\varepsilon)} \rightarrow \overline{B(\mathbf{0}, r-\varepsilon)}$ is a contraction $\forall \mathbf{y} \in B(\mathbf{0}, \frac{r}{2})$. Since $\overline{B(\mathbf{0}, r-\varepsilon)}$ is complete, $\exists \mathbf{x}^* \in \overline{B(\mathbf{0}, r-\varepsilon)}$ s.t. $\mathbf{x}^* = F(\mathbf{x}^*)$. This shows $F(\mathbf{x}^*) = \mathbf{y} - g(\mathbf{x}^*) = \mathbf{x}^*$, and hence $\mathbf{x}^* + g(\mathbf{x}^*) = \mathbf{y}$. ■

Theorem 1.13 (Inverse Function Theorem)

Let E be open in $\mathbb{R}^n$ and $f : E \rightarrow \mathbb{R}^n \in C^1(E)$. Suppose the linear transformation $f'(\mathbf{x}_0) : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is invertible, then there exists an open set $U \subseteq E$ containing $\mathbf{x}_0$ and an open set $V \subseteq \mathbb{R}^n$ containing $f(\mathbf{x}_0)$ such that $f : U \rightarrow V$ is a bijection. Moreover, the inverse map $f^{-1} : V \rightarrow U$ is also continuously differentiable, and

$$(f^{-1})'(f(\mathbf{x}_0)) = (f'(\mathbf{x}_0))^{-1}.$$

Proof.

Let $A = f'(\mathbf{x}_0)$. Firstly, f is differentiable at $\mathbf{x}_0$, so:

$$\lim_{\mathbf{x} \rightarrow \mathbf{0}} \frac{\|f(\mathbf{x}_0 + \mathbf{x}) - f(\mathbf{x}_0) - A\mathbf{x}\|}{\|\mathbf{x}\|} = 0.$$

Given A is invertible, as $\mathbf{x} \rightarrow \mathbf{0}$, we have:

$$\frac{\|A^{-1}(f(\mathbf{x}_0 + \mathbf{x}) - f(\mathbf{x}_0) - A\mathbf{x})\|}{\|\mathbf{x}\|} \leq \|A^{-1}\| \frac{\|f(\mathbf{x}_0 + \mathbf{x}) - f(\mathbf{x}_0) - A\mathbf{x}\|}{\|\mathbf{x}\|} \rightarrow 0.$$

Step 1: Construct the auxiliary function g and show it is a contraction

Let $g(\mathbf{x}) = A^{-1}(f(\mathbf{x}_0 + \mathbf{x}) - f(\mathbf{x}_0)) - \mathbf{x}$. It is clear that $g(\mathbf{0}) = \mathbf{0}$. Substituting this into our limit condition yields $\lim_{\mathbf{x} \rightarrow \mathbf{0}} \frac{\|g(\mathbf{x})\|}{\|\mathbf{x}\|} = 0$. Rewrite the limit condition as:

$$\lim_{\mathbf{x} \rightarrow \mathbf{0}} \frac{\|g(\mathbf{x}) - g(\mathbf{0}) - g'(\mathbf{0})\mathbf{x}\|}{\|\mathbf{x}\|} = 0,$$

which shows that g is differentiable at $\mathbf{0}$ with $g'(\mathbf{0}) = \mathbf{0}$.

Since both f and A are C^1 , g is also C^1 . By definition, there exists an $r > 0$ such that for all $\mathbf{x} \in B(\mathbf{0}, r)$:

$$\left\| \frac{\partial g}{\partial x_j}(\mathbf{x}) - \frac{\partial g}{\partial x_j}(\mathbf{0}) \right\| = \left\| \frac{\partial g}{\partial x_j}(\mathbf{x}) \right\| \leq \frac{1}{2n^2}.$$

For $\mathbf{x} \in B(\mathbf{0}, r)$ and $\mathbf{v} \in \mathbb{R}^n$,

$$\|D_{\mathbf{v}}g(\mathbf{x})\| = \left\| \sum_{j=1}^n v_j \frac{\partial g}{\partial x_j}(\mathbf{x}) \right\| \leq \sum_{j=1}^n |v_j| \left\| \frac{\partial g}{\partial x_j}(\mathbf{x}) \right\| \leq \sum_{j=1}^n \|\mathbf{v}\| \frac{1}{2n^2} = \frac{1}{2n} \|\mathbf{v}\| \leq \frac{1}{2} \|\mathbf{v}\|.$$

For $\mathbf{x}, \mathbf{y} \in B(\mathbf{0}, r)$, consider:

$$\begin{aligned} g(\mathbf{y}) - g(\mathbf{x}) &= \int_0^1 \frac{d}{dt} g(\mathbf{x} + t(\mathbf{y} - \mathbf{x})) dt \quad (\text{FTC(1)}) \\ &= \int_0^1 Dg(\mathbf{x} + t(\mathbf{y} - \mathbf{x}))(\mathbf{y} - \mathbf{x}) dt \quad (\text{Chain rule}) \\ &= \int_0^1 D_{\mathbf{y}-\mathbf{x}}g(\mathbf{x} + t(\mathbf{y} - \mathbf{x})) dt \quad (\text{Since } g \text{ is } C^1) \end{aligned}$$

Therefore,

$$\|g(\mathbf{y}) - g(\mathbf{x})\| = \left\| \int_0^1 D_{\mathbf{y}-\mathbf{x}}g(\mathbf{x} + t(\mathbf{y} - \mathbf{x})) dt \right\| \leq \int_0^1 \frac{1}{2} \|\mathbf{y} - \mathbf{x}\| dt \leq \frac{1}{2} \|\mathbf{y} - \mathbf{x}\|,$$

which means g is a contraction.

Step 2: Define Φ , establish local bijection, and show Φ^{-1} is continuous

Define $\Phi(\mathbf{x}) = g(\mathbf{x}) + \mathbf{x} = A^{-1}(f(\mathbf{x}_0 + \mathbf{x}) - f(\mathbf{x}_0))$, where $g(\mathbf{0}) = \mathbf{0}$ and g is a contraction. Apply the lemma (1.2), Φ is one-to-one on $B(\mathbf{0}, r)$ and $B(\mathbf{0}, \frac{r}{2}) \subseteq \Phi(B(\mathbf{0}, r))$. Set $W = \Phi^{-1}(B(\mathbf{0}, \frac{r}{2}))$, then $\Phi|_W : W \rightarrow B(\mathbf{0}, \frac{r}{2})$ is a bijection.

Clearly, Φ is continuous. We claim that Φ^{-1} is also continuous. For $\mathbf{x}, \mathbf{y} \in B(\mathbf{0}, r)$:

$$\|\Phi(\mathbf{x}) - \Phi(\mathbf{y})\| = \|(\mathbf{x} - \mathbf{y}) + (g(\mathbf{x}) - g(\mathbf{y}))\| \geq \|\mathbf{x} - \mathbf{y}\| - \|g(\mathbf{x}) - g(\mathbf{y})\| \geq \frac{1}{2} \|\mathbf{x} - \mathbf{y}\|.$$

Let $\mathbf{x} = \Phi^{-1}(\mathbf{u})$ and $\mathbf{y} = \Phi^{-1}(\mathbf{v})$, with $\mathbf{u}, \mathbf{v} \in B(\mathbf{0}, \frac{r}{2}) \subseteq \Phi(B(\mathbf{0}, r))$, then:

$$\begin{aligned} \|\Phi(\Phi^{-1}(\mathbf{u})) - \Phi(\Phi^{-1}(\mathbf{v}))\| &\geq \frac{1}{2} \|\Phi^{-1}(\mathbf{u}) - \Phi^{-1}(\mathbf{v})\| \\ \|\Phi^{-1}(\mathbf{u}) - \Phi^{-1}(\mathbf{v})\| &\leq 2 \|\mathbf{u} - \mathbf{v}\|. \end{aligned}$$

This shows that Φ^{-1} is Lipschitz continuous and hence continuous.

Step 3: Prove Φ^{-1} is differentiable at $\mathbf{0}$

We claim that the derivative of Φ^{-1} at $\mathbf{0}$ is the identity map id . Let $\mathbf{z} \in B(\mathbf{0}, \frac{r}{2}) \setminus \{\mathbf{0}\}$.

Since $\Phi|_W$ is bijective, let $\mathbf{z} = \Phi(\mathbf{x}) = \mathbf{x} + g(\mathbf{x})$, which means $\mathbf{x} = \Phi^{-1}(\mathbf{z})$ and $\mathbf{x} \neq \mathbf{0}$.

We want to show:

$$\lim_{\mathbf{z} \rightarrow \mathbf{0}} \frac{\|\Phi^{-1}(\mathbf{z}) - \Phi^{-1}(\mathbf{0}) - \text{id}(\mathbf{z})\|}{\|\mathbf{z}\|} = 0.$$

Since $\Phi^{-1}(\mathbf{0}) = \mathbf{0}$, the expression inside the limit simplifies to:

$$\frac{\|\Phi^{-1}(\mathbf{z}) - \mathbf{z}\|}{\|\mathbf{z}\|} = \frac{\|\mathbf{x} - (\mathbf{x} + g(\mathbf{x}))\|}{\|\mathbf{z}\|} = \frac{\| -g(\mathbf{x}) \|}{\|\mathbf{z}\|} = \frac{\|g(\mathbf{x})\|}{\|\mathbf{x}\|} \cdot \frac{\|\mathbf{x}\|}{\|\mathbf{z}\|}.$$

From the Lipschitz continuity in Step 2, taking $\mathbf{v} = \mathbf{0}$, we have $\|\Phi^{-1}(\mathbf{z})\| \leq 2\|\mathbf{z}\|$, which implies $\frac{\|\mathbf{x}\|}{\|\mathbf{z}\|} \leq 2$. Furthermore, as $\mathbf{z} \rightarrow \mathbf{0}$, sequential continuity dictates that $\mathbf{x} = \Phi^{-1}(\mathbf{z}) \rightarrow \mathbf{0}$. Using the Squeeze Theorem:

$$\lim_{\mathbf{z} \rightarrow \mathbf{0}} \frac{\|\Phi^{-1}(\mathbf{z}) - \mathbf{z}\|}{\|\mathbf{z}\|} = \lim_{\mathbf{x} \rightarrow \mathbf{0}} \left(\frac{\|g(\mathbf{x})\|}{\|\mathbf{x}\|} \cdot \frac{\|\mathbf{x}\|}{\|\mathbf{z}\|} \right) \leq \lim_{\mathbf{x} \rightarrow \mathbf{0}} \left(\frac{\|g(\mathbf{x})\|}{\|\mathbf{x}\|} \cdot 2 \right) = 0 \cdot 2 = 0.$$

This proves that Φ^{-1} is differentiable at $\mathbf{0}$ with $(\Phi^{-1})'(\mathbf{0}) = \text{id}$.

Step 4: Prove f^{-1} is differentiable at $f(\mathbf{x}_0)$

Recall the definition $\Phi(\mathbf{x}) = A^{-1}(f(\mathbf{x}_0 + \mathbf{x}) - f(\mathbf{x}_0))$. Multiplying by A and rearranging yields:

$$f(\mathbf{x} + \mathbf{x}_0) = A\Phi(\mathbf{x}) + f(\mathbf{x}_0).$$

Since Φ is a local bijection, f is also one-to-one and onto near $\mathbf{x}_0$. Let $\mathbf{y} = f(\mathbf{x} + \mathbf{x}_0)$, then $\mathbf{y} = A\Phi(\mathbf{x}) + f(\mathbf{x}_0)$. We can express $\mathbf{x}$ as:

$$\mathbf{x} = \Phi^{-1}(A^{-1}(\mathbf{y} - f(\mathbf{x}_0))).$$

Thus, the inverse function f^{-1} is given by:

$$f^{-1}(\mathbf{y}) = \mathbf{x} + \mathbf{x}_0 = \Phi^{-1}(A^{-1}(\mathbf{y} - f(\mathbf{x}_0))) + \mathbf{x}_0.$$

We apply the Chain Rule to differentiate f^{-1} at $f(\mathbf{x}_0)$:

$$\begin{aligned} (f^{-1})'(f(\mathbf{x}_0)) &= (\Phi^{-1})'(A^{-1}(f(\mathbf{x}_0) - f(\mathbf{x}_0))) \circ (A^{-1})'(f(\mathbf{x}_0) - f(\mathbf{x}_0)) \\ &= (\Phi^{-1})'(\mathbf{0}) \circ A^{-1}(\mathbf{0}) \\ &= \text{id} \circ A^{-1} = A^{-1} = (f'(\mathbf{x}_0))^{-1}. \end{aligned}$$

■

1.3 Implicit Function Theorem

Definition 1.14 (Graph)

Given a function $f : U \subseteq X \rightarrow Y$, the graph of f is defined as the following subset of Cartesian product space:

$$\mathcal{G}(f) = \{(x, f(x)) : x \in X\}.$$

Example 1.1

Let $f : U \subseteq \mathbb{R}^{n-1} \rightarrow \mathbb{R}$ be a function. Define the graph of f as a $(n-1)$ -dimension subset in $\mathbb{R}^n$:

$$\{(x_1, \dots, x_{n-1}, f(x_1, \dots, x_{n-1})) \in \mathbb{R}^n \mid (x_1, \dots, x_{n-1}) \in U\}.$$

Definition 1.15 (Level Set)

Let $U \subseteq \mathbb{R}^n$ be an open set and $f : U \rightarrow \mathbb{R}$ be a real-valued function. For a given constant $c \in \mathbb{R}$, the set of all points $\mathbf{x} \in U$ that satisfy the equation $f(\mathbf{x}) = c$ forms a subset of U . This subset is formally called the **level set** of f at level c , denoted by:

$$L_c = \{\mathbf{x} \in U \mid f(\mathbf{x}) = c\}.$$

Example 1.2 1. **Level Curve** ($n = 2$): A 1D level set in a 2D space is defined by a 2-variable equation $f(x, y) = c$. We call this a **level curve / contour line / isoline** of f at level c .

2. **Level Surface** ($n = 3$): A 2D level set in a 3D space is defined by a 3-variable equation $f(x, y, z) = c$. We call this a **level surface / isosurface** of f at level c .

3. **Level Hypersurface** ($n > 3$): An $(n-1)$ -dimensional level set in an n -dimensional space is defined by $f(x_1, \dots, x_n) = c$. We simply call this a **level hypersurface**.

Remark. Let $U \subseteq \mathbb{R}^3$ and a three-variable function $f : U \rightarrow \mathbb{R}$ defined by $w = f(x, y, z)$.

1. The graph of f is defined as $\{(x, y, z, f(x, y, z)) : (x, y, z) \in U\}$, which is a 3-dimensional subset in $\mathbb{R}^4$.
2. Set $w = c$. Given this constant c , we have a level surface of f at level c , which is a 2-dimensional subset $S = \{(x, y, z) \in U : f(x, y, z) = c\}$ in $\mathbb{R}^3$.
3. Locally, S can be expressed as the graph of a 2-variable function, $S' = \{(x, y, z(x, y)) : (x, y) \in \mathbb{R}^2\}$, which is a 2-dimensional subset in $\mathbb{R}^3$, where $S' \subseteq S$.

Remark. A related but distinct concept is the intersection of two surfaces. For example, suppose we have a double cone and a cutting plane, both of which are level surfaces defined by 3-variable equations. Their intersection forms a conic section (a quadratic curve). Intuitively, a system of 3 variables restricted by 2 independent equations reduces the degrees of freedom to 1 ($3 - 2 = 1$), resulting in a 1-dimensional curve. However, this is not the same concept as a "level curve." A standard level curve is defined by a single equation in a 2D space (e.g., $f(x, y) = c$), whereas this intersection is a 1D curve living in a 3D space, defined by a system of two equations.

Theorem 1.14 (Implicit Function Theorem)

Let E be an open set in $\mathbb{R}^n$, and $f : E \rightarrow \mathbb{R}$ is C^1 . Suppose $y \in E$ such that $f(y) = 0$ and $f_{x_n} = \frac{\partial f}{\partial x_n} \neq 0$, then:

1. There exists an open subset $U \subseteq \mathbb{R}^{n-1}$ containing $(y_1, y_2, \dots, y_{n-1})$, and a map $g : U \rightarrow \mathbb{R}$ such that $g(y_1, y_2, \dots, y_{n-1}) = y_n$.

2. The level set $\{x \in V : f(x) = 0\}$ is a graph of a function of g over U , i.e. $\{x \in V : f(x) = 0\} = \{(x_1, x_2, \dots, x_{n-1}, g(x_1, x_2, \dots, x_{n-1})) : (x_1, x_2, \dots, x_{n-1}) \in U\}$.

3. For $j = 1, 2, \dots, n-1$,

$$\frac{\partial g}{\partial x_j}(y_1, y_2, \dots, y_{n-1}) = -\frac{\frac{\partial f}{\partial x_j}(y)}{\frac{\partial f}{\partial x_n}(y)}.$$

Proof.

Step 1: Define a new function F and check its Jacobian

Define $F : E \rightarrow \mathbb{R}^n$ by $(x_1, x_2, \dots, x_n) \mapsto (x_1, x_2, \dots, x_{n-1}, f(x_1, x_2, \dots, x_n))$.

Since $f \in C^1(E)$, it follows that $F \in C^1(E)$. The Jacobian matrix of F at x is:

$$DF(x) = \begin{pmatrix} - & \nabla x_1 & - \\ - & \nabla x_2 & - \\ & \vdots & \\ - & \nabla f & - \end{pmatrix} = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f}{\partial x_1}(x) & \frac{\partial f}{\partial x_2}(x) & \cdots & \frac{\partial f}{\partial x_n}(x) \end{pmatrix}$$

We are given that $\frac{\partial f}{\partial x_n}(y) \neq 0$. Therefore, the determinant evaluated at y is:

$$\det(DF(y)) = \frac{\partial f}{\partial x_n}(y) \neq 0,$$

which means $DF(y)$ is invertible.

Step 2: Apply the Inverse Function Theorem to construct local inverse

By the Inverse Function Theorem, there exists an open set $V \subseteq E$ and $W \subseteq \mathbb{R}^n$ such that $y \in V$, $F(y) \in W$, and $F : V \rightarrow W$ is a C^1 diffeomorphism. Write the inverse in coordinates as $F^{-1}(u) = (h_1(u), h_2(u), \dots, h_n(u))$ for $u = (u_1, u_2, \dots, u_n) \in W$. By construction, $F(F^{-1}(u)) = u$, which implies:

$$F(h_1(u), h_2(u), \dots, h_n(u)) = (u_1, u_2, \dots, u_n).$$

Using the definition of F , the left-hand side expands to:

$$(h_1(u), h_2(u), \dots, h_{n-1}(u), f(h_1(u), \dots, h_n(u))).$$

Equating the components element by element, we obtain:

$$h_1(u) = u_1, \quad h_2(u) = u_2, \quad \dots, \quad h_{n-1}(u) = u_{n-1},$$

$$f(h_1(u), h_2(u), \dots, h_n(u)) = u_n \implies f(u_1, u_2, \dots, u_{n-1}, h_n(u)) = u_n \quad (*).$$

Thus, the inverse function takes the form:

$$F^{-1}(u_1, u_2, \dots, u_n) = (u_1, u_2, \dots, u_{n-1}, h_n(u_1, u_2, \dots, u_n)).$$

Step 3: Restrict to $u_n = 0$ and define the implicit function g

If we restrict the domain to $u_n = 0$, by (*), we have:

$$f(u_1, \dots, u_{n-1}, h_n(u_1, \dots, u_{n-1}, 0)) = 0.$$

Set an open set $U = \{x' = (x_1, x_2, \dots, x_{n-1}) \in \mathbb{R}^{n-1} \mid (x', 0) \in W\}$. Since $F(y_1, \dots, y_n) = (y_1, \dots, y_{n-1}, f(y_1, \dots, y_n)) = (y_1, \dots, y_{n-1}, 0) \in W$, U is an open set containing $(y_1, y_2, \dots, y_{n-1}) = y'$.

Define $g : U \rightarrow \mathbb{R}$ by $g(x') = g(x_1, \dots, x_{n-1}) = h_n(x_1, \dots, x_{n-1}, 0)$. Note that

$$y = \underbrace{(y', y_n)}_{\in V} = F^{-1}F(y) = F^{-1}(y', 0) = (y', h_n(y', 0)) = (y', g(y')).$$

So $y_n = g(y')$.

Step 4: Show the zero set is locally the graph of g

Claim: $\{x \in V : f(x) = 0\} = \{(x', g(x')) : x' \in U\}$. Namely, we want to show the solution set of the equation $f(x) = 0$ (A) is exactly the graph of the function $g(x')$ (B). Recall $U = \{x' \in \mathbb{R}^{n-1} : (x', 0) \in W\}$.

($\subseteq$) Take $x = (x', x_n) \in A$, where $x \in V$ and $f(x) = 0$.

Apply F first, $F(x) = (x', f(x)) = (x', 0)$, where $F(x) \in W$, so $x' \in U$.

Apply F^{-1} sequentially, $\underbrace{F^{-1}F(x)}_x = F^{-1}(x', 0) = (x', h_n(x', 0)) = (x', g(x'))$.

So $x \in B$.

($\supseteq$) Take $x = (x', x_n) \in B$, where $x' \in U$ and $x = (x', g(x'))$.

Since $x' \in U$, $(x', 0) \in W$. Apply F^{-1} first, $F^{-1}(x', 0) = (x', h_n(x', 0)) = (x', g(x')) = x$, where $x \in V$.

Apply F sequentially, $\underbrace{FF^{-1}(x', 0)}_{(x', 0)} = F(x) = (x', f(x))$. So $x \in A$.

Step 5: Obtain IFT formula

Now we have $(x_1, \dots, x_{n-1}) \in U \subseteq \mathbb{R}^{n-1}$ (where U is open), $f(x', g(x')) = 0$, and $y_n = g(y')$.

Since U is open, there exists $\delta > 0$ such that if $|t| < \delta$, then $y' + te_j \in U$, which implies:

$$f(y' + te_j, g(y' + te_j)) = 0$$

Let $\phi(t) = f(y' + te_j, g(y' + te_j))$. Thus, $\phi(t) = 0$ when $|t| < \delta$. Therefore, $\phi'(t) = 0$ for $|t| < \delta$.

Differentiating with respect to t on both sides by the chain rule:

$$\phi'(t) = \frac{\partial f}{\partial x_j}(y) \frac{dx_j}{dt}(t) + \frac{\partial f}{\partial x_n}(y) \cdot \frac{\partial g}{\partial x_j}(y') \frac{dx_j}{dt}(t).$$

Since $\frac{dx_j}{dt} = 1$, we evaluate at $t = 0$:

$$\begin{aligned} 0 &= \frac{\partial f}{\partial x_j}(y) + \underbrace{\frac{\partial f}{\partial x_n}(y)}_{\neq 0} \frac{\partial g}{\partial x_j}(y') \\ \therefore \frac{\partial g}{\partial x_j}(y') &= -\frac{\frac{\partial f}{\partial x_j}(y)}{\frac{\partial f}{\partial x_n}(y)} \end{aligned}$$

■

Definition 1.16 (Manifold)

A subset $M \subseteq \mathbb{R}^n$ is called a k -dimensional manifold if, for every point $p \in M$, there exists a neighborhood V of p in M that is **diffeomorphic** to an open set U in $\mathbb{R}^k$.

Remark. In advanced calculus (e.g., Spivak's *Calculus on Manifolds*), being a manifold is often established through equivalent conditions. For a subset $M \subseteq \mathbb{R}^n$ to be a k -dimensional manifold locally around a point p , the following three perspectives are mathematically equivalent:

1. **The Geometric View (Diffeomorphism):** Locally, M is diffeomorphic to an open set in $\mathbb{R}^k$.
2. **The Analytical View (Graph of a Function):** Locally, M can be written as the **graph** of a C^1 function (by expressing $n - k$ variables in terms of the other k variables).
3. **The Algebraic View (Level Set / Zero Set):** Locally, M is the solution set of $n - k$ equations where the Jacobian matrix has full rank.

Applications in Economics

1. Indifference Curves and Utility Functions

Let $U(x, y)$ be a utility function representing a consumer's preferences over two goods. An **indifference curve** is fundamentally a 1D **level curve** defined by the equation $U(x, y) = c$, where c is a constant utility level. By the IFT, provided

the marginal utility $\frac{\partial U}{\partial y} \neq 0$, we can locally express y as a smooth function of x (i.e., $y = g(x)$). Most importantly, the theorem gives us the exact slope of this curve, which economists call the **Marginal Rate of Substitution (MRS)**:

$$\frac{dy}{dx} = -\frac{\frac{\partial U}{\partial x}}{\frac{\partial U}{\partial y}} = -\frac{MU_x}{MU_y}$$

The IFT guarantees we can compute this trade-off without ever needing to explicitly isolate y .

2. Isoquant Curves and Production Functions

Similarly, in producer theory, a production function is given by $Q = F(K, L)$, where K is capital and L is labor. An **isoquant** is the level curve $F(K, L) = c$. Applying the IFT, we can find the slope of the isoquant, known as the **Marginal Rate of Technical Substitution (MRTS)**:

$$\frac{dK}{dL} = -\frac{\frac{\partial F}{\partial L}}{\frac{\partial F}{\partial K}} = -\frac{MP_L}{MP_K}$$

This tells the firm exactly how much capital can be replaced by one unit of labor while maintaining the same output level c .

3. Comparative Statics

Beyond 2D level curves, the generalized n -dimensional IFT is the foundation of **Comparative Statics** in macroeconomics and general equilibrium theory. Suppose an economic equilibrium is defined by a complex system of equations $F(\mathbf{y}, \mathbf{x}) = \mathbf{0}$, where $\mathbf{y}$ are endogenous variables (e.g., prices, GDP) and $\mathbf{x}$ are exogenous policy parameters (e.g., tax rates, money supply). Even if the system is too complex to explicitly solve for the equilibrium state $\mathbf{y}^* = g(\mathbf{x})$, the IFT guarantees that this relationship exists locally, provided the Jacobian matrix $D_{\mathbf{y}}F$ is invertible. Furthermore, it allows economists to calculate $\frac{\partial \mathbf{y}^*}{\partial \mathbf{x}}$, precisely predicting how a shift in government policy will ripple through the equilibrium variables.

2 Measure Theory

2.1 Outer Measure

Definition 2.1 (Open box)

An open box $B \subseteq \mathbb{R}^n$ is any of the form

$$B = \prod_{i=1}^n (a_i, b_i) = \{(x_1, x_2, \dots, x_n) \in \mathbb{R}^n : a_i < x_i < b_i, i = 1, 2, \dots, n\}.$$

We define the volume of B to be

$$\text{Vol}(B) = \prod_{i=1}^n (b_i - a_i).$$

Definition 2.2

Let $\Omega \in \mathbb{R}^n$. We say a collection $(B_j)_{j \in \mathcal{J}}$ of boxes covering Ω if $\Omega \subseteq \bigcup_{j \in \mathcal{J}} B_j$.

Definition 2.3 (Outer Measure)

If $\Omega \in \mathbb{R}^n$, we define the outer measure of Ω to be the quantity

$$m^*(\Omega) = \inf \left\{ \sum_{j \in \mathcal{J}} \text{Vol}(B_j) : (B_j)_{j \in \mathcal{J}} \text{ is a countable collection of boxes covering } \Omega \right\}.$$

Remark. $0 \leq m^*(\Omega) \leq \infty$.

Theorem 2.1

1. (Empty set) $m^*(\emptyset) = 0$.
2. (Monotonicity) If $A \subseteq B \subseteq \mathbb{R}^n$, then $m^*(A) \leq m^*(B)$.
3. (Countable sub-additivity) For a countable collection $(A_j)_{j \in \mathcal{J}}$ in $\mathbb{R}^n$, we have $m^*(\bigcup_{j \in \mathcal{J}} A_j) \leq \sum_{j \in \mathcal{J}} m^*(A_j)$.
4. (Translation Invariance) For $\Omega \subseteq \mathbb{R}^n$ and $x \in \mathbb{R}^n$, we have $m^*(\Omega + x) = m^*(\Omega)$.

Proof.

For 1, given any $\epsilon > 0$, $(0, \epsilon^{\frac{1}{n}})^n$ is an open boxes covering $\emptyset$. By definition, $0 \leq m^*(\emptyset) \leq \text{Vol}((0, \epsilon^{\frac{1}{n}})^n) = \epsilon$. So $m^*(\emptyset) = 0$.

For 2, if $m^*(B) = \infty$, there is nothing to prove. Now assume $m^*(B) < \infty$, given any $\epsilon > 0$, there exists a countable collection of open boxes $(B_j)_{j \in \mathcal{J}}$ of B s.t. $\sum_{j \in \mathcal{J}} \text{Vol}(B_j) \leq m^*(B) + \epsilon$. Since $A \subseteq B \subseteq \bigcup_{j \in \mathcal{J}} B_j$,

$m^*(A) \leq \sum_{j \in \mathcal{J}} \text{Vol}(B_j) \leq m^*(B) + \epsilon$. Take $\epsilon \rightarrow 0$, we have $m^*(A) \leq m^*(B)$.

For 3, if $m^*(A_j) = \infty$ for some $j \in \mathcal{J}$, then we are done. Now assume $m^*(A_j) < \infty \forall j \in \mathcal{J}$. Given any $\epsilon > 0$, there exist a countable collection of open boxes $\{B_{j,k}\}_{k \in \mathbb{N}}$ of B_j s.t. $\sum_{k=1}^{\infty} \text{Vol}(B_{j,k}) \leq m^*(A_j) + \frac{\epsilon}{2^j}$. Since

$$\bigcup_{j \in \mathbb{N}} A_j \subseteq \bigcup_{j \in \mathbb{N}} \bigcup_{k \in \mathbb{N}} B_{j,k}, \quad m^*\left(\bigcup_{j \in \mathbb{N}} A_j\right) \leq \sum_{j \in \mathbb{N}} \sum_{k \in \mathbb{N}} \text{Vol}(B_{j,k}) \leq \sum_{j \in \mathbb{N}} m^*(A_j) + \epsilon.$$

Take $\epsilon \rightarrow 0$, then we have

$$m^*\left(\bigcup_{j \in \mathcal{J}} A_j\right) < \sum_{j \in \mathcal{J}} m^*(A_j).$$

For 4, since $\sum_{j \in \mathbb{N}} \text{Vol}(B_j) = \sum_{j \in \mathbb{N}} \text{Vol}(B_j + x)$, by the definition of outer measure, we have $m^*(x + \Omega) = m^*(\Omega)$. ■

Proposition 1

Let $B = \prod_{i=1}^n [a_i, b_i] = \{(x_1, x_2, \dots, x_n) \in \mathbb{R}^n : a_i \leq x_i \leq b_i, i = 1, 2, \dots, n\}$, we have $m^*(B) = \prod_{i=1}^n (b_i - a_i)$.

Proof.

($\leq$) Let an open box $U_\epsilon = \prod_{i=1}^n (a_i + \epsilon, b_i - \epsilon)$ covering B . By definition, $m^*(B) \leq \text{Vol}(U_\epsilon) = \prod_{i=1}^n (b_i - a_i + 2\epsilon)$. Take $\epsilon \rightarrow 0$, then we have $m^*(B) \leq \prod_{i=1}^n (b_i - a_i)$.

($\geq$) Let $\text{Vol}(B) = \prod_{i=1}^n (b_i - a_i)$ and any countable collection of open boxes $\{B_j\}_{j \in \mathcal{J}}$ covering B . If we can show $\text{Vol}(B) \leq \sum_{j \in \mathcal{J}} \text{Vol}(B_j)$, then $\prod_{i=1}^n (b_i - a_i) \leq m^*(B)$. Since B is compact, there exists an finite subcover of open boxes of $\{B_j\}_{j \in \mathcal{J}}$, denoted by $\{U_j\}_{j=1}^k$ s.t. $B \subseteq \bigcup_{j=1}^k U_j$ and $\sum_{j=1}^k \text{Vol}(U_j) \leq \sum_{j=1}^{\infty} \text{Vol}(B_j)$. Choose a closed box Q containing B and U_j for $j = 1, 2, \dots, k$. Define a function f on Q s.t.

$$f(x) = \begin{cases} 1 & \text{if } x \in B \\ 0 & \text{if } x \in Q \setminus B, \end{cases} \quad f_j(x) = \begin{cases} 1 & \text{if } x \in U_j \\ 0 & \text{if } x \in Q \setminus U_j, \end{cases}$$

where $\text{Vol}(B) = \int_Q f(x) dx$ and $\text{Vol}(U_j) = \int_Q f_j(x) dx$. If $x \in B$, $f(x) = 1$, and there exists an integer ℓ s.t. $f_\ell(x) = 1$. If $x \in Q \setminus B$, $f(x) = 0$. So we have $f(x) \leq \sum_{j=1}^k f_j(x)$. By the comparison property of Riemann integral, $\int_Q f(x) dx \leq \int_Q \sum_{j=1}^k f_j(x) dx = \sum_{j=1}^k \int_Q f_j(x) dx$, which means $\text{Vol}(B) \leq \sum_{j=1}^k \text{Vol}(B_j) \leq$

$\sum_{j=1}^{\infty} \text{Vol}(B_j)$. Hence $\text{Vol}(B) = \prod_{i=1}^n (b_i - a_i) \leq m^*(B)$. ■

Example 2.1

Consider the m_1^* measure on $\mathbb{R}$.

1. $m^*(\mathbb{R}) = \infty$.

2. $m^*(\mathbb{Q}) = 0$.

We know $\mathbb{Q}$ is countable. Let $\mathbb{Q} = \cup_{q \in \mathbb{Q}} \{q\}$, $m^*(\mathbb{Q}) \leq \sum_{q \in \mathbb{Q}} m^*(\{q\}) = 0$.

3. $m^*(\mathbb{R} \setminus \mathbb{Q}) = \infty$.

Observe that $\mathbb{R} = (\mathbb{R} \setminus \mathbb{Q}) \cup (\mathbb{Q})$, so $m^*(\mathbb{R}) \leq m^*(\mathbb{R} \setminus \mathbb{Q}) + m^*(\mathbb{Q})$, by 1 and 2, we have $m^*(\mathbb{R} \setminus \mathbb{Q}) = \infty$. This shows that $\mathbb{R} \setminus \mathbb{Q}$ is uncountable, since the set of irrational numbers has measure nonzero.

Remark. Every countable set has measure zero.

Lemma (Two Properties of Cosets)

For any $x, y \in \mathbb{R}$, let $x + \mathbb{Q} = \{x + q : q \in \mathbb{Q}\}$ be a coset of $\mathbb{Q}$ in $\mathbb{R}$. The following two properties hold:

1. If $(x + \mathbb{Q}) \cap (y + \mathbb{Q}) \neq \emptyset$, then $x + \mathbb{Q} = y + \mathbb{Q}$.

2. Every coset meets the interval $[0, 1]$, i.e., $(x + \mathbb{Q}) \cap [0, 1] \neq \emptyset$.

Proof.

Proof of 1 (Disjoint or Identical): Suppose $(x + \mathbb{Q}) \cap (y + \mathbb{Q}) \neq \emptyset$. Let z be an element in this intersection. Then $z = x + q_1$ and $z = y + q_2$ for some $q_1, q_2 \in \mathbb{Q}$. This implies $x - y = q_2 - q_1$. Since the rationals are closed under subtraction, $x - y \in \mathbb{Q}$. Now, for any element $a \in x + \mathbb{Q}$, we can write $a = x + r$ for some $r \in \mathbb{Q}$. Substituting $x = y + (q_2 - q_1)$, we have:

$$a = y + (q_2 - q_1) + r = y + (q_2 - q_1 + r).$$

Since $q_2 - q_1 + r \in \mathbb{Q}$, we get $a \in y + \mathbb{Q}$. This shows $x + \mathbb{Q} \subseteq y + \mathbb{Q}$. By symmetry, $y + \mathbb{Q} \subseteq x + \mathbb{Q}$, so the two cosets are exactly equal.

Proof of 2 (Intersection with $[0, 1]$): Let $x + \mathbb{Q}$ be any coset. We want to find a rational number q such that $x + q \in [0, 1]$. This is equivalent to requiring $0 \leq x + q \leq 1$, or simply:

$$-x \leq q \leq 1 - x.$$

Because the set of rational numbers $\mathbb{Q}$ is dense in $\mathbb{R}$, any interval with positive length contains a rational number. Since the interval $[-x, 1 - x]$ has length $1 > 0$,

it must contain at least one rational number q . For such q , we have $x + q \in [0, 1]$. Therefore, $(x + \mathbb{Q}) \cap [0, 1] \neq \emptyset$. ■

Remark. The properties of these cosets stem directly from the fact that they are equivalence classes. Define a relation $\sim$ on $\mathbb{R}$ by:

$$x \sim y \iff x - y \in \mathbb{Q}.$$

This is an equivalence relation because it satisfies:

- **Reflexivity:** $x - x = 0 \in \mathbb{Q} \implies x \sim x$.
- **Symmetry:** $x \sim y \implies x - y \in \mathbb{Q} \implies -(x - y) = y - x \in \mathbb{Q} \implies y \sim x$.
- **Transitivity:** $x \sim y$ and $y \sim z \implies x - y \in \mathbb{Q}$ and $y - z \in \mathbb{Q} \implies (x - y) + (y - z) = x - z \in \mathbb{Q} \implies x \sim z$.

The equivalence class of an element $x \in \mathbb{R}$ under this relation is exactly its coset:

$$[x] = \{y \in \mathbb{R} : y \sim x\} = \{y \in \mathbb{R} : y - x \in \mathbb{Q}\} = \{x + q : q \in \mathbb{Q}\} = x + \mathbb{Q}.$$

A fundamental theorem of equivalence relations states that equivalence classes form a **partition** of the underlying set ($\mathbb{R}$). This means that any two equivalence classes (cosets) are either completely disjoint or exactly identical, which is precisely the underlying mechanism of **Property 1**.

Proposition 2 (Failure of Countable Additivity)

There exists a countable collection $(A_j)_{j \in J}$ of disjoint subsets of $\mathbb{R}$ such that

$$m^*\left(\bigcup_{j \in J} A_j\right) \neq \sum_{j \in J} m^*(A_j).$$

Proof.

We construct such a family from the cosets of $\mathbb{Q}$ in $\mathbb{R}$.

Step 1 (Choose one representative): Let $\mathbb{R}/\mathbb{Q}$ denote the collection of all cosets. By Property 2 of our previous Lemma, we know that $A \cap [0, 1] \neq \emptyset$ for every coset A . Thus, by the Axiom of Choice, we can choose exactly one representative $x_A \in A \cap [0, 1]$ for each coset $A \in \mathbb{R}/\mathbb{Q}$. Define $E = \{x_A : A \in \mathbb{R}/\mathbb{Q}\}$. By construction, $E \subseteq [0, 1]$.

Step 2 (Rational translates of E): Consider the set $X = \bigcup_{q \in \mathbb{Q} \cap [-1, 1]} (q + E)$.

Claim: $[0, 1] \subseteq X \subseteq [-1, 2]$. Let $y \in [0, 1]$. There exists a coset A such that $y \in A$. Since $x_A \in A$ is the representative of the same coset, we have $y - x_A = q$

for some $q \in \mathbb{Q}$. Since $y \in [0, 1]$ and $x_A \in [0, 1]$, we get $q = y - x_A \in [-1, 1]$. Thus, $y = q + x_A \in q + E \subseteq X$. So $[0, 1] \subseteq X$. Also, since $E \subseteq [0, 1]$ and $q \in [-1, 1]$, $q + E \subseteq [-1, 2]$. Thus $X \subseteq [-1, 2]$.

Step 3 (Disjointness): The collection $\{q + E : q \in \mathbb{Q} \cap [-1, 1]\}$ is pairwise disjoint. Suppose by contradiction that $(q + E) \cap (q' + E) \neq \emptyset$ for $q \neq q'$. Then $q + x_A = q' + x_{A'}$ for some $x_A, x_{A'} \in E$. This implies $x_A - x_{A'} = q' - q \in \mathbb{Q}$. By Property 1 of our Lemma, since their difference is rational, x_A and $x_{A'}$ belong to the same coset. However, our construction in Step 1 chose exactly one representative per coset, so we must have $x_A = x_{A'}$. This forces $q = q'$, which is a contradiction.

Step 4 (Measure contradiction): By Step 2, $m^*([0, 1]) \leq m^*(X) \leq m^*([-1, 2])$, which implies $1 \leq m^*(X) \leq 3$. Assume countable additivity holds. Since m^* is translation-invariant, $m^*(q + E) = m^*(E)$. Thus,

$$m^*(X) = m^*\left(\bigcup_{q \in \mathbb{Q} \cap [-1, 1]} q + E\right) = \sum_{q \in \mathbb{Q} \cap [-1, 1]} m^*(E).$$

If $m^*(E) = 0$, then $\sum m^*(E) = 0$. If $m^*(E) > 0$, then $\sum m^*(E) = \infty$. Both cases contradict $1 \leq m^*(X) \leq 3$. Hence, countable additivity fails for outer measure m^* . ■

Proposition 3 (Failure of Finite Additivity)

There exists a finite collection of disjoint subsets of $\mathbb{R}$ such that the outer measure of their union is not equal to the sum of their outer measures.

Proof.

We use the same Vitali set E constructed previously. We know $m^*(E) > 0$ because if $m^*(E) = 0$, countable subadditivity would force

$$m^*(X) \leq \sum_{q \in \mathbb{Q} \cap [-1, 1]} m^*(q + E) = 0,$$

which contradicts $m^*(X) \geq 1$.

Since $m^*(E) > 0$, by the Archimedean property, there exists an integer $n \geq 1$ such that $m^*(E) > \frac{1}{n}$. Let I be a finite subset of $\mathbb{Q} \cap [-1, 1]$ consisting of exactly $3n$ distinct rational numbers. Consider the finite disjoint union $\bigcup_{q \in I} (q + E)$. Because this union is a subset of X , and we already established $X \subseteq [-1, 2]$, we must have:

$$m^*\left(\bigcup_{q \in I} (q + E)\right) \leq m^*([-1, 2]) = 3.$$

Assume for the sake of contradiction that finite additivity holds for outer mea-

sure. Then the measure of the union equals the sum of the measures:

$$m^* \left(\bigcup_{q \in I} (q + E) \right) = \sum_{q \in I} m^*(q + E) = \sum_{q \in I} m^*(E) = 3n \cdot m^*(E).$$

Substituting our bound $m^*(E) > \frac{1}{n}$ into the equation, we get:

$$3n \cdot m^*(E) > 3n \left(\frac{1}{n} \right) = 3.$$

This implies $3 < 3$, which is a blatant contradiction. Thus, finite additivity fails for the Lebesgue outer measure m^* . ■

2.2 Lebesgue Measure

Definition 2.4 (Lebesgue Measurability)

Let E be a subset of $\mathbb{R}^n$. We say that E is (Lebesgue) measurable if we have the identity:

$$m^*(A) = m^*(A \cap E) + m^*(A \setminus E) \quad \text{for all } A \subseteq \mathbb{R}^n.$$

In this case, we define the Lebesgue measure of E as $m^*(E) = m(E)$.

Remark.

1. This definition is proposed by Constantin Carathéodory.
2. Recall $A = (A \cap E) \cup (A \setminus E)$. By subadditivity, we always have $m^*(A) \leq m^*(A \cap E) + m^*(A \setminus E)$. So, E is measurable if and only if $m^*(A) \geq m^*(A \cap E) + m^*(A \setminus E)$.

Example 2.2

$\mathbb{R}^n$ and $\emptyset$ are measurable (Easy exercise).

Proposition 4

The half-space $H = \{\mathbf{x} = (x_1, x_2, \dots, x_n) \in \mathbb{R}^n : x_n \geq 0\}$ is Lebesgue measurable.

Proof.

Let $E = H$. For any test set $A \subseteq \mathbb{R}^n$, it suffices to show that $m^*(A) \geq m^*(A \cap E) + m^*(A \setminus E)$.

If $m^*(A) = \infty$, the inequality holds trivially, so there is nothing to prove. Next, we assume $m^*(A) < \infty$. Given any $\epsilon > 0$, by the definition of outer measure, there exists a countable collection of open boxes $\{B_k\}_{k=1}^{\infty}$ such that $A \subseteq \bigcup_{k=1}^{\infty} B_k$ and

$$\sum_{k=1}^{\infty} \text{Vol}(B_k) \leq m^*(A) + \epsilon.$$

For each open box B_k , the hyperplane $x_n = 0$ splits it into two pieces: $B_k \cap E$ and $B_k \setminus E$. Since these two pieces are essentially boxes (or empty) and their volumes naturally add up to the volume of the original box B_k , we have:

$$m^*(B_k) = m^*(B_k \cap E) + m^*(B_k \setminus E).$$

(Recall that the outer measure of a box is exactly its volume).

Summing this over all k , we get:

$$\sum_{k=1}^{\infty} \text{Vol}(B_k) = \sum_{k=1}^{\infty} m^*(B_k \cap E) + \sum_{k=1}^{\infty} m^*(B_k \setminus E).$$

By the countable subadditivity of the outer measure, we know:

$$\sum_{k=1}^{\infty} m^*(B_k \cap E) \geq m^*\left(\bigcup_{k=1}^{\infty} (B_k \cap E)\right) = m^*\left(\left(\bigcup_{k=1}^{\infty} B_k\right) \cap E\right).$$

Since $A \subseteq \bigcup B_k$, by monotonicity we have $(\bigcup B_k) \cap E \supseteq A \cap E$. Therefore:

$$m^*\left(\left(\bigcup_{k=1}^{\infty} B_k\right) \cap E\right) \geq m^*(A \cap E).$$

Similarly, for the outside part:

$$\sum_{k=1}^{\infty} m^*(B_k \setminus E) \geq m^*(A \setminus E).$$

Substituting these lower bounds back into our volume sum, we obtain:

$$m^*(A) + \epsilon \geq \sum_{k=1}^{\infty} \text{Vol}(B_k) \geq m^*(A \cap E) + m^*(A \setminus E).$$

Since $\epsilon > 0$ is arbitrary, we let $\epsilon \rightarrow 0$ to conclude:

$$m^*(A) \geq m^*(A \cap E) + m^*(A \setminus E).$$

Hence, the half-space H is measurable. ■

Remark. A similar argument can apply to any half-space of the form $\{\mathbf{x} \in \mathbb{R}^n : x_j \geq a\}$ or $\{\mathbf{x} \in \mathbb{R}^n : x_j \leq a\}$ for any coordinate $1 \leq j \leq n$ and any constant $a \in \mathbb{R}$.

Theorem 2.2 (Properties of Measurable Sets)

- (a) If E is measurable, then $\mathbb{R}^n \setminus E$ is also measurable.
- (b) If E is measurable and $x \in \mathbb{R}^n$, then $x + E$ is also measurable.

- (c) If E_1 and E_2 are measurable, then $E_1 \cap E_2$ and $E_1 \cup E_2$ are measurable.
- (d) If $E_1, E_2, \dots, E_N$ are measurable, then $\bigcup_{i=1}^N E_i$ and $\bigcap_{i=1}^N E_i$ are measurable.
- (e) Every open box and closed box is measurable.
- (f) Any set E of outer measure zero is measurable.

Proof.

Proof of (a): Since $A \setminus (\mathbb{R}^n \setminus E) = A \cap E$ and $A \cap (\mathbb{R}^n \setminus E) = A \setminus E$, the definition of measurability is symmetric with respect to E and E^c .

Proof of (b): Since E is measurable, to show $x + E$ is measurable, we need to show that for any $A \subseteq \mathbb{R}^n$,

$$m^*(A) = m^*(A \cap (x + E)) + m^*(A \setminus (x + E)).$$

First, observe the following set identities when shifted by $-x$:

$$(A \cap (x + E)) - x = (A - x) \cap E$$

$$(A \setminus (x + E)) - x = (A - x) \setminus E$$

By the translation invariance of the outer measure m^* , we have:

$$m^*(A \cap (x + E)) = m^*((A \cap (x + E)) - x) = m^*((A - x) \cap E)$$

$$m^*(A \setminus (x + E)) = m^*((A \setminus (x + E)) - x) = m^*((A - x) \setminus E)$$

Let $B = A - x$. Since E is measurable, testing with the set B yields:

$$m^*(B) = m^*(B \cap E) + m^*(B \setminus E).$$

Substituting back $B = A - x$ and using the translation invariance $m^*(A - x) = m^*(A)$ on the left side, we get:

$$m^*(A) = m^*((A - x) \cap E) + m^*((A - x) \setminus E).$$

Replacing the terms on the right side with the equalities we found earlier:

$$m^*(A) = m^*(A \cap (x + E)) + m^*(A \setminus (x + E)).$$

This holds for all $A \subseteq \mathbb{R}^n$, thus $x + E$ is measurable.

Proof of (c):

Part 1: Union is measurable.

We want to show that $E_1 \cup E_2$ is measurable. For any test set $A \subseteq \mathbb{R}^n$, since E_1 is measurable, we can split A using E_1 :

$$m^*(A) = m^*(A \cap E_1) + m^*(A \setminus E_1).$$

Next, we apply the measurability of E_2 to the test set $A \setminus E_1$:

$$m^*(A \setminus E_1) = m^*((A \setminus E_1) \cap E_2) + m^*((A \setminus E_1) \setminus E_2).$$

Substitute this back into the first equation:

$$m^*(A) = m^*(A \cap E_1) + m^*((A \setminus E_1) \cap E_2) + m^*((A \setminus E_1) \setminus E_2).$$

Now, we make two observations from set algebra:

- First, the remaining part outside both sets is $(A \setminus E_1) \setminus E_2 = A \cap E_1^c \cap E_2^c = A \cap (E_1 \cup E_2)^c = A \setminus (E_1 \cup E_2)$.
- Second, the parts intersecting the sets combine to $(A \cap E_1) \cup ((A \setminus E_1) \cap E_2) = A \cap (E_1 \cup E_2)$.

By the subadditivity of the outer measure m^* , the sum of the measures of the first two terms is bounded below by the measure of their union:

$$m^*(A \cap E_1) + m^*((A \setminus E_1) \cap E_2) \geq m^*((A \cap E_1) \cup ((A \setminus E_1) \cap E_2)) = m^*(A \cap (E_1 \cup E_2)).$$

Replacing these terms in our expanded equation, we obtain:

$$m^*(A) \geq m^*(A \cap (E_1 \cup E_2)) + m^*(A \setminus (E_1 \cup E_2)).$$

Since the reverse inequality is always true by subadditivity, $E_1 \cup E_2$ is measurable.

Part 2: Intersection is measurable.

(Method 1: By De Morgan's Laws)

By De Morgan's laws, we can express the intersection as $E_1 \cap E_2 = (E_1^c \cup E_2^c)^c$. Since E_1 and E_2 are measurable, their complements E_1^c and E_2^c are measurable by Property (a). By Part 1 of this proof, the union of two measurable sets is measurable, so $E_1^c \cup E_2^c$ is measurable. Finally, taking the complement once more, we conclude that $(E_1^c \cup E_2^c)^c = E_1 \cap E_2$ is measurable.

(Method 2: Direct Carathéodory Approach)

Alternatively, we can prove it directly. Since E_1 is measurable, split A :

$$m^*(A) = m^*(A \cap E_1) + m^*(A \setminus E_1).$$

Apply the measurability of E_2 to the test set $A \cap E_1$:

$$m^*(A \cap E_1) = m^*(A \cap E_1 \cap E_2) + m^*((A \cap E_1) \setminus E_2).$$

Substitute this into the first equation:

$$m^*(A) = m^*(A \cap E_1 \cap E_2) + m^*((A \cap E_1) \setminus E_2) + m^*(A \setminus E_1).$$

Observe that the last two terms cover the part of A outside the intersection: $((A \cap E_1) \setminus E_2) \cup (A \setminus E_1) = A \setminus (E_1 \cap E_2)$. By subadditivity, $m^*((A \cap E_1) \setminus E_2) + m^*(A \setminus E_1) \geq m^*(A \setminus (E_1 \cap E_2))$. This yields:

$$m^*(A) \geq m^*(A \cap (E_1 \cap E_2)) + m^*(A \setminus (E_1 \cap E_2)),$$

which proves $E_1 \cap E_2$ is measurable.

Proof of (d): Follows from (c) by mathematical induction.

Proof of (e):

Let $B = \{\mathbf{x} \in \mathbb{R}^n : a_i < x_i < b_i \text{ for } i = 1, 2, \dots, n\}$ be an open box. We can express B as the intersection of $2n$ half-spaces:

$$B = \bigcap_{i=1}^n P_i$$

where each $P_i = \{\mathbf{x} \in \mathbb{R}^n : x_i > a_i\} \cap \{\mathbf{x} \in \mathbb{R}^n : x_i < b_i\}$. From the previous proposition and remark, every such half-space is Lebesgue measurable. By Property (d) proved above, any finite intersection of measurable sets is also measurable. Since the open box B is a finite intersection of measurable half-spaces, B is measurable. For any closed box, we can apply the same argument in proof of (c) via De Morgan's Laws, and this is left as an exercise.

Proof of (f): Suppose $m^*(E) = 0$. For any A , $A \cap E \subseteq E \implies m^*(A \cap E) = 0$. Also $A \setminus E \subseteq A \implies m^*(A \setminus E) \leq m^*(A)$. Thus, $m^*(A \cap E) + m^*(A \setminus E) \leq 0 + m^*(A) = m^*(A)$. E is measurable. ■

Theorem 2.3 (Finite Additivity)

If $(E_j)_{j=1}^N$ is a finite collection of disjoint measurable sets, then for any $A \subseteq \mathbb{R}^n$:

$$m^*\left(A \cap \bigcup_{j=1}^N E_j\right) = \sum_{j=1}^N m^*(A \cap E_j).$$

Moreover, taking $A = \mathbb{R}^n$, we have $m\left(\bigcup_{j=1}^N E_j\right) = \sum_{j=1}^N m(E_j)$.

Proof.

We proceed by induction. For $N = 2$, let E, F be disjoint measurable sets. Test

the set $A \cap (E \cup F)$ with the measurable set E :

$$\begin{aligned} m^*(A \cap (E \cup F)) &= m^*((A \cap (E \cup F)) \cap E) + m^*((A \cap (E \cup F)) \setminus E) \\ &= m^*(A \cap E) + m^*(A \cap F). \end{aligned}$$

Suppose it holds for k sets. Let $U_k = \bigcup_{j=1}^k E_j$. Since $U_k \cap E_{k+1} = \emptyset$ and U_k is measurable:

$$m^*(A \cap (U_k \cup E_{k+1})) = m^*(A \cap U_k) + m^*(A \cap E_{k+1}) = \sum_{j=1}^k m^*(A \cap E_j) + m^*(A \cap E_{k+1}).$$

■

Corollary

If $A \subseteq B$ are two measurable sets, then $m(B) = m(A) + m(B \setminus A)$.

Proof.

Observe that $B \setminus A = B \cap (\mathbb{R}^n \setminus A)$, so $B \setminus A$ is measurable. Also, A and $B \setminus A$ are disjoint measurable sets, by finite additivity, we obtain the identity. ■

Theorem 2.4 (Countable Additivity)

If $(E_j)_{j=1}^\infty$ is a countable collection of disjoint measurable sets, then $E = \bigcup_{j=1}^\infty E_j$ is measurable and $m(E) = \sum_{j=1}^\infty m(E_j)$.

Proof.

Let $F_N = \bigcup_{k=1}^N E_k$. F_N is measurable. For any $A \subseteq \mathbb{R}^n$,

$$m^*(A) = m^*(A \cap F_N) + m^*(A \setminus F_N) \geq \sum_{k=1}^N m^*(A \cap E_k) + m^*(A \setminus E).$$

Letting $N \rightarrow \infty$, we obtain $m^*(A) \geq \sum_{k=1}^\infty m^*(A \cap E_k) + m^*(A \setminus E)$. By subadditivity, $\sum_{k=1}^\infty m^*(A \cap E_k) \geq m^*(A \cap E)$. Thus $m^*(A) \geq m^*(A \cap E) + m^*(A \setminus E)$, proving E is measurable. By taking $A = \mathbb{R}^n$, we get $m(F_N) = \sum_{k=1}^N m(E_k) \leq m(E)$. Letting $N \rightarrow \infty$ gives $\sum_{k=1}^\infty m(E_k) \leq m(E)$. The reverse inequality holds by countable subadditivity. ■

Theorem 2.5 (σ -algebra)

If $(\Omega_j)_{j=1}^\infty$ is any countable collection of measurable sets, then $\bigcup_{j=1}^\infty \Omega_j$ and $\bigcap_{j=1}^\infty \Omega_j$ are also measurable.

Proof.

We can disjointify the union. Let $E_1 = \Omega_1$, and for $N \geq 2$, define

$$E_N = \Omega_N \setminus \left(\bigcup_{k=1}^{N-1} \Omega_k \right).$$

Since finite unions and complements of measurable sets are measurable, each E_N is measurable. The sets (E_N) are pairwise disjoint, and $\bigcup_{j=1}^{\infty} \Omega_j = \bigcup_{N=1}^{\infty} E_N$. By the previous theorem, the countable union of disjoint measurable sets is measurable. The intersection is measurable by De Morgan's laws: $\bigcap \Omega_j = \left(\bigcup \Omega_j^c \right)^c$. ■

Remark. Every σ -algebra is a Boolean algebra (finite unions and complements), but the converse is not true.

Theorem 2.6

Every open set can be written as a countable union of rational open boxes.

Proof.

We say an open box is rational if $B = \prod_{i=1}^n (a_i, b_i)$ where $a_i, b_i \in \mathbb{Q}$. The set of all rational boxes is countable. First, we show that for any open ball $B(x, r)$, there exists a rational box B such that $x \in B \subseteq B(x, r)$. Let a_i, b_i be rational numbers such that $x_i - \frac{r}{n} < a_i < x_i < b_i < x_i + \frac{r}{n}$. Then $x \in B = \prod (a_i, b_i)$. For any $y \in B$, $|y_i - x_i| < \frac{r}{n}$, which implies $\|y - x\| \leq \sum |y_i - x_i| < r$, so $B \subseteq B(x, r)$.

Let E be an open set in $\mathbb{R}^n$. For every $x \in E$, there exists $r_x > 0$ such that $B(x, r_x) \subseteq E$. By the above claim, there is a rational box B_x such that $x \in B_x \subseteq B(x, r_x) \subseteq E$. Thus $E = \bigcup_{x \in E} B_x$. Since there are only countably many rational boxes, this union is countable. ■

Theorem 2.7 (Borel property)

Any open set and closed set is Lebesgue measurable.

Proof.

Any open set is the countable union of rational open boxes. Since boxes are measurable and countable unions of measurable sets are measurable, every open set is measurable. A closed set is the complement of an open set, hence it is also measurable. ■

2.3 Measurable Functions

Definition 2.5 (σ -Algebra)

Let X be a set. A collection $\mathcal{A}$ of subsets of X is called a σ -algebra if it satisfies the following three properties:

1. $X \in \mathcal{A}$.
2. Closed under complement: If $E \in \mathcal{A}$, then $X \setminus E \in \mathcal{A}$.
3. Closed under countable unions: If $E_1, E_2, \dots \in \mathcal{A}$, then $\bigcup_{i=1}^{\infty} E_i \in \mathcal{A}$.

Remark. The definition of a σ -algebra inherently implies two additional properties:

1. Contains the empty set: Since $X \in \mathcal{A}$ and it is closed under complement, $X \setminus X = \emptyset \in \mathcal{A}$.
2. Closed under countable intersections: If $E_1, E_2, \dots \in \mathcal{A}$, their complements $X \setminus E_i$ are also in $\mathcal{A}$. By the countable union property, $\bigcup_{i=1}^{\infty} (X \setminus E_i) \in \mathcal{A}$. Taking the complement of this union and applying De Morgan's laws yields:

$$X \setminus \left(\bigcup_{i=1}^{\infty} (X \setminus E_i) \right) = \bigcap_{i=1}^{\infty} E_i \in \mathcal{A}.$$

Definition 2.6 (Borel σ -Algebra)

Let $(X, \mathcal{T})$ be a topological space. The Borel σ -algebra $\mathcal{B}(X)$ is the σ -algebra generated by the collection of all open sets $\mathcal{T}$, i.e., $\mathcal{B}(X) = \sigma(\mathcal{T})$.

A subset $E \subseteq X$ is a **Borel set** if $E \in \mathcal{B}(X)$.

Remark. Borel σ -algebra is the smallest σ -algebra contains all open sets. The term “smallest” refers to the intersection of all σ -algebras on X that contain $\mathcal{T}$:

$$\sigma(\mathcal{T}) = \bigcap \{ \mathcal{A} : \mathcal{A} \text{ is a } \sigma\text{-algebra on } X \text{ and } \mathcal{T} \subseteq \mathcal{A} \}.$$

This ensures that $\mathcal{B}(X)$ contains no more sets than are strictly necessary to satisfy the σ -algebra axioms while including all open sets.

Example 2.3

Let $X = \{1, 2, 3\}$ and $\mathcal{T} = \{\{1\}\}$. We want to find the smallest σ -algebra on X containing $\mathcal{T}$, denoted by $\sigma(\mathcal{T})$.

According to the axioms of a σ -algebra, we can construct it step-by-step:

1. **Include the empty set and the whole space:** A σ -algebra must contain $\emptyset$ and X .

Current collection: $\{\emptyset, \{1, 2, 3\}\}$

2. **Include the generating sets:** The σ -algebra must contain all elements of $\mathcal{T}$.

Current collection: $\{\emptyset, \{1, 2, 3\}, \{1\}\}$

3. **Close under complements:** The complement of $\{1\}$ is $X \setminus \{1\} = \{2, 3\}$. This must also be included.

Current collection: $\{\emptyset, \{1, 2, 3\}, \{1\}, \{2, 3\}\}$

4. **Close under countable unions:** We check all possible unions of the sets in our collection. For instance, $\{1\} \cup \{2, 3\} = \{1, 2, 3\}$, which is already in our collection. No new sets can be generated.

Therefore, the smallest σ -algebra generated by $\mathcal{T}$ is:

$$\sigma(\mathcal{T}) = \{\emptyset, \{1, 2, 3\}, \{1\}, \{2, 3\}\}.$$

Definition 2.7 (Measurable Function)

Let Ω be a measurable subset of $\mathbb{R}^n$ and let $f : \Omega \rightarrow \mathbb{R}^m$ be a function. We say f is measurable if $f^{-1}(V)$ is measurable for every open set $V \subseteq \mathbb{R}^m$.

Remark. In particular, we call f is Borel measurable since $f^{-1}(B)$ is measurable for every Borel set $B \in \mathcal{B}(\mathbb{R}^m)$.

Theorem 2.8

Let $f : \Omega \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a continuous function, where Ω is measurable. Then f is measurable.

Proof.

Let W be an open set in $\mathbb{R}^m$. Since f is continuous, $f^{-1}(W) = \Omega \cap U$, where U is an open set in $\mathbb{R}^n$. Because U is open, it is measurable. Since Ω is measurable, their intersection $\Omega \cap U$ is also measurable. Thus, $f^{-1}(W)$ is measurable, which means f is measurable. ■

Theorem 2.9

Let Ω be a measurable subset in $\mathbb{R}^n$ and $f : \Omega \rightarrow \mathbb{R}^m$ be a function. f is measurable if and only if $f^{-1}(B)$ is measurable for every open box B .

Proof.

($\Rightarrow$) An open box B is an open set, so $f^{-1}(B)$ is measurable by definition.

($\Leftarrow$) Given any open set V , it can be written as a countable union of open boxes: $V = \bigcup_{i=1}^{\infty} B_i$. Recall that $f^{-1}(\bigcup_{i=1}^{\infty} B_i) = \bigcup_{i=1}^{\infty} f^{-1}(B_i)$. Since each $f^{-1}(B_i)$ is measurable by assumption, and countable union of measurable sets is measurable, $f^{-1}(V)$ is measurable. ■

Corollary

Let Ω be a measurable subset of $\mathbb{R}^n$, and let $f : \Omega \rightarrow \mathbb{R}^m$ be a function such that $f = (f_1, f_2, \dots, f_m)$, where $f_j : \Omega \rightarrow \mathbb{R}$ is the j -th coordinate of f . Then f is measurable if and only if all of the f_j are individually measurable.

Proof.

($\Rightarrow$) Fix $j \in \{1, 2, \dots, m\}$ and consider $f_j : \Omega \rightarrow \mathbb{R}$. Let $I \subseteq \mathbb{R}$ be an open interval. Consider the open set $B_j = \mathbb{R} \times \mathbb{R} \times \dots \times I \times \dots \times \mathbb{R}$ (where I is at the j -th position). Note that $x \in f^{-1}(B_j) \Leftrightarrow f(x) \in B_j \Leftrightarrow f_j(x) \in I \Leftrightarrow x \in f_j^{-1}(I)$. If f is measurable, $f^{-1}(B_j)$ is measurable, so $f_j^{-1}(I)$ is measurable for any open interval I , hence f_j is measurable.

($\Leftarrow$) Let $B = I_1 \times I_2 \times \dots \times I_m$ be an open box in $\mathbb{R}^m$. $x \in f^{-1}(B) \Leftrightarrow f(x) \in B \Leftrightarrow f_j(x) \in I_j$ for $j = 1, \dots, m \Leftrightarrow x \in f_j^{-1}(I_j)$ for $j = 1, \dots, m$. Thus $f^{-1}(B) = \bigcap_{j=1}^m f_j^{-1}(I_j)$. Since each f_j is measurable, $f_j^{-1}(I_j)$ is measurable, so their finite intersection $f^{-1}(B)$ is measurable. ■

Theorem 2.10

Let Ω be a measurable subset of $\mathbb{R}^n$, and let W be an open subset of $\mathbb{R}^m$. If $f : \Omega \rightarrow W$ is measurable, and $g : W \rightarrow \mathbb{R}^p$ is continuous, then $g \circ f : \Omega \rightarrow \mathbb{R}^p$ is measurable.

Proof.

Let V be an open set in $\mathbb{R}^p$. We have $(g \circ f)^{-1}(V) = f^{-1}(g^{-1}(V))$. Since g is continuous, $g^{-1}(V) = W \cap U$, where U is open by the continuity of g . So $g^{-1}(V)$ is open in W . Because f is measurable, $f^{-1}(g^{-1}(V))$ is measurable. Thus $(g \circ f)^{-1}(V)$ is measurable. ■

Corollary

Let $\Omega \subseteq \mathbb{R}^n$ be measurable. If $f : \Omega \rightarrow \mathbb{R}$ is measurable, then $|f|, \max\{f, 0\}, \min\{f, 0\}$ is measurable.

Proof.

Let

1. $g_1 : \mathbb{R} \rightarrow \mathbb{R}$ defined by $g_1(x) = |x|$,

2. $g_2 : \mathbb{R} \rightarrow \mathbb{R}$ defined by $g_2(x) = \frac{x+|x|}{2}$,

3. $g_3 : \mathbb{R} \rightarrow \mathbb{R}$ defined by $g_3(x) = \frac{x-|x|}{2}$.

where g_1, g_2, g_3 are continuous on their open domains. By theorem (2.10), we are done. ■

Corollary

If $f, g : \Omega \rightarrow \mathbb{R}$ are measurable functions, then $f + g, f - g, fg, f/g$ (where $g \neq 0$), $\max\{f, g\}$, and $\min\{f, g\}$ are measurable.

Proof.

Consider $F : \Omega \rightarrow \mathbb{R}^2$ defined by $F(x) = (f(x), g(x))$. By Corollary (2.3), f, g are measurable $\Rightarrow F$ is measurable. Define

1. $h_{\pm} : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by $h_{\pm}(u, v) = u \pm v$,

2. $h_* : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by $h_*(u, v) = uv$,

3. $h_{\div} : \mathbb{R}^2 \setminus \{0\} \rightarrow \mathbb{R}$ defined by $h_{\div}(u, v) = u/v$,

4. $h_{\max} : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by $h_{\max}(u, v) = \frac{u+v+|u-v|}{2}$,

5. $h_{\min} : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by $h_{\min}(u, v) = \frac{u+v-|u-v|}{2}$.

The above functions are all continuous on their respective open domains. By theorem (2.10), we are done. ■

Proposition 5

Let Ω be a measurable subset of $\mathbb{R}^n$, $f : \Omega \rightarrow \mathbb{R}$ be a function. f is measurable if and only if $f^{-1}((\alpha, \infty))$ is measurable for any $\alpha \in \mathbb{R}$.

Proof.

($\Rightarrow$) Trivial, because (α, ∞) is an open set.

($\Leftarrow$) Let $I = (\alpha, \beta)$ be an open interval. We can write $(\alpha, \beta) = (\alpha, \infty) \cap (\mathbb{R} \setminus [\beta, \infty))$. Note that $[\beta, \infty) = \bigcap_{m=1}^{\infty} (\beta - \frac{1}{m}, \infty)$. Therefore, $f^{-1}([\beta, \infty)) = f^{-1}\left(\bigcap_{m=1}^{\infty} (\beta - \frac{1}{m}, \infty)\right) = \bigcap_{m=1}^{\infty} f^{-1}((\beta - \frac{1}{m}, \infty))$. By assumption, $f^{-1}((\beta - \frac{1}{m}, \infty))$ is measurable, so the countable intersection is measurable. Thus, $f^{-1}((\alpha, \beta)) = f^{-1}((\alpha, \infty)) \cap f^{-1}(\mathbb{R} \setminus [\beta, \infty))$ is the intersection of two measurable sets, hence measurable. ■

Definition 2.8 (Measurable Functions in the Extended Reals)

Let Ω be a measurable subset of $\mathbb{R}^n$. The set of **extended real numbers** is defined as:

$$\mathbb{R}^* = \mathbb{R} \cup \{\infty, -\infty\}.$$

A function $f : \Omega \rightarrow \mathbb{R}^*$ is called **measurable** if the set $f^{-1}((a, \infty])$ is measurable for all $a \in \mathbb{R}$.

Remark. Note that $f^{-1}((a, \infty]) = f^{-1}((a, \infty)) \cup f^{-1}(\{\infty\})$. This definition implies that $f^{-1}((a, b))$ is measurable for any $a < b$, because we can write the interval as:

$$(a, b) = (a, \infty] \cap [-\infty, b) = (a, \infty] \cap (\mathbb{R}^* \setminus [b, \infty]).$$

The set $[b, \infty]$ is measurable because it can be expressed as a countable intersection:

$$[b, \infty] = \bigcap_{n=1}^{\infty} \left(b - \frac{1}{n}, \infty \right].$$

Theorem 2.11

Let $\Omega \subseteq \mathbb{R}^n$ be measurable. For each positive integer n , let $f_n : \Omega \rightarrow \mathbb{R}^*$ be a measurable function. Then $\sup f_n$, $\inf f_n$, $\limsup_{n \rightarrow \infty} f_n$, and $\liminf_{n \rightarrow \infty} f_n$ are measurable.

Proof.

Let $g(x) = \sup_{n \geq 1} f_n(x)$. We check if $g^{-1}((a, \infty])$ is measurable for any $a \in \mathbb{R}$. $x \in g^{-1}((a, \infty]) \Leftrightarrow \sup f_n(x) > a \Leftrightarrow \exists k \in \mathbb{N} \text{ s.t. } f_k(x) > a \Leftrightarrow x \in f_k^{-1}((a, \infty])$.

Therefore, $g^{-1}((a, \infty]) = \bigcup_{n=1}^{\infty} f_n^{-1}((a, \infty])$, which is a countable union of measurable sets, so it is measurable.

Let $h(x) = \inf_{n \geq 1} f_n(x)$. $x \in h^{-1}([a, \infty]) \Leftrightarrow \inf f_n(x) \geq a \Leftrightarrow \forall n \in \mathbb{N}, f_n(x) \geq a \Leftrightarrow x \in \bigcap_{n=1}^{\infty} f_n^{-1}([a, \infty])$. This is a countable intersection of measurable sets, so it is measurable.

Since $\limsup_{n \rightarrow \infty} f_n(x) = \inf_{N \geq 1} \sup_{n \geq N} f_n(x)$, let $s_N(x) = \sup_{n \geq N} f_n(x)$, we know s_N is measurable for each N . After taking the infimum, $\limsup_{n \rightarrow \infty} f_n(x)$ is measurable. Similarly for $\liminf$. ■

Remark. In particular, if a sequence of measurable functions $\{f_n\}$ converges pointwise to a function f (i.e., $\lim_{n \rightarrow \infty} f_n(x) = f(x)$ for all $x \in \Omega$), then f is also measurable. This follows immediately from the theorem, because if the limit exists, we have:

$$f(x) = \limsup_{n \rightarrow \infty} f_n(x) = \liminf_{n \rightarrow \infty} f_n(x).$$

This is one of the most powerful properties of Lebesgue measurable functions: the pointwise limit of a sequence of measurable functions is always measurable (unlike continuous functions, where a pointwise limit may fail to be continuous).

2.4 Lebesgue Integration

Definition 2.9 (Characteristic Function)

Let Ω be a set and $E \subseteq \Omega$ be a subset. The **characteristic function** of E , denoted by $\chi_E : \Omega \rightarrow \mathbb{R}$, is defined as:

$$\chi_E(x) = \begin{cases} 1 & \text{if } x \in E \\ 0 & \text{if } x \in \Omega \setminus E \end{cases}$$

Remark. In some contexts, the characteristic function is also called the **indicator function**, often denoted by $\mathbb{1}_E(x)$. Note that χ_E maps the membership of a set to the numerical values 1 and 0, which allows us to convert set operations into algebraic operations. For example:

1. $\chi_{A \cap B}(x) = \chi_A(x) \cdot \chi_B(x)$
2. $\chi_{A \cup B}(x) = \chi_A(x) + \chi_B(x) - \chi_{A \cap B}(x)$
3. $\chi_{\Omega \setminus E}(x) = 1 - \chi_E(x)$

Proposition 6

Let $\Omega \subseteq \mathbb{R}^n$ be a measurable set and let E be a measurable subset of Ω . The characteristic function $\chi_E : \Omega \rightarrow \mathbb{R}$ is measurable.

Proof.

Let I be an open interval in $\mathbb{R}$. If $0 \notin I, 1 \notin I$, then $\chi_E^{-1}(I) = \emptyset$. If $0 \in I, 1 \notin I$, then $\chi_E^{-1}(I) = \Omega \setminus E$. If $0 \notin I, 1 \in I$, then $\chi_E^{-1}(I) = E$. If $0 \in I, 1 \in I$, then $\chi_E^{-1}(I) = \Omega$. In all cases, the resulting set is measurable. Thus χ_E is measurable. ■

Definition 2.10 (Simple Function)

Let $\Omega \subseteq \mathbb{R}^n$ be a measurable set, and $f : \Omega \rightarrow \mathbb{R}$ be a measurable function. We say f is a **simple function** if the image $f(\Omega)$ is finite.

Proposition 7

Let $f : \Omega \rightarrow \mathbb{R}$ be a simple function (a measurable function where $f(\Omega)$ is finite). Then there exist finitely many real numbers $c_1, \dots, c_N$ and pairwise disjoint measurable sets

$$E_1, \dots, E_N \text{ such that } f = \sum_{j=1}^N c_j \chi_{E_j}.$$

Proof.

By definition of simple function, $f(\Omega)$ is finite. WLOG, we may write the image as $f(\Omega) = \{c_1, c_2, \dots, c_N\}$ where c_j are distinct. Define $E_j = f^{-1}(\{c_j\})$ for each

$1 \leq j \leq N$. Suppose $E_m \cap E_n \neq \emptyset$ for $m \neq n$. Then $\exists p \in E_m \cap E_n$ such that $f(p) = c_m$ and $f(p) = c_n$, which violates the definition of a function since $c_m \neq c_n$. So E_j are disjoint. Also, $\{c_j\} = \bigcap_{k=1}^{\infty} \left(c_j - \frac{1}{k}, c_j + \frac{1}{k}\right)$, so $E_j = f^{-1}(\{c_j\}) = \bigcap_{k=1}^{\infty} f^{-1}\left(\left(c_j - \frac{1}{k}, c_j + \frac{1}{k}\right)\right)$, which is a countable intersection of measurable sets and thus measurable. Note that $\Omega = f(\cup\{c_j\}) = \cup f(\{c_j\}) = \cup E_j$. For any $x \in \Omega$, $x \in E_k$ for some k , so $f(x) = c_k$. Simultaneously, $\sum_{j=1}^N c_j \chi_{E_j}(x) = c_k \chi_{E_k}(x) = c_k$. ■

Theorem 2.12

Let Ω be a measurable subset of $\mathbb{R}^n$, and let $f : \Omega \rightarrow [0, \infty]$ be a non-negative measurable function. Then there exists a sequence of simple functions $f_1, f_2, \dots$ such that each f_n is non-negative, the sequence is increasing ($0 \leq f_1 \leq f_2 \leq \dots$), and $\lim_{n \rightarrow \infty} f_n(x) = f(x)$ for all $x \in \Omega$.

Proof.

For each integer n , define $E_{n,k} = \{x \in \Omega : \frac{k}{2^n} \leq f(x) < \frac{k+1}{2^n}\}$ for $k = 0, 1, \dots, n2^n - 1$, and $F_n = \{x \in \Omega : f(x) \geq n\}$. Both $E_{n,k}$ and F_n are pre-images of intervals, hence measurable. Define $f_n(x) = \sum_{k=0}^{n2^n-1} \frac{k}{2^n} \chi_{E_{n,k}}(x) + n \chi_{F_n}(x)$. To show f_n is increasing, look at $f_{n+1}(x)$: If $f(x) \geq n$, then $f_n(x) = n \leq f_{n+1}(x)$. If $f(x) < n$, then for some k , $\frac{k}{2^n} \leq f(x) < \frac{k+1}{2^n}$, so $f_n(x) = \frac{k}{2^n}$. In the $n+1$ grid, this interval is split into $\left[\frac{2k}{2^{n+1}}, \frac{2k+1}{2^{n+1}}\right)$ and $\left[\frac{2k+1}{2^{n+1}}, \frac{2(k+1)}{2^{n+1}}\right)$. So $f_{n+1}(x)$ is either $\frac{2k}{2^{n+1}} = \frac{k}{2^n}$ or $\frac{2k+1}{2^{n+1}} > \frac{k}{2^n}$. Thus $f_n(x) \leq f_{n+1}(x)$. As $n \rightarrow \infty$, if $f(x) = \infty$, $f_n(x) = n \rightarrow \infty$.

If $f(x) < \infty \exists N \in \mathbb{N}$ s.t. $f(x) < N$. Let $n \geq f(x)$, there exists some k s.t.

$$\begin{aligned} \frac{k}{2^n} \leq f(x) < \frac{k+1}{2^n} &\Rightarrow f_n(x) \leq f(x) < f_n(x) + \frac{1}{2^n} \\ &\Rightarrow 0 \leq |f(x) - f_n(x)| < \frac{1}{2^n}, \end{aligned}$$

so $\lim_{n \rightarrow \infty} f_n(x) = f(x)$. ■

Definition 2.11 (Lebesgue Integral of Non-Negative Simple Function)

Let $\Omega \subseteq \mathbb{R}^n$ be measurable, and $f : \Omega \rightarrow \mathbb{R}$ be a simple function, where $f \geq 0$. Thus, f is measurable and the image $f(\Omega)$ is finite and contained in $[0, \infty)$. We define the Lebesgue integral of f on Ω by

$$\int_{\Omega} f = \sum_{\lambda \in f(\Omega), \lambda > 0} \lambda m(\{x \in \Omega : f(x) = \lambda\}).$$

Remark. By definition, a non-negative simple function is a finite linear combination of

characteristic functions:

$$f = \sum_{j=1}^N c_j \chi_{E_j},$$

where the coefficients c_j are distinct non-negative real numbers and the corresponding measurable sets $E_j = \{x \in \Omega : f(x) = c_j\}$ are pairwise disjoint. Using this explicit form, the Lebesgue integral of f evaluates directly to:

$$\int_{\Omega} f = \sum_{j=1}^N c_j m(E_j)$$

Note that if any coefficient $c_j = 0$, its contribution to the sum is $0 \cdot m(E_j) = 0$.

Proposition 8

Let $\Omega \subseteq \mathbb{R}^n$ be measurable, and let $E_1, \dots, E_N$ be pairwise disjoint measurable subsets of Ω . Let $c_1, \dots, c_N$ be non-negative real numbers (may repeat). Then

$$\int_{\Omega} \sum_{j=1}^N c_j \chi_{E_j} = \sum_{j=1}^N c_j m(E_j).$$

Proof.

Let $f = \sum_{j=1}^N c_j \chi_{E_j}$. Let $d_1, d_2, \dots, d_L$ be the distinct positive numbers among $c_1, \dots, c_N$. Define $F_l = \bigcup_{\{j: c_j = d_l\}} E_j$. Since E_j are disjoint and measurable,

$m(F_l) = \sum_{\{j: c_j = d_l\}} m(E_j)$. We can rewrite f as $f = \sum_{l=1}^L d_l \chi_{F_l}$, where d_l are distinct.

By the definition of the integral of a simple function,

$$\int_{\Omega} f = \sum_{l=1}^L d_l m(F_l) = \sum_{l=1}^L d_l \sum_{\{j: c_j = d_l\}} m(E_j) = \sum_{l=1}^L \sum_{\{j: c_j = d_l\}} d_l m(E_j) = \sum_{j=1}^N c_j m(E_j).$$

■

Theorem 2.13

Let $\Omega \subseteq \mathbb{R}^n$ be a measurable set, and let f and g be non-negative simple functions. The following properties hold:

- (a) $0 \leq \int_{\Omega} f \leq \infty$. Furthermore, $\int_{\Omega} f = 0$ if and only if $m(\{x \in \Omega : f(x) \neq 0\}) = 0$.
- (b) $\int_{\Omega} (f + g) = \int_{\Omega} f + \int_{\Omega} g$.
- (c) $\int_{\Omega} c f = c \int_{\Omega} f$ for any constant $c \geq 0$.
- (d) If $f(x) \leq g(x)$ for all $x \in \Omega$, then $\int_{\Omega} f \leq \int_{\Omega} g$.

(e) If $f(x) = g(x)$ almost everywhere, then $\int_{\Omega} f = \int_{\Omega} g$.

Proof.

Proof of (a): If $f = 0$ on Ω , then clearly $\int_{\Omega} f = 0$. In general, we may assume $f = \sum_{j=1}^N c_j \chi_{E_j}$ with $c_j \geq 0$. By definition, $\int_{\Omega} f = \sum_{j=1}^N c_j m(E_j)$, which implies

$0 \leq \int_{\Omega} f \leq \infty$ (the integral reaches infinity if $m(E_j) = \infty$ for some j where $c_j > 0$).

($\Rightarrow$) Suppose $\int_{\Omega} f = 0$. Since all terms $c_j m(E_j)$ are non-negative, we must have $c_j m(E_j) = 0$ for all $j = 1, \dots, N$. For any $c_k > 0$, it follows that $m(E_k) = 0$.

Therefore, $m(\{x \in \Omega : f(x) \neq 0\}) = m\left(\bigcup_{\{k:c_k>0\}} E_k\right) = \sum_{\{k:c_k>0\}} m(E_k) = 0$.

($\Leftarrow$) Conversely, suppose $m(\{x \in \Omega : f(x) \neq 0\}) = 0$. For any k such that $c_k > 0$, we have $E_k \subseteq \{x \in \Omega : f(x) \neq 0\}$, which implies $m(E_k) = 0$. Thus, every term in the sum $\sum_{j=1}^N c_j m(E_j)$ is zero, yielding $\int_{\Omega} f = 0$.

Proof of (b): Let $f = \sum_{j=0}^N c_j \chi_{E_j}$ and $g = \sum_{k=0}^M d_k \chi_{F_k}$, where $\{c_0, \dots, c_N\}$ and $\{d_0, \dots, d_M\}$ are distinct non-negative real numbers, and $\{E_j\}_{j=0}^N$ and $\{F_k\}_{k=0}^M$ are disjoint measurable sets forming partitions of Ω . The intersection $\{E_j \cap F_k\}$ forms a finite, pairwise disjoint measurable partition of Ω . We can express the indicator functions as $\chi_{E_j} = \sum_{k=0}^M \chi_{E_j \cap F_k}$ and $\chi_{F_k} = \sum_{j=0}^N \chi_{E_j \cap F_k}$. This allows us to

write $f + g = \sum_{j=0}^N \sum_{k=0}^M (c_j + d_k) \chi_{E_j \cap F_k}$. Taking the integral yields:

$$\begin{aligned} \int_{\Omega} (f + g) &= \sum_{j=0}^N \sum_{k=0}^M (c_j + d_k) m(E_j \cap F_k) \\ &= \sum_{j=0}^N c_j m\left(\bigcup_{k=0}^M (E_j \cap F_k)\right) + \sum_{k=0}^M d_k m\left(\bigcup_{j=0}^N (E_j \cap F_k)\right) \\ &= \sum_{j=0}^N c_j m(E_j) + \sum_{k=0}^M d_k m(F_k) = \int_{\Omega} f + \int_{\Omega} g. \end{aligned}$$

Proof of (c): By direct computation,

$$\int_{\Omega} c f = \sum_{j=1}^N (c \cdot c_j) m(E_j) = c \sum_{j=1}^N c_j m(E_j) = c \int_{\Omega} f.$$

Proof of (d): Since $f(x) \leq g(x)$ for all $x \in \Omega$, the difference $g - f$ is a non-negative simple function. We can write $g = (g - f) + f$. Applying property (b), we obtain

$\int_{\Omega} g = \int_{\Omega} (g - f) + \int_{\Omega} f$. Since the integral of a non-negative simple function is non-negative, $\int_{\Omega} g \geq \int_{\Omega} f$.

Proof of (e): Let $N = \{x \in \Omega : f(x) \neq g(x)\} = \{x \in \Omega : (f - g)(x) \neq 0\}$. By the given assumption, $m(N) = 0$, $f = g$ almost everywhere. Since f and g are simple functions, $f - g$ is also a simple function. Applying the result from (a), since the measure of the set where $f - g \neq 0$ is zero (i.e. $f - g = 0$ a.e.), we have $\int_{\Omega} f - g = 0$. Consequently, $\int_{\Omega} f = \int_{\Omega} g$ by (b). ■

Definition 2.12 (Lebesgue Integral for Non-Negative Measurable Functions)

Let $\Omega \subseteq \mathbb{R}^n$ be a measurable set and let $f : \Omega \rightarrow [0, \infty]$ be a measurable and non-negative function. We define the Lebesgue integral of f over Ω as the supremum of the integrals of all non-negative simple functions bounded by f :

$$\int_{\Omega} f = \sup \left\{ \int_{\Omega} s : 0 \leq s \leq f, \text{ where } s \text{ is a simple function} \right\}$$

Remark.

1. If $\Omega' \subseteq \Omega$ is a measurable subset, then the integral over Ω' is defined by $\int_{\Omega'} f := \int_{\Omega} f|_{\Omega'}$.

2. If f itself is a non-negative simple function such that $f = \sum_{j=1}^N c_j \chi_{E_j}$, this supremum definition coincides with the standard simple function integral $\int_{\Omega} f = \sum_{j=1}^N c_j m(E_j)$.

3. By definition, the value of the integral satisfies $0 \leq \int_{\Omega} f \leq \infty$.

3. By definition, the value of the integral satisfies $0 \leq \int_{\Omega} f \leq \infty$.

Theorem 2.14

Let $\Omega \subseteq \mathbb{R}^n$ be measurable, and let $f, g : \Omega \rightarrow [0, \infty]$ be non-negative measurable functions. The following properties hold:

(a) $0 \leq \int_{\Omega} f \leq \infty$. Furthermore, $\int_{\Omega} f = 0$ iff $f(x) = 0$ a.e.

(b) $\int_{\Omega} c f = c \int_{\Omega} f$ for any constant $c \geq 0$.

(c) If $f(x) \leq g(x)$ for all $x \in \Omega$, then $\int_{\Omega} f \leq \int_{\Omega} g$.

(d) If $f(x) = g(x)$ almost everywhere in Ω , then $\int_{\Omega} f = \int_{\Omega} g$.

(e) If Ω' is a measurable subset of Ω , then $\int_{\Omega'} f = \int_{\Omega} f \chi_{\Omega'} \leq \int_{\Omega} f$.

Proof.

Proof of (a): By definition, $\int_{\Omega} f = \sup\{\int_{\Omega} s : 0 \leq s \leq f\}$. Since s is non-negative, $0 \leq \int_{\Omega} s$, which directly implies $0 \leq \sup \int_{\Omega} s \leq \infty$.

($\Rightarrow$) Assume $\int_{\Omega} f = 0$. We wish to show that $f = 0$ almost everywhere, which is equivalent to proving that the measure of the set $E = \{x \in \Omega : f(x) > 0\}$ is zero.

We can write E as a countable union $E = \bigcup_{k=1}^{\infty} A_k$, where $A_k = \{x \in \Omega : f(x) \geq \frac{1}{k}\}$.

Because f is measurable, each $A_k = f^{-1}([\frac{1}{k}, \infty))$ is a measurable set. Define a simple function $s_k = \frac{1}{k}\chi_{A_k}$. It is clear that $s_k \leq f$. Thus, $\int_{\Omega} s_k \leq \int_{\Omega} f = 0$. Since $\int_{\Omega} s_k = \frac{1}{k}m(A_k)$, it must be that $\frac{1}{k}m(A_k) = 0$, implying $m(A_k) = 0$ for all positive integers k . By countable subadditivity, $m(E) \leq \sum_{k=1}^{\infty} m(A_k) = 0$. Therefore, $f(x) = 0$ almost everywhere.

($\Leftarrow$) Assume $f(x) = 0$ almost everywhere. Let $E = \{x \in \Omega : f(x) > 0\}$, so $m(E) = 0$. Let s be any non-negative simple function such that $s \leq f$. For any $x \in \Omega \setminus E$, we have $f(x) = 0$, and therefore $s(x) = 0$. We can express $s = \sum_{i=1}^N c_i \chi_{E_i}$.

If $c_i > 0$, then E_i must be a subset of E , which means $m(E_i) \leq m(E) = 0$. Thus, $\int_{\Omega} s = \sum c_i m(E_i) = 0$. Since the integral of every minorizing simple function is zero, the supremum $\int_{\Omega} f$ must also be zero.

Proof of (b): Let $c > 0$. We first show $\int_{\Omega} cf \geq c \int_{\Omega} f$. Let s be any simple function with $0 \leq s \leq f$. Multiplying by c yields $0 \leq cs \leq cf$. Since cs is a simple function bounded by cf , we have $\int_{\Omega} cs \leq \int_{\Omega} cf$. Therefore, $c \int_{\Omega} s \leq \int_{\Omega} cf$. Taking the supremum over all such s yields $c \int_{\Omega} f \leq \int_{\Omega} cf$. Next, we show $\int_{\Omega} cf \leq c \int_{\Omega} f$. Let t be any simple function such that $0 \leq t \leq cf$. Dividing by c , we get $0 \leq \frac{t}{c} \leq f$. Thus, $\int_{\Omega} \frac{t}{c} \leq \int_{\Omega} f$, which implies $\int_{\Omega} t \leq c \int_{\Omega} f$. Taking the supremum over all such t yields $\int_{\Omega} cf \leq c \int_{\Omega} f$. This establishes equality.

Proof of (c): Consider any simple function s such that $0 \leq s \leq f$. Since $f \leq g$, it follows that $0 \leq s \leq g$. Therefore, $\int_{\Omega} s \leq \int_{\Omega} g$. Because this holds for all simple functions bounded by f , taking the supremum yields $\int_{\Omega} f \leq \int_{\Omega} g$.

Proof of (d): Let $N = \{x \in \Omega : f(x) \neq g(x)\}$. By assumption, $m(N) = 0$. Consequently, $f = g$ on the set $\Omega \setminus N$. We first prove $\int_{\Omega} f \leq \int_{\Omega} g$. Let s be any simple function bounded by f ($0 \leq s \leq f$). Define a new simple function $t = s\chi_{\Omega \setminus N}$. For any $x \in \Omega \setminus N$, $t(x) = s(x) \leq f(x) = g(x)$. For any $x \in N$, $t(x) = 0 \leq g(x)$. Thus, $0 \leq t \leq g$ everywhere on Ω , which implies $\int_{\Omega} t \leq \int_{\Omega} g$. Since $t = s$ almost everywhere (they differ only on N), we know from the simple function properties that $\int_{\Omega} t = \int_{\Omega} s$. Hence, $\int_{\Omega} s \leq \int_{\Omega} g$. Taking the supremum

over s yields $\int_{\Omega} f \leq \int_{\Omega} g$. A completely symmetric argument proves $\int_{\Omega} g \leq \int_{\Omega} f$, thus confirming equality.

Proof of (e): We want to show $\int_{\Omega'} f = \int_{\Omega} f \chi_{\Omega'}$. Let s be any simple function on Ω' such that $0 \leq s \leq f|_{\Omega'}$. We can extend s to a function $\tilde{s}$ on all of Ω by setting $\tilde{s}(x) = s(x)$ on Ω' and 0 on $\Omega \setminus \Omega'$. It is clear that $\tilde{s} \leq f \chi_{\Omega'}$, which implies $\int_{\Omega} \tilde{s} \leq \int_{\Omega} f \chi_{\Omega'}$. Since $\int_{\Omega} \tilde{s} = \int_{\Omega'} s$, we have $\int_{\Omega'} s \leq \int_{\Omega} f \chi_{\Omega'}$. Taking the supremum over s gives $\int_{\Omega'} f \leq \int_{\Omega} f \chi_{\Omega'}$. Conversely, let t be any simple function on Ω such that $0 \leq t \leq f \chi_{\Omega'}$. We notice that t restricted to Ω' satisfies $t = t|_{\Omega'}$ (Why?), $t|_{\Omega'} \leq f \chi_{\Omega'} = f|_{\Omega'}$. Therefore, $\int_{\Omega} t = \int_{\Omega'} t|_{\Omega'} \leq \int_{\Omega'} f|_{\Omega'} = \int_{\Omega'} f$. Taking the supremum over t yields $\int_{\Omega} f \chi_{\Omega'} \leq \int_{\Omega'} f$. Finally, since $f \chi_{\Omega'} \leq f$, property (c) guarantees that $\int_{\Omega} f \chi_{\Omega'} \leq \int_{\Omega} f$, giving us the final bound $\int_{\Omega'} f \leq \int_{\Omega} f$. ■

Remark. It is very difficult to prove that $\int_{\Omega} (f + g) = \int_{\Omega} f + \int_{\Omega} g$ directly. We require a big theorem to prove this later.

Definition 2.13 (Absolutely Integrable Function)

Let $\Omega \subseteq \mathbb{R}^n$ be measurable and $f : \Omega \rightarrow \mathbb{R}^*$ be a measurable function. We say f is absolutely integrable if the integral of its absolute value is finite, i.e., $\int_{\Omega} |f| < \infty$.

Definition 2.14

Let $\Omega \subseteq \mathbb{R}^n$ be a measurable set. We define $\mathcal{L}^p(\Omega)$ as the space of all measurable functions $f : \Omega \rightarrow \mathbb{R}^*$ such that $\int_{\Omega} |f|^p d\mu < \infty$.

However, $\mathcal{L}^p(\Omega)$ is only a seminormed space, not a true normed space, because $\int_{\Omega} |f|^p d\mu = 0$ only implies $f = 0$ almost everywhere, not everywhere. To resolve this, we introduce an equivalence relation $f \sim g$ if $f = g$ a.e.

Definition 2.15

The Lebesgue space $L^p(\Omega)$ is defined as the quotient space $\mathcal{L}^p(\Omega)/\sim$, which consists of equivalence classes of functions:

$$L^p(\Omega) = \{[f] \mid f \in \mathcal{L}^p(\Omega)\}$$

equipped with the norm $\|f\|_{L^p} = \left(\int_{\Omega} |f|^p d\mu \right)^{1/p}$.

Remark.

1. In particular, for $p = 1$, the L^1 space is the set of all equivalence classes of absolutely integrable functions, with the norm $\|f\|_{L^1} = \int_{\Omega} |f| d\mu$.
2. Strictly speaking, the elements of $L^1(\Omega)$ are equivalence classes $[f]$, not individual functions f . However, it is a universal and standard abuse of terminology in

mathematical analysis to simply write $f \in L^1(\Omega)$ instead of $[f] \in L^1(\Omega)$, and to treat the elements of L^p spaces as functions. We do this with the implicit understanding that we consider two functions to be "the exact same" if they are equal almost everywhere.

If $f : \Omega \rightarrow \mathbb{R}$ is a generic measurable function, it might take negative values. We can decompose it into a positive part and a negative part: $f^+(x) = \max\{f(x), 0\}$, and $f^-(x) = \max\{-f(x), 0\}$. Both f^+ and f^- are non-negative. Obviously, we can write $f(x) = f^+(x) - f^-(x)$ and the absolute value as $|f(x)| = f^+(x) + f^-(x)$. Since f is measurable, f^+ , f^- , and $|f|$ are all measurable. By the additivity of non-negative integrals, $\int_{\Omega} |f| = \int_{\Omega} f^+ + \int_{\Omega} f^-$. If a function is absolutely integrable ($\int_{\Omega} |f| < \infty$), both $\int_{\Omega} f^+ < \infty$ and $\int_{\Omega} f^- < \infty$ must hold. This allows us to define the Lebesgue integral of f unambiguously.

Definition 2.16 (Lebesgue Integral of an Absolutely Integrable Function)

Let $f \in L^1(\Omega)$. We define the Lebesgue integral of f as $\int_{\Omega} f = \int_{\Omega} f^+ - \int_{\Omega} f^-$. This prevents the indeterminate form $\infty - \infty$.

Theorem 2.15

Let $\Omega \subseteq \mathbb{R}^n$ be a measurable set, $c \in \mathbb{R}$ a constant, and let $f, g \in L^1(\Omega)$. The following properties hold:

- (a) $cf \in L^1(\Omega)$ and $\int_{\Omega} cf = c \int_{\Omega} f$
- (b) $f + g \in L^1(\Omega)$ and $\int_{\Omega} (f + g) = \int_{\Omega} f + \int_{\Omega} g$
- (c) $f - g \in L^1(\Omega)$ and $\int_{\Omega} (f - g) = \int_{\Omega} f - \int_{\Omega} g$
- (d) If $f(x) \leq g(x)$ for all $x \in \Omega$, then $\int_{\Omega} f \leq \int_{\Omega} g$
- (e) $|\int_{\Omega} f| \leq \int_{\Omega} |f|$
- (f) If $f(x) = g(x)$ almost everywhere, then $\int_{\Omega} f = \int_{\Omega} g$

Proof.

Proof of (a): We check absolute integrability: $\int_{\Omega} |cf| = \int_{\Omega} |c||f| = |c| \int_{\Omega} |f| < \infty$. Hence, $cf \in L^1(\Omega)$. If $c \geq 0$, then the positive and negative parts scale cleanly: $(cf)^+ = cf^+$ and $(cf)^- = cf^-$. Thus, $\int_{\Omega} cf = \int_{\Omega} (cf)^+ - \int_{\Omega} (cf)^- = c \int_{\Omega} f^+ - c \int_{\Omega} f^- = c \int_{\Omega} f$. If $c < 0$, the scaling reverses the parts: $(cf)^+ = (-c)f^-$ and $(cf)^- = (-c)f^+$. Thus, $\int_{\Omega} cf = \int_{\Omega} (-c)f^- - \int_{\Omega} (-c)f^+ = (-c) \int_{\Omega} f^- - (-c) \int_{\Omega} f^+ = c \int_{\Omega} f^+ - c \int_{\Omega} f^- = c \int_{\Omega} f$.

Proof of (b): By the triangle inequality, $|f + g| \leq |f| + |g|$. Integrating gives

$\int_{\Omega} |f+g| \leq \int_{\Omega} (|f|+|g|) = \int_{\Omega} |f| + \int_{\Omega} |g| < \infty$, guaranteeing $f+g \in L^1(\Omega)$. Observe the algebraic identity $f+g = (f^+ - f^-) + (g^+ - g^-) = (f^+ + g^+) - (f^- + g^-)$. All terms in parentheses are non-negative. By definition, $\int_{\Omega} (f+g) = \int_{\Omega} (f^+ + g^+) - \int_{\Omega} (f^- + g^-)$. Utilizing the additivity for non-negative functions, we get $\int_{\Omega} f^+ + \int_{\Omega} g^+ - \int_{\Omega} f^- - \int_{\Omega} g^-$, which regroups exactly to $\int_{\Omega} f + \int_{\Omega} g$.

Proof of (c): This is a direct consequence of (a) and (b), since $f-g = f + (-1)g$.

Proof of (d): Since $g-f \in L^1(\Omega)$ and $g-f \geq 0$, its integral must be non-negative: $\int_{\Omega} (g-f) \geq 0$. By linearity, $\int_{\Omega} g - \int_{\Omega} f \geq 0$, which means $\int_{\Omega} f \leq \int_{\Omega} g$.

Proof of (e): Since $f \in L^1(\Omega)$, both $|f|$ and $-|f|$ are in $L^1(\Omega)$. Because $-|f| \leq f \leq |f|$, applying property (d) preserves the inequalities under the integral sign: $-\int_{\Omega} |f| \leq \int_{\Omega} f \leq \int_{\Omega} |f|$. This is the definition of absolute value bounds, giving $|\int_{\Omega} f| \leq \int_{\Omega} |f|$.

Proof of (f): Define $h = f-g \in L^1(\Omega)$. Since $f = g$ almost everywhere, $h = 0$ a.e. and $|h| = 0$ a.e. Therefore, $\int_{\Omega} |h| = 0$ implies $\int_{\Omega} h = 0$. By linearity, $\int_{\Omega} f - \int_{\Omega} g = 0$. ■

2.5 LMCT, Fatuo and LDCT

Theorem 2.16 (Lebesgue Monotone Convergence Theorem)

Let $\Omega \subseteq \mathbb{R}^n$ be a measurable set, and let $\{f_n\}$ be a sequence of non-negative measurable functions, $f_n : \Omega \rightarrow [0, \infty]$. Suppose the sequence is monotonically increasing, meaning $f_n(x) \leq f_{n+1}(x)$ for all $x \in \Omega$ and all positive integers n . Then, the limit function $f(x) = \lim_{n \rightarrow \infty} f_n(x)$ is measurable, and we can interchange the limit and the integral:

$$\int_{\Omega} \lim_{n \rightarrow \infty} f_n = \lim_{n \rightarrow \infty} \int_{\Omega} f_n$$

Proof.

Let $f(x) = \sup_n f_n(x) = \lim_{n \rightarrow \infty} f_n(x)$ (the limit exists because the sequence is monotonically increasing). Note that $f : \Omega \rightarrow [0, \infty]$ is non-negative and measurable. We want to show that $\int_{\Omega} f = \sup_n \int_{\Omega} f_n$.

Step 1: Bounding the sequence of integrals. Since $f_n(x) \leq f_{n+1}(x)$, property (c) tells us that $\int_{\Omega} f_n \leq \int_{\Omega} f_{n+1}$. Thus, the sequence of real numbers $\{\int_{\Omega} f_n\}$ is monotonically increasing. Furthermore, for all n , $f_n \leq f$, which implies $\int_{\Omega} f_n \leq \int_{\Omega} f$. Taking the supremum over n , we obtain $\sup_n \int_{\Omega} f_n \leq \int_{\Omega} f$.

Step 2: Proving the reverse inequality. Recall that $\int_{\Omega} f = \sup_{0 \leq s \leq f} \int_{\Omega} s$, where s is a simple function. It suffices to show that for any simple function s with $0 \leq s \leq f$, we have $\int_{\Omega} s \leq \sup_n \int_{\Omega} f_n$. Let $0 < \epsilon < 1$ be given. For each

positive integer n , define the set $E_n := \{x \in \Omega : f_n(x) \geq (1 - \epsilon)s(x)\}$. Since f_n and s are measurable functions, E_n is a measurable set. Because the sequence $\{f_n\}$ is monotonically increasing, it follows that $E_n \subseteq E_{n+1}$.

We claim that the union $\bigcup_{n=1}^{\infty} E_n = \Omega$. To prove this claim, fix any $x \in \Omega$. By definition, $0 \leq s(x) \leq f(x) = \sup_n f_n(x)$. We need to show there exists an index N such that $f_N(x) \geq (1 - \epsilon)s(x)$. Suppose by contradiction that $f_n(x) < (1 - \epsilon)s(x)$ for all $n \in \mathbb{N}$. If $s(x) = 0$, this yields $0 \leq f_n(x) < 0$, which is impossible. If $s(x) > 0$, taking the supremum yields $f(x) = \sup_n f_n(x) \leq (1 - \epsilon)s(x) < s(x)$. This contradicts the fact that $s(x) \leq f(x)$. Hence, the claim holds.

Step 3: Convergence of the simple function integral. Our goal is to show that $\lim_{n \rightarrow \infty} \int_{E_n} s = \int_{\Omega} s$. If $s = 0$, the equality is trivial. Assume $s \neq 0$, and write it

in its standard representation $s = \sum_{j=1}^N c_j \chi_{F_j}$. The integral of s over E_n is $\int_{E_n} s =$

$\int_{\Omega} s \chi_{E_n} = \int_{\Omega} \sum_{j=1}^N c_j \chi_{F_j \cap E_n} = \sum_{j=1}^N c_j m(F_j \cap E_n)$. Notice that the sets $F_j \cap E_n$ are nested ($F_j \cap E_1 \subseteq F_j \cap E_2 \subseteq \dots$), and their union over n is $F_j \cap \Omega = F_j$. By the continuity of measure from below, $\lim_{n \rightarrow \infty} m(F_j \cap E_n) = m\left(\bigcup_{n=1}^{\infty} (F_j \cap E_n)\right) = m(F_j)$.

Therefore, $\lim_{n \rightarrow \infty} \int_{E_n} s = \sum_{j=1}^N c_j m(F_j) = \int_{\Omega} s$.

Step 4: Final comparison. By the definition of E_n , we know that on E_n , $f_n(x) \geq (1 - \epsilon)s(x)$. Integrating over E_n gives $(1 - \epsilon) \int_{E_n} s \leq \int_{E_n} f_n$. Since f_n is non-negative, $\int_{E_n} f_n \leq \int_{\Omega} f_n$. Taking the supremum over all n on both sides: $(1 - \epsilon) \lim_{n \rightarrow \infty} \int_{E_n} s \leq \sup_n \int_{\Omega} f_n$. Using the result from Step 3, we have $(1 - \epsilon) \int_{\Omega} s \leq \sup_n \int_{\Omega} f_n$. Since this inequality holds for any $0 < \epsilon < 1$, we can let $\epsilon \rightarrow 0$ to obtain $\int_{\Omega} s \leq \sup_n \int_{\Omega} f_n$. Finally, since this holds for every simple function $0 \leq s \leq f$, taking the supremum over all such s yields $\int_{\Omega} f \leq \sup_n \int_{\Omega} f_n$. This completes the proof. ■

Remark. The Monotone Convergence Theorem implies that we can interchange limit and integration if the sequence is increasing. In general, we cannot arbitrarily interchange limit and integration. For example, let $f_n(x) = \chi_{[n, n+1]}$. For all n , $\int_{\mathbb{R}} f_n = 1$, which means $\lim_{n \rightarrow \infty} \int_{\mathbb{R}} f_n = 1$. However, the pointwise limit is $\lim_{n \rightarrow \infty} f_n(x) = 0$, so $\int_{\mathbb{R}} (\lim_{n \rightarrow \infty} f_n) = 0$. The reason MCT fails here is that the sequence f_n is not monotonically increasing.

Theorem 2.17

Let $\Omega \subseteq \mathbb{R}^n$ be measurable, and let $f, g : \Omega \rightarrow [0, \infty]$ be measurable functions. Then $\int_{\Omega} (f + g) = \int_{\Omega} f + \int_{\Omega} g$.

Proof.

Because f and g are non-negative measurable functions, there exist sequences of simple functions $\{s_n\}$ and $\{t_n\}$ such that $s_n \uparrow f$ and $t_n \uparrow g$. It follows that the sequence $(s_n + t_n)$ is monotonically increasing and converges pointwise to $(f + g)$. Applying the Lebesgue Monotone Convergence Theorem, we have $\lim_{n \rightarrow \infty} \int_{\Omega} (s_n + t_n) = \int_{\Omega} (f + g)$. Since s_n and t_n are simple functions, we already know that $\int_{\Omega} (s_n + t_n) = \int_{\Omega} s_n + \int_{\Omega} t_n$ for all n . Taking the limit: $\lim_{n \rightarrow \infty} \left(\int_{\Omega} s_n + \int_{\Omega} t_n \right) = \lim_{n \rightarrow \infty} \int_{\Omega} s_n + \lim_{n \rightarrow \infty} \int_{\Omega} t_n = \int_{\Omega} f + \int_{\Omega} g$. Thus, the equality holds. ■

Corollary

If $\Omega \subseteq \mathbb{R}^n$ is measurable, and $g_1, g_2, \dots$ are non-negative measurable functions from $\Omega \rightarrow [0, \infty]$, then the integral of the infinite sum equals the infinite sum of the integrals: $\int_{\Omega} \sum_{n=1}^{\infty} g_n = \sum_{n=1}^{\infty} \int_{\Omega} g_n$.

Proof.

Let the partial sum be $G_N = \sum_{n=1}^N g_n$. The sequence G_N is non-negative, monotonically increasing, and measurable. By the Lebesgue Monotone Convergence Theorem, $\lim_{N \rightarrow \infty} \int_{\Omega} G_N = \int_{\Omega} \lim_{N \rightarrow \infty} G_N = \int_{\Omega} \sum_{n=1}^{\infty} g_n$. On the other hand, evaluating the limit directly using finite additivity: $\lim_{N \rightarrow \infty} \int_{\Omega} G_N = \lim_{N \rightarrow \infty} \int_{\Omega} \sum_{n=1}^N g_n = \lim_{N \rightarrow \infty} \sum_{n=1}^N \int_{\Omega} g_n = \sum_{n=1}^{\infty} \int_{\Omega} g_n$. ■

Lemma (Fatou)

Let $\Omega \subseteq \mathbb{R}^n$ be a measurable set, and let $f_1, f_2, \dots : \Omega \rightarrow [0, \infty]$ be a sequence of non-negative measurable functions. Then,

$$\int_{\Omega} \liminf_{n \rightarrow \infty} f_n \leq \liminf_{n \rightarrow \infty} \int_{\Omega} f_n$$

Proof.

Define a new sequence of functions $g_N(x) = \inf_{n \geq N} f_n(x)$. As N increases, the set

of functions we are taking the infimum over shrinks, which means g_N is monotonically increasing. By the Lebesgue Monotone Convergence Theorem, we can interchange the limit and the integral for g_N : $\lim_{N \rightarrow \infty} \int_{\Omega} g_N = \int_{\Omega} \lim_{N \rightarrow \infty} g_N$. By definition, the limit of g_N is exactly the limit infimum: $\lim_{N \rightarrow \infty} g_N = \liminf_{n \rightarrow \infty} f_n$. Thus,

$$\int_{\Omega} \liminf_{n \rightarrow \infty} f_n = \lim_{N \rightarrow \infty} \int_{\Omega} g_N.$$

Now, fix a specific $N \in \mathbb{N}$. For all $n \geq N$, we have $g_N(x) = \inf_{k \geq N} f_k(x) \leq f_n(x)$. Integrating both sides gives $\int_{\Omega} g_N(x) \leq \int_{\Omega} f_n(x)$ for all $n \geq N$. Taking the infimum over $n \geq N$ on the right side yields $\int_{\Omega} g_N(x) \leq \inf_{n \geq N} \int_{\Omega} f_n(x)$. Finally, taking the limit as $N \rightarrow \infty$ on both sides: $\lim_{N \rightarrow \infty} \int_{\Omega} g_N(x) \leq \lim_{N \rightarrow \infty} \inf_{n \geq N} \int_{\Omega} f_n(x) = \liminf_{n \rightarrow \infty} \int_{\Omega} f_n$. Combining this with our earlier equality completes the proof. ■

Lemma

Let $\Omega \subseteq \mathbb{R}^n$ be measurable, and let $f : \Omega \rightarrow [0, \infty]$ be a measurable function such that its integral is finite ($\int_{\Omega} f < \infty$). Then f is finite almost everywhere, meaning the set $\{x \in \Omega : f(x) = \infty\}$ has measure zero.

Proof.

For each integer $k \geq 1$, define the set $A_k := \{x \in \Omega : f(x) > k\}$. Because f is measurable, the preimage $A_k = f^{-1}((k, \infty])$ is a measurable set. The set of points where f is infinity can be written as an infinite intersection: $\{x \in \Omega : f(x) = \infty\} = \bigcap_{k=1}^{\infty} A_k$. Define the characteristic function scaled by k : $k\chi_{A_k}$. If $x \in A_k$, then $k\chi_{A_k}(x) = k < f(x)$. If $x \notin A_k$, then $k\chi_{A_k}(x) = 0 \leq f(x)$. In either case, the simple function $k\chi_{A_k}$ is bounded by f . Therefore, $\int_{\Omega} k\chi_{A_k} \leq \int_{\Omega} f$, which simplifies to $k \cdot m(A_k) \leq \int_{\Omega} f$, or $m(A_k) \leq \frac{1}{k} \int_{\Omega} f$. Since the integral $\int_{\Omega} f$ is a finite constant, taking the limit as $k \rightarrow \infty$ gives $\lim_{k \rightarrow \infty} m(A_k) = 0$. Since the sets are nested ($A_{k+1} \subset A_k$), the measure of their intersection is the limit of their measures: $m\left(\bigcap_{k=1}^{\infty} A_k\right) = \lim_{k \rightarrow \infty} m(A_k) = 0$. ■

Lemma (Borel-Cantelli)

Let $\Omega_1, \Omega_2, \dots$ be measurable subsets of $\mathbb{R}^n$ such that the sum of their measures is finite ($\sum_{n=1}^{\infty} m(\Omega_n) < \infty$). Then the set $N = \{x \in \mathbb{R}^n : x \in \Omega_n \text{ for infinitely many } n\}$ has measure zero.

Proof.

Let $G_N(x) = \sum_{n=1}^N \chi_{\Omega_n}(x)$ be the number of sets among the first N sets that contain x . G_N is a non-negative, monotonically increasing, measurable function. Integrating G_N yields $\int_{\Omega} G_N = \sum_{n=1}^N \int_{\Omega} \chi_{\Omega_n} = \sum_{n=1}^N m(\Omega_n) < \infty$. By the Lebesgue Monotone Convergence Theorem, we can pass the limit inside the integral: $\lim_{N \rightarrow \infty} \int_{\Omega} G_N = \int_{\Omega} \lim_{N \rightarrow \infty} G_N = \int_{\Omega} \sum_{n=1}^{\infty} \chi_{\Omega_n} = \sum_{n=1}^{\infty} m(\Omega_n) < \infty$. Let $G(x) = \lim_{N \rightarrow \infty} G_N(x) = \sum_{n=1}^{\infty} \chi_{\Omega_n}(x)$. The function $G(x)$ counts the total number of sets Ω_n that contain x . If a point x belongs to infinitely many sets Ω_n , then $G(x) = \infty$. Thus, the set N is contained in the set $\{x \in \mathbb{R}^n : G(x) = \infty\}$. Because we have established that $\int_{\Omega} G < \infty$, the previous lemma guarantees that $G(x)$ can only be infinity on a set of measure zero. Therefore, N must have measure zero. ■

Theorem 2.18 (Lebesgue Dominated Convergence Theorem)

Let $\Omega \subseteq \mathbb{R}^n$ be measurable, and let $\{f_n\}$ be a sequence of measurable functions from Ω to $\mathbb{R}^*$ such that the pointwise limit $\lim_{n \rightarrow \infty} f_n(x) = f(x)$ exists for almost every $x \in \Omega$. Furthermore, suppose there exists an absolutely integrable function $F : \Omega \rightarrow [0, \infty]$ that strictly dominates the sequence: $|f_n(x)| \leq F(x)$ for almost every $x \in \Omega$. Then we can interchange the limit and the integral:

$$\int_{\Omega} f = \lim_{n \rightarrow \infty} \int_{\Omega} f_n$$

Proof.

First, we discard the problematic sets. Let $E = \{x : F(x) = \infty\}$. Since F is absolutely integrable ($\int_{\Omega} |F| < \infty$), the measure $m(E) = 0$. We can restrict our attention to the domain $\Omega \setminus E$, where $F(x) < \infty$. The integrals over $\Omega \setminus E$ are identical to the integrals over Ω . On this restricted domain, f_n takes values in $\mathbb{R}$. The domination condition $|f_n(x)| \leq F(x)$ is equivalent to $-F(x) \leq f_n(x) \leq F(x)$. This gives us two non-negative sequences of functions: $F(x) + f_n(x) \geq 0$ and $F(x) - f_n(x) \geq 0$.

We apply Fatou's Lemma to the non-negative sequence $\{F + f_n\}$:

$$\int_{\Omega} \liminf_{n \rightarrow \infty} (F + f_n) \leq \liminf_{n \rightarrow \infty} \int_{\Omega} (F + f_n)$$

Because F is independent of n and integrable, we can pull it out:

$$\int_{\Omega} (F + f) \leq \int_{\Omega} F + \liminf_{n \rightarrow \infty} \int_{\Omega} f_n$$

Subtracting the finite value $\int_{\Omega} F$ from both sides yields: $\int_{\Omega} f \leq \liminf_{n \rightarrow \infty} \int_{\Omega} f_n$.
Next, we apply Fatou's Lemma to the non-negative sequence $\{F - f_n\}$:

$$\int_{\Omega} \liminf_{n \rightarrow \infty} (F - f_n) \leq \liminf_{n \rightarrow \infty} \int_{\Omega} (F - f_n)$$

Notice that $\liminf (F - f_n) = F - \limsup f_n = F - f$. The right side becomes $\liminf (\int F - \int f_n) = \int F - \limsup \int f_n$. Substituting these into the inequality:

$$\int_{\Omega} (F - f) \leq \int_{\Omega} F - \limsup_{n \rightarrow \infty} \int_{\Omega} f_n$$

Subtracting $\int_{\Omega} F$ and multiplying by -1 reverses the inequality: $\int_{\Omega} f \geq \limsup_{n \rightarrow \infty} \int_{\Omega} f_n$.

Combining our two results, we have established:

$$\limsup_{n \rightarrow \infty} \int_{\Omega} f_n \leq \int_{\Omega} f \leq \liminf_{n \rightarrow \infty} \int_{\Omega} f_n$$

Since the limit infimum can never exceed the limit supremum ($\liminf \int f_n \leq \limsup \int f_n$), they must all be strictly equal. Therefore, the limit exists and $\lim_{n \rightarrow \infty} \int_{\Omega} f_n = \int_{\Omega} f$. ■

Definition 2.17 (Upper and Lower Lebesgue Integral)

Let $\Omega \subseteq \mathbb{R}^n$ be measurable, and let $f : \Omega \rightarrow \mathbb{R}$ be a function (which is NOT necessarily measurable). We define the upper Lebesgue integral as the infimum of the integrals of all bounding L^1 functions from above:

$$\overline{\int}_{\Omega} f := \inf \left\{ \int_{\Omega} g : g \in L^1(\Omega), \text{ and } g \geq f \right\}$$

Similarly, the lower Lebesgue integral is defined using functions bounding from below:

$$\underline{\int}_{\Omega} f := \sup \left\{ \int_{\Omega} h : h \in L^1(\Omega), \text{ and } h \leq f \right\}$$

Remark. By definition, the lower integral is always less than or equal to the upper integral: $\underline{\int}_{\Omega} f \leq \overline{\int}_{\Omega} f$. If $f \in L^1(\Omega)$, then we can just pick f itself as the bounding function, meaning $\underline{\int}_{\Omega} f = \int_{\Omega} f = \overline{\int}_{\Omega} f$.

Theorem 2.19

Let $\Omega \subseteq \mathbb{R}^n$ be measurable and let $f : \Omega \rightarrow \mathbb{R}$ be a function. If the upper and lower integrals match ($\overline{\int}_{\Omega} f = \underline{\int}_{\Omega} f$), then $f \in L^1(\Omega)$ (meaning f is guaranteed to be measurable and absolutely integrable), and $\int_{\Omega} f = \overline{\int}_{\Omega} f$.

Proof.

Assume the integrals match and denote the common value by $A = \overline{\int}_{\Omega} f = \underline{\int}_{\Omega} f$. By the infimum definition of the upper integral, for each positive integer $n \geq 1$, there exists an upper bounding function $\tilde{f}_n \in L^1(\Omega)$ such that $\tilde{f}_n \geq f$ and $\int_{\Omega} \tilde{f}_n \leq A + \frac{1}{n}$. Similarly, by the supremum definition of the lower integral, there exists a lower bounding function $\underline{f}_n \in L^1(\Omega)$ such that $\underline{f}_n \leq f$ and $A - \frac{1}{n} \leq \int_{\Omega} \underline{f}_n$.

We construct the pointwise infimum and supremum functions: define $\tilde{F} = \inf_n \tilde{f}_n$ and $\underline{F} = \sup_n \underline{f}_n$. Because taking the infimum or supremum preserves inequalities, we have the tight sandwich: $\underline{F} \leq f \leq \tilde{F}$. Since countable infimums and supremums of measurable functions are measurable, $\tilde{F}$ and $\underline{F}$ are measurable. Furthermore, they are absolutely integrable because they are bounded by the first terms of our sequences: $|\tilde{F}| \leq |\tilde{f}_1|$ and $|\underline{F}| \leq |\underline{f}_1|$, which guarantees $\int_{\Omega} |\tilde{F}| \leq \int_{\Omega} |\tilde{f}_1| < \infty$. Thus $\tilde{F}, \underline{F} \in L^1(\Omega)$.

Since $\tilde{F} \leq \tilde{f}_n$, we know $\int_{\Omega} \tilde{F} \leq A + \frac{1}{n}$ for all n . Letting $n \rightarrow \infty$ gives $\int_{\Omega} \tilde{F} \leq A$. Similarly, since $\underline{F} \geq \underline{f}_n$, we know $A - \frac{1}{n} \leq \int_{\Omega} \underline{F}$ for all n , yielding $A \leq \int_{\Omega} \underline{F}$. Because $\underline{F} \leq \tilde{F}$, we must also have $\int_{\Omega} \underline{F} \leq \int_{\Omega} \tilde{F}$. The only way to satisfy all these inequalities is if $\int_{\Omega} \tilde{F} = \int_{\Omega} \underline{F} = A$.

This implies $\int_{\Omega} (\tilde{F} - \underline{F}) = 0$. Since $\tilde{F} - \underline{F} \geq 0$, the function must be zero almost everywhere, meaning $\tilde{F} = \underline{F}$ almost everywhere. Since f is sandwiched between them ($\underline{F} \leq f \leq \tilde{F}$), it must be that $\underline{F} = f = \tilde{F}$ almost everywhere. Because f is equal to an L^1 function almost everywhere, f is measurable, $f \in L^1(\Omega)$, and its integral is exactly A . ■

Theorem 2.20

Let I be a bounded interval and $f : I \rightarrow \mathbb{R}$ be a Riemann integrable function. Then f is also (absolutely) Lebesgue integrable, and the Lebesgue integral equals the Riemann integral: $\int_I f d\mu = \mathcal{R} \int_I f dx$.

Proof.

Let $A = \mathcal{R} \int_I f$. Because f is Riemann integrable, for any $\epsilon > 0$, there exists a partition $\mathcal{P}$ of the interval I into finitely many subintervals J such that the lower and upper Darboux sums are bounded tight around A :

$$A - \epsilon \leq \sum_{J \in \mathcal{P}} |J| \inf_{x \in J} f(x) \leq A \leq \sum_{J \in \mathcal{P}} |J| \sup_{x \in J} f(x) \leq A + \epsilon$$

We can construct step functions (which are essentially simple functions) based on this partition. Define the lower step function $f_{\epsilon} = \sum_{J \in \mathcal{P}} \inf_{x \in J} f(x) \chi_J$ and the up-

per step function $\tilde{f}_\epsilon = \sum_{J \in \mathcal{P}} \sup_{x \in J} f(x) \chi_J$. Obviously, $f_\epsilon \leq f \leq \tilde{f}_\epsilon$. Their Lebesgue integrals perfectly match the Darboux sums, yielding:

$$A - \epsilon \leq \int_I f_\epsilon \leq A \leq \int_I \tilde{f}_\epsilon \leq A + \epsilon$$

Because f_ϵ bounds f from below, its integral bounds the lower Lebesgue integral: $\int_I f_\epsilon \leq \underline{\int}_I f$. Because $\tilde{f}_\epsilon$ bounds f from above, its integral bounds the upper Lebesgue integral: $\overline{\int}_I f \leq \int_I \tilde{f}_\epsilon$. Stringing these inequalities together gives $A - \epsilon \leq \underline{\int}_I f \leq \overline{\int}_I f \leq A + \epsilon$. Since this bound holds for any arbitrary $\epsilon > 0$, we conclude that the upper and lower integrals are exactly equal to A ($\underline{\int}_I f = \overline{\int}_I f = A$). By the previous theorem, f is Lebesgue measurable, and its Lebesgue integral equals A . ■

Lastly, we shall prove a big theorem in Calculus. Suppose f is an absolutely integrable function on $\mathbb{R}^2$. We would like to calculate the double integral by evaluating iterated integrals. Fubini's Theorem guarantees that under absolute integrability, the order of integration does not matter:

$$\int_{\mathbb{R}^2} f = \int_{\mathbb{R}} \int_{\mathbb{R}} f(x, y) dx dy = \int_{\mathbb{R}} \int_{\mathbb{R}} f(x, y) dy dx$$

Lemma (Fubini's Theorem for Characteristic Functions)

Let $Q = [-N, N]^2$ for some integer $N \in \mathbb{N}$, and let $E \subseteq Q$ be a measurable set. Then for almost every $x \in [-N, N]$, the slice function $\chi_E(x, y)$ is integrable with respect to y , and for almost every $y \in [-N, N]$, $\chi_E(x, y)$ is integrable with respect to x . Furthermore, we have:

$$\int_{[-N, N]} \left(\int_{[-N, N]} \chi_E(x, y) dy \right) dx = \int_{[-N, N]} \left(\int_{[-N, N]} \chi_E(x, y) dx \right) dy = \int_Q \chi_E = m(E)$$

Proof.

Step 1: The key identity for characteristic functions. Because χ_E is not guaranteed to be measurable over individual slices initially, we define the upper and lower integrals over the y -slice. For each $x \in [-N, N]$, define $U(x) = \overline{\int}_{[-N, N]} \chi_E(x, y) dy$ and $L(x) = \underline{\int}_{[-N, N]} \chi_E(x, y) dy$. We claim that the upper integral of $U(x)$ is bounded by the measure of E : $\overline{\int}_{[-N, N]} U(x) dx \leq m(E)$. (This will be proven in Step 2).

If this claim is true, the same logic applied to the complement $Q \setminus E$ gives $\overline{\int}_{[-N, N]} \overline{\int}_{[-N, N]} \chi_{Q \setminus E}(x, y) dy dx \leq m(Q \setminus E) = 4N^2 - m(E)$. However, the left-hand side can be rewritten: $\overline{\int} \overline{\int} (1 - \chi_E) dy dx = 4N^2 - \underline{\int} \underline{\int} \chi_E dy dx = 4N^2 - \underline{\int} L(x) dx$. This

algebraic manipulation yields $m(E) \leq \int L(x)dx \leq \overline{\int} L(x)dx \leq \overline{\int} U(x)dx \leq m(E)$. The squeeze theorem forces $m(E) = \overline{\int} L(x)dx = \int U(x)dx$. Since $0 \leq L \leq U$ and $\int (U - L) = 0$, we must have $U = L$ almost everywhere. This implies $\chi_E(x, y)$ is integrable for almost every x , and the iterated integral equals $m(E)$.

Step 2: Proof of the bounding claim. By the definition of Lebesgue outer measure, $m(E) = \inf \sum m(B_j)$ over all countable covers of open boxes B_j such that $E \subseteq \bigcup B_j$. Let $B_j = I_j \times J_j$. The measure of a box trivially respects Fubini: $m(B_j) = m(I_j)m(J_j) = \int_{\mathbb{R}} \left(\int_{\mathbb{R}} \chi_{B_j}(x, y)dy \right) dx$.

Let $H(x, y) = \sum \chi_{B_j}(x, y)$ be the sum of indicator functions of the covering boxes. Since $E \subseteq \bigcup B_j$, we have $\chi_E \leq H$. Therefore, $U(x) \leq \overline{\int} H(x, y)dy$. Integrating $U(x)$ over the bounds gives:

$$\overline{\int}_{[-N, N]} U(x)dx \leq \overline{\int}_{[-N, N]} \int_{[-N, N]} H(x, y)dydx = \int_{[-N, N]} \sum \chi_{I_j}(x)m(J_j)dx$$

By the Monotone Convergence Theorem for sums, we pull the sum outside:

$$= \sum \int \chi_{I_j}(x)m(J_j)dx = \sum m(I_j)m(J_j) = \sum m(B_j) \leq m(E) + \epsilon$$

Since this bound holds for any optimal cover and any $\epsilon > 0$, we conclude that $\overline{\int} U(x)dx \leq m(E)$.

Step 3: Conclusion by symmetry. We have established that $\int_{[-N, N]} \left(\int_{[-N, N]} \chi_E(x, y)dy \right) dx = m(E)$. By symmetry, exchanging the roles of x and y in the definitions of the upper and lower integrals yields the exact same bound and squeeze result. Thus, $\int_{[-N, N]} \left(\int_{[-N, N]} \chi_E(x, y)dx \right) dy = m(E)$ as well, completing the proof. ■

Theorem 2.21 (Tonelli)

Let $f : \mathbb{R}^2 \rightarrow [0, \infty]$ be a non-negative measurable function. Then the slice functions $f(x, y)$ are measurable for almost every x and almost every y . The functions $F(x) = \int_{\mathbb{R}} f(x, y)dy$ and $G(y) = \int_{\mathbb{R}} f(x, y)dx$ are measurable, and:

$$\int_{\mathbb{R}} \left(\int_{\mathbb{R}} f(x, y)dy \right) dx = \int_{\mathbb{R}} \left(\int_{\mathbb{R}} f(x, y)dx \right) dy = \int_{\mathbb{R}^2} f$$

Proof.

Step 1: Reduction to simple and characteristic functions on a bounded square. Assume first that f is supported on a bounded square $[-N, N] \times [-N, N]$. Since f is measurable and non-negative, there exists a sequence of simple functions s_k increasing to f , with their support also restricted to $[-N, N]^2$.

A simple function is a finite sum $s_k = \sum c_l \chi_{E_l}$. By the preceding Lemma, the theorem holds for the characteristic function of any measurable set $E_l \subseteq [-N, N]^2$ for both integration orders. By linearity of the integral, the theorem also holds for any simple function s_k .

Fixing x , the sequence $s_k(x, \cdot)$ increases to $f(x, \cdot)$. By the Lebesgue Monotone Convergence Theorem (MCT) applied to the inner integral, $\lim_{k \rightarrow \infty} \int_{\mathbb{R}} s_k(x, y) dy = \int_{\mathbb{R}} f(x, y) dy$. Applying the MCT a second time to the outer integral, we can exchange the limit and integration:

$$\int_{\mathbb{R}} \left(\lim_{k \rightarrow \infty} \int_{\mathbb{R}} s_k(x, y) dy \right) dx = \lim_{k \rightarrow \infty} \int_{\mathbb{R}} \int_{\mathbb{R}} s_k(x, y) dy dx = \lim_{k \rightarrow \infty} \int_{\mathbb{R}^2} s_k = \int_{\mathbb{R}^2} f$$

Step 2: Reduction to a bounded interval for general non-negative functions. Now let $f \geq 0$ be a general measurable function on $\mathbb{R}^2$. For any integer $N \in \mathbb{N}$, define the localized function $f_N := f \chi_{[-N, N] \times [-N, N]}$. These localized functions form a monotonically increasing sequence $0 \leq f_1 \leq f_2 \leq \dots \leq f$ that converges pointwise to f . By Step 1, Tonelli's theorem holds for all f_N , so $\int_{\mathbb{R}^2} f_N = \int_{\mathbb{R}} \int_{\mathbb{R}} f_N(x, y) dy dx$.

By the MCT applied over $\mathbb{R}^2$, $\lim_{N \rightarrow \infty} \int_{\mathbb{R}^2} f_N = \int_{\mathbb{R}^2} f$. Fixing x , the sequence $f_N(x, \cdot)$ increases to $f(x, \cdot)$. Applying the MCT twice exactly as we did in Step 1 allows us to exchange the limit and integration again:

$$\int_{\mathbb{R}} \left(\lim_{N \rightarrow \infty} \int_{\mathbb{R}} f_N(x, y) dy \right) dx = \lim_{N \rightarrow \infty} \int_{\mathbb{R}} \int_{\mathbb{R}} f_N(x, y) dy dx = \lim_{N \rightarrow \infty} \int_{\mathbb{R}^2} f_N = \int_{\mathbb{R}^2} f$$

Step 3: Conclusion by symmetry. By symmetry, repeating the identical limiting processes for the reverse integration order (dx inner, dy outer) yields $\int_{\mathbb{R}} \left(\int_{\mathbb{R}} f(x, y) dx \right) dy = \int_{\mathbb{R}^2} f$. This completes the proof of Tonelli's Theorem. ■

Theorem 2.22 (Fubini)

Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be an absolutely integrable function ($f \in L^1(\mathbb{R}^2)$). Then for almost every x , $f(x, y)$ is absolutely integrable with respect to y , and for almost every y , $f(x, y)$ is absolutely integrable with respect to x . Furthermore, the functions $F(x) = \int_{\mathbb{R}} f(x, y) dy$ and $G(y) = \int_{\mathbb{R}} f(x, y) dx$ are absolutely integrable, and we have:

$$\int_{\mathbb{R}} F(x) dx = \int_{\mathbb{R}} \left(\int_{\mathbb{R}} f(x, y) dy \right) dx = \int_{\mathbb{R}} \left(\int_{\mathbb{R}} f(x, y) dx \right) dy = \int_{\mathbb{R}} G(y) dy = \int_{\mathbb{R}^2} f$$

Proof.

Reduction to non-negative functions. Let $f \in L^1(\mathbb{R}^2)$. We can decompose

$f = f^+ - f^-$, where $f^+, f^- \in L^1(\mathbb{R}^2)$ are non-negative measurable functions.

By Tonelli's Theorem (proven above), there exist functions $F^+, F^- : \mathbb{R} \rightarrow [0, \infty]$ such that for almost every x , $F^+(x) = \int_{\mathbb{R}} f^+(x, y) dy$ and $F^-(x) = \int_{\mathbb{R}} f^-(x, y) dy$. We also know that $\int_{\mathbb{R}} F^+(x) dx = \int_{\mathbb{R}^2} f^+$ and $\int_{\mathbb{R}} F^-(x) dx = \int_{\mathbb{R}^2} f^-$.

Since $f \in L^1(\mathbb{R}^2)$, both $\int_{\mathbb{R}^2} f^+$ and $\int_{\mathbb{R}^2} f^-$ are finite. Thus, for almost every x , the absolute value integral is finite:

$$\int_{\mathbb{R}} |f(x, y)| dy = \int_{\mathbb{R}} f^+(x, y) dy + \int_{\mathbb{R}} f^-(x, y) dy < \infty$$

This means $f(x, y)$ is absolutely integrable with respect to y for almost every x .

We can safely define $F(x) := \int_{\mathbb{R}} f(x, y) dy = F^+(x) - F^-(x)$. Since both F^+ and F^- are in $L^1(\mathbb{R})$, their difference $F \in L^1(\mathbb{R})$. Integrating $F(x)$ over $\mathbb{R}$ yields:

$$\int_{\mathbb{R}} F(x) dx = \int_{\mathbb{R}} F^+(x) dx - \int_{\mathbb{R}} F^-(x) dx = \int_{\mathbb{R}^2} f^+ - \int_{\mathbb{R}^2} f^- = \int_{\mathbb{R}^2} f$$

By symmetry, applying the exact same decomposition $f = f^+ - f^-$ and utilizing Tonelli's theorem for the dx inner integral proves that $G(y) = \int_{\mathbb{R}} f(x, y) dx \in L^1(\mathbb{R})$, and integrating $G(y)$ with respect to y yields exactly $\int_{\mathbb{R}^2} f$. This establishes Fubini's Theorem in full. ■

3 Approximation Theory

3.1 The Stone-Weierstrass Theorem

Definition 3.1

Let X be a compact metric space. A subset $A \subseteq C(X)$ is called an algebra if:

1. A is a vector subspace of $C(X)$ ($\alpha f + \beta g \in A$).
2. A is closed under pointwise multiplication ($f, g \in A \implies fg \in A$).

For $f, g \in C(X)$, we define:

$$(f \vee g)(x) := \max\{f(x), g(x)\} = \frac{1}{2}(f(x) + g(x)) + \frac{1}{2}|f(x) - g(x)|$$

$$(f \wedge g)(x) := \min\{f(x), g(x)\} = \frac{1}{2}(f(x) + g(x)) - \frac{1}{2}|f(x) - g(x)|$$

Definition 3.2

A subset $L \subseteq C(X)$ is called a vector lattice if it is a vector subspace, and for any $f, g \in L$, we have $f \vee g \in L$ and $f \wedge g \in L$.

Proposition 9

A vector subspace $L \subseteq C(X)$ is a vector lattice if and only if $f \in L \implies |f| \in L$.

Proof. ($\implies$) Take $g = -f \in L$. Then $f \vee (-f) = \max\{f, -f\} = |f| \in L$. ($\impliedby$) For $f, g \in L$, since L is a subspace, $f - g \in L$, implying $|f - g| \in L$. Thus $f \vee g = \frac{1}{2}(f + g) + \frac{1}{2}|f - g| \in L$, and similarly for $f \wedge g$. ■

Definition 3.3

Let S be a set of functions on X .

- We say S "separates points" if for every distinct $x, y \in X$, there exists $f \in S$ such that $f(x) \neq f(y)$.
- We say S "vanishes at $x_0 \in X$ " if $f(x_0) = 0$ for all $f \in S$.

Lemma

Let $n \in \mathbb{N}$ and $0 \leq x \leq 1$. Then $\sum_{k=0}^n \left(\frac{k}{n} - x\right)^2 \binom{n}{k} x^k (1-x)^{n-k} = \frac{x(1-x)}{n}$.

Proof. Recall the binomial theorem: $(x+y)^n = \sum_{k=0}^n \binom{n}{k} x^k y^{n-k}$. Taking the partial derivative with respect to x and multiplying by x gives:

$$nx(x+y)^{n-1} = \sum_{k=0}^n k \binom{n}{k} x^k y^{n-k}$$

Plugging in $y = 1 - x$, we get $\sum_{k=0}^n k \binom{n}{k} x^k (1-x)^{n-k} = nx$. Taking a second partial derivative of the original identity with respect to x and multiplying by x^2 yields:

$$n(n-1)x^2(x+y)^{n-2} = \sum_{k=0}^n k(k-1) \binom{n}{k} x^k y^{n-k}$$

Setting $y = 1 - x$ gives $\sum_{k=0}^n k(k-1) \binom{n}{k} x^k (1-x)^{n-k} = n(n-1)x^2$. Now we expand the target summation:

$$\sum_{k=0}^n \left(\frac{k^2}{n^2} - \frac{2kx}{n} + x^2 \right) \binom{n}{k} x^k (1-x)^{n-k}$$

Using $k^2 = k(k-1) + k$ and substituting our established identities simplifies the expression exactly to $\frac{x(1-x)}{n}$. ■

Theorem 3.1 (Weierstrass Approximation Theorem)

Let $h : [a, b] \rightarrow \mathbb{R}$ be continuous. There exists a sequence of polynomials $\{p_n\}_{n \in \mathbb{N}}$ such that $p_n \rightarrow h$ uniformly on $[a, b]$.

Proof. We first work on $[0, 1]$. Define the Bernstein polynomials $B_n h(x) := \sum_{k=0}^n h\left(\frac{k}{n}\right) \binom{n}{k} x^k (1-x)^{n-k}$. We claim $\lim_{n \rightarrow \infty} \|B_n h - h\|_\infty = 0$. Since $\sum_{k=0}^n \binom{n}{k} x^k (1-x)^{n-k} = 1$, we write:

$$B_n h(x) - h(x) = \sum_{k=0}^n \left(h\left(\frac{k}{n}\right) - h(x) \right) \binom{n}{k} x^k (1-x)^{n-k}$$

Since h is continuous on a compact set, it is uniformly continuous and bounded (say by M). For any $\epsilon > 0$, there exists $\delta > 0$ such that $|u - v| < \delta \implies |h(u) - h(v)| < \frac{\epsilon}{2}$. Split the sum into indices where $|\frac{k}{n} - x| < \delta$ (call this A_n) and $|\frac{k}{n} - x| \geq \delta$ (call this A_n^c). For $k \in A_n$, the difference is bounded by $\frac{\epsilon}{2}$, so the sum is bounded by $\frac{\epsilon}{2}$. For $k \in A_n^c$, we have $1 \leq \frac{(\frac{k}{n} - x)^2}{\delta^2}$. The difference is bounded by $2M$, so this part of the sum is bounded by:

$$\frac{2M}{\delta^2} \sum_{k \in A_n^c} \left(\frac{k}{n} - x \right)^2 \binom{n}{k} x^k (1-x)^{n-k} \leq \frac{2M}{\delta^2} \frac{x(1-x)}{n} \leq \frac{2M}{4n\delta^2}$$

Choosing n large enough such that $\frac{M}{2n\delta^2} < \frac{\epsilon}{2}$, the total error is bounded by ϵ . Thus $B_n h \rightarrow h$ uniformly. For a general interval $[a, b]$, use the transformation $t = \frac{x-a}{b-a}$ and define $\phi(t) = h(a + (b-a)t)$ to shift the domain to $[0, 1]$, run the approximation, and shift back. ■

Lemma

Let $A \subseteq C(X)$ be an algebra. Then its uniform closure $\overline{A}$ is a closed algebra and a vector lattice.

Proof. A is a vector subspace, so standard limit arguments show $\overline{A}$ is also a vector subspace. To show it is an algebra, take $f, g \in \overline{A}$. There exist sequences $f_n, g_n \in A$ such that $f_n \rightarrow f$ and $g_n \rightarrow g$ uniformly. Since continuous functions on a compact set are bounded, standard analysis shows $f_n g_n \rightarrow fg$ uniformly. Since A is an algebra, $f_n g_n \in A$, meaning $fg \in \overline{A}$. To show $\overline{A}$ is a vector lattice, it suffices to prove $f \in \overline{A} \implies |f| \in \overline{A}$. Let $h(t) = |t|$. By Weierstrass Approximation, there exists a sequence of polynomials $p_n(t)$ converging uniformly to $h(t)$ on $[-\|f\|_\infty, \|f\|_\infty]$. By defining $q_n(t) = p_n(t) - p_n(0)$, we ensure $q_n(0) = 0$ while still converging uniformly to $|t|$. Since q_n has no constant term, $q_n(f)$ is a polynomial in f without constant term, so $q_n(f) \in \overline{A}$ (since $\overline{A}$ is an algebra). As $q_n(f) \rightarrow |f|$ uniformly, $|f| \in \overline{A}$. ■

Lemma

Let $A \subseteq C(X)$ be an algebra that separates points and does not vanish at any point. Then for any distinct $x, y \in X$ and any $\alpha, \beta \in \mathbb{R}$, there exists a function $h \in A$ such that $h(x) = \alpha$ and $h(y) = \beta$.

Proof. Since A separates points, there is $f \in A$ with $f(x) \neq f(y)$. If neither is zero, we construct $h = af + bf^2 \in A$. We solve the linear system for a and b which is guaranteed to have a unique solution since the determinant $f(x)f(y)(f(y) - f(x)) \neq 0$. If one is zero, say $f(x) = 0, f(y) \neq 0$, we find $g \in A$ such that $g(x) \neq 0$ (since A doesn't vanish at x) and construct a combination that works. ■

Theorem 3.2 (Stone-Weierstrass Theorem)

Let X be a compact metric space, and $A \subseteq C(X)$ be an algebra. If A separates points and does not vanish at any point of X , then A is dense in $C(X)$ under the uniform topology, i.e., $\overline{A} = C(X)$.

Proof. Let $f \in C(X)$. We want to show that for any $\epsilon > 0$, there exists $h \in A$ (or $\overline{A}$) such that $\|f - h\|_\infty < \epsilon$. Fix $y \in X$. For each $x \in X$ with $x \neq y$, by the previous lemma, there exists $h_{x,y} \in A$ such that $h_{x,y}(x) = f(x)$ and $h_{x,y}(y) = f(y)$. Define the open set $U_{x,y} = \{z \in X : h_{x,y}(z) > f(z) - \epsilon\}$. Since $h_{x,y}(x) = f(x) > f(x) - \epsilon$, $x \in U_{x,y}$. Also $y \in U_{x,y}$. The sets $\{U_{x,y}\}_{x \in X}$ form an open cover of X . By compactness, there is a finite subcover $U_{x_1,y}, \dots, U_{x_k,y}$. Define $H_y := h_{x_1,y} \vee \dots \vee h_{x_k,y}$. Since $\overline{A}$ is a vector lattice, $H_y \in \overline{A}$. By construction, $H_y(z) > f(z) - \epsilon$ for all $z \in X$, and $H_y(y) = f(y)$. Now define $V_y = \{z \in X : H_y(z) < f(z) + \epsilon\}$. Since $H_y(y) = f(y) < f(y) + \epsilon$, $y \in V_y$. The sets $\{V_y\}_{y \in X}$ form an open cover of X . By compactness, there is a finite subcover $V_{y_1}, \dots, V_{y_l}$. Define $h = H_{y_1} \wedge \dots \wedge H_{y_l} \in \overline{A}$. By construction, since every $H_{y_i}(z) > f(z) - \epsilon$, we have $h(z) > f(z) - \epsilon$ everywhere. Furthermore, for any $z, z \in V_{y_j}$ for some j , meaning $H_{y_j}(z) < f(z) + \epsilon$. The minimum $h(z)$ must therefore be strictly less than $f(z) + \epsilon$. Thus, $f(z) - \epsilon < h(z) < f(z) + \epsilon$ for all $z \in X$, which means $\|f - h\|_\infty < \epsilon$. ■

Corollary

Let $X \subseteq \mathbb{R}^n$ be compact. The algebra of all polynomials in the coordinate functions is dense in $C(X)$.

Corollary

The set of trigonometric polynomial functions is dense in $C_{per}([-\pi, \pi])$.

4 Basic Functional Analysis

Definition 4.1 (Normed Vector Space)

Let V be a vector space over $\mathcal{F} = \mathbb{R}$ (or $\mathbb{C}$). A norm is a function $\|\cdot\| : V \rightarrow [0, \infty)$ satisfying the following properties for all $\mathbf{x}, \mathbf{y} \in V$ and $\alpha \in \mathcal{F}$:

1. $\|\mathbf{x}\| \geq 0$ and $\|\mathbf{x}\| = 0 \iff \mathbf{x} = \mathbf{0}$.
2. $\|\alpha\mathbf{x}\| = |\alpha|\|\mathbf{x}\|$
3. $\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$

A vector space equipped with a norm is called a normed vector space and is denoted by $(V, \|\cdot\|)$.

Remark. Every normed space $(V, \|\cdot\|)$ induces a natural metric $d(\mathbf{x}, \mathbf{y}) := \|\mathbf{x} - \mathbf{y}\|$.

Definition 4.2 (Banach Space)

A normed space $(V, \|\cdot\|)$ is called a Banach space if it is complete with respect to the metric induced by the norm. That is, every Cauchy sequence $\{\mathbf{v}_i\}_{i=1}^{\infty} \subseteq V$ converges to an element $\mathbf{v} \in V$.

Definition 4.3 (Hilbert Space)

We say $(V, \|\cdot\|)$ is a Hilbert space if

1. V is an inner product space.
2. V is a Banach space.

Remark.

1. Every inner product induces a norm via $\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}$. However, not every norm is induced by an inner product. For example, the ℓ^1 -norm on $\mathbb{R}^n$ is not induced by any inner product.
2. Hence, inner product space is also a normed space, if it is complete, it is a Hilbert space.

Definition 4.4

Let (X, d_X) and (Y, d_Y) be two metric spaces. Define $\mathcal{B}(X \rightarrow Y) := \{f \mid f : X \rightarrow Y \text{ is a bounded function}\}$. We can define a uniform / sup norm metric $d_{\infty}(f, g) := \sup_{x \in X} d_Y(f(x), g(x))$.

One can show that $(\mathcal{B}(X \rightarrow Y), d_{\infty})$ is a metric space. Next, define $C(X \rightarrow Y) := \{f \in \mathcal{B}(X \rightarrow Y) \mid f : X \rightarrow Y \text{ is continuous}\}$. Clearly, $C(X \rightarrow Y) \subset \mathcal{B}(X \rightarrow Y)$.

Remark. Let $\{f_n\}_{n \in \mathbb{N}}$ be a sequence in $\mathcal{B}(X \rightarrow Y)$ and another function $f \in \mathcal{B}(X \rightarrow Y)$.
TFAE:

1. $f_n \rightarrow f$ converges in the d_∞ -metric.
2. f_n converges to f uniformly.

Theorem 4.1

Let (X, d_X) be a metric space and (Y, d_Y) a complete metric space. Then $C(X \rightarrow Y)$ equipped with the metric d_∞ is a complete metric subspace of $\mathcal{B}(X \rightarrow Y)$.

Proof.

Let $\{f_n\} \subseteq C(X \rightarrow Y)$ be a Cauchy sequence in the d_∞ metric. We will show that it converges to a function $f \in C(X \rightarrow Y)$.

Step 1: Pointwise convergence.

Let $\epsilon > 0$. Since $\{f_n\}$ is Cauchy, there exists an $N \in \mathbb{N}$ such that for all $n, m \geq N$, we have $d_\infty(f_n, f_m) < \epsilon$. By the definition of the supremum metric, for any fixed $x \in X$,

$$d_Y(f_n(x), f_m(x)) \leq d_\infty(f_n, f_m) < \epsilon$$

This shows that for each $x \in X$, the sequence $\{f_n(x)\}$ is a Cauchy sequence in Y . Since (Y, d_Y) is complete, this sequence converges to some point in Y . We define the limit function $f : X \rightarrow Y$ by $f(x) := \lim_{n \rightarrow \infty} f_n(x)$.

Step 2: Uniform convergence.

We need to show that $f_n \rightarrow f$ uniformly. From Step 1, we know that for all $n, m \geq N$ and all $x \in X$, $d_Y(f_n(x), f_m(x)) < \epsilon$. Fixing $n \geq N$ and taking the limit as $m \rightarrow \infty$, the continuity of the metric d_Y gives:

$$d_Y(f_n(x), f(x)) = \lim_{m \rightarrow \infty} d_Y(f_n(x), f_m(x)) \leq \epsilon \quad \text{for all } x \in X$$

Taking the supremum over all $x \in X$, we get $d_\infty(f_n, f) \leq \epsilon$. Thus, f_n converges to f uniformly. (This also implies f is bounded, so $f \in \mathcal{B}(X \rightarrow Y)$).

Step 3: Continuity of the limit function (The $\epsilon/3$ argument).

To show $f \in C(X \rightarrow Y)$, we must prove f is continuous at any arbitrary point $x_0 \in X$. Let $\epsilon > 0$. First, by uniform convergence, choose an integer K such that for all $x \in X$,

$$d_Y(f_K(x), f(x)) < \frac{\epsilon}{3}$$

Second, since f_K is continuous at x_0 , there exists a $\delta > 0$ such that for all $x \in X$ with $d_X(x, x_0) < \delta$,

$$d_Y(f_K(x), f_K(x_0)) < \frac{\epsilon}{3}$$

Now, for any $x \in X$ with $d_X(x, x_0) < \delta$, we apply the triangle inequality:

$$\begin{aligned} d_Y(f(x), f(x_0)) &\leq d_Y(f(x), f_K(x)) + d_Y(f_K(x), f_K(x_0)) + d_Y(f_K(x_0), f(x_0)) \\ &< \frac{\epsilon}{3} + \frac{\epsilon}{3} + \frac{\epsilon}{3} = \epsilon \end{aligned}$$

Therefore, f is continuous at x_0 . Since x_0 was arbitrary, $f \in C(X \rightarrow Y)$. This concludes the proof that $C(X \rightarrow Y)$ is a complete metric space. ■

Proposition 10

Let (K, d) be a non-empty compact metric space. Let $\mathcal{B}(K) := \mathcal{B}(K \rightarrow \mathbb{R})$ and $C(K) := C(K \rightarrow \mathbb{R})$. Then $\mathcal{B}(K)$ is a vector space over $\mathbb{R}$, and $C(K)$ is a closed vector subspace in $\mathcal{B}(K)$.

Proof.

We can check that $(f + g)(x) = f(x) + g(x)$ and $(cf)(x) = cf(x)$ are well-defined. Also, $|(\alpha f + \beta g)(x)| \leq |\alpha||f(x)| + |\beta||g(x)|$, which guarantees closure under linear combinations. Furthermore, since K is compact, any continuous function $f \in C(K)$ attains its maximum and minimum, meaning f is bounded ($C(K) \subset \mathcal{B}(K)$). By uniform convergence, the limit of any convergent sequence in $C(K)$ is also continuous on K , proving closedness. ■

Theorem 4.2

On $\mathcal{B}(K)$ or $C(K)$, we define the sup-norm $\|f\|_\infty := \sup_{x \in K} |f(x)|$. Both $(\mathcal{B}(K), \|\cdot\|_\infty)$ and $(C(K), \|\cdot\|_\infty)$ are Banach spaces.

Proof. This is the direct consequence of the previous proposition and theorem (4.1). ■

4.1 Arzelà-Ascoli Theorem

Definition 4.5

Let $\mathcal{F} \subset C(K)$.

1. The family $\mathcal{F}$ is called *pointwise bounded* if for all $x \in K$, there exists a constant $M_x < \infty$ such that $|f(x)| \leq M_x$ for every $f \in \mathcal{F}$ (independent of f).
2. The family $\mathcal{F}$ is called *uniformly bounded* if there exists a constant $M < \infty$ such that $|f(x)| \leq M$ for every $x \in K$ and $f \in \mathcal{F}$ (independent of f and x).
3. The family $\mathcal{F}$ is called *equicontinuous* if for all $\epsilon > 0$, there exists $\delta > 0$ such that $d(x, y) < \delta \Rightarrow |f(x) - f(y)| < \epsilon$ for all $x, y \in K$ and $f \in \mathcal{F}$ (independent of f and x).

Theorem 4.3

If $\mathcal{F} \subset C(K)$ is pointwise bounded and equicontinuous, then $\mathcal{F}$ is uniformly bounded.

Proof.

Since $\mathcal{F}$ is equicontinuous, take $\epsilon = 1$, then there exists $\delta > 0$ such that $d(x, y) < \delta \Rightarrow |f(x) - f(y)| < 1$ for all $f \in \mathcal{F}$. Let $\bigcup_{x \in K} B(x, \delta)$ be an open cover of K .

Since K is compact, there exists a finite subcover such that $K \subseteq \bigcup_{i=1}^r B(x_i, \delta)$. By pointwise boundedness, $|f(x_i)| \leq M_i$ for each $i = 1, \dots, r$ and for all $f \in \mathcal{F}$. Choose $M = \max_{1 \leq i \leq r} (M_i + 1)$. Given any $f \in \mathcal{F}$ and $x \in K$, there exists x_i such that $d(x, x_i) < \delta$. Therefore,

$$|f(x)| \leq |f(x) - f(x_i)| + |f(x_i)| \leq 1 + M_i \leq M.$$

Hence, $\mathcal{F}$ is uniformly bounded. ■

Theorem 4.4 (Arzelà-Ascoli I)

Let (K, d) be a compact metric space. If $\{f_n\} \subset C(K)$ is bounded and equicontinuous, then $\{f_n\}$ has a subsequence which converges uniformly to a continuous function on K .

Proof.

Step 1: A compact set is separable (has a countable, dense subset). For each $k \in \mathbb{N}$, $\bigcup_{x \in K} B(x, \frac{1}{k})$ forms an open cover of K . Since K is compact, there

exists a finite subcover $\{B(x_{k,j}, \frac{1}{k})\}_{j=1}^{N_k}$. Let $S_k = \{x_{k,j}\}_{j=1}^{N_k}$. Define $S := \bigcup_{k=1}^{\infty} S_k$.

Clearly, S is countable. We claim S is dense. Fix $\epsilon > 0$. There exists k_0 such that $\frac{1}{k_0} < \epsilon$. Since $K \subseteq \bigcup_{j=1}^{N_{k_0}} B(x_{k_0,j}, \frac{1}{k_0})$, for any $x \in K$, there exists $x_{k_0,j} \in S$ such that $d(x, x_{k_0,j}) < \frac{1}{k_0} < \epsilon$. Thus, S is dense.

Step 2: Diagonal Process. Let $S = \{p_1, p_2, \dots\}$. We construct a uniformly convergent subsequence. Consider $\{f_n(p_1)\}_{n=1}^{\infty}$. Since $\{f_n\}$ is uniformly bounded, this scalar sequence is bounded in $\mathbb{R}$. By Bolzano-Weierstrass, there exists a convergent subsequence $\{f_n^{(1)}(p_1)\}$. Next, consider $\{f_n^{(1)}(p_2)\}_{n=1}^{\infty}$. By BW again, there exists a subsequence $\{f_n^{(2)}\}$ such that $\{f_n^{(2)}(p_2)\}$ converges. Note that $\{f_n^{(2)}(p_1)\}$ still converges as it is a subsequence of the first. By induction, there exists $\{f_n^{(k)}\}_{n=1}^{\infty}$ which is a subsequence of $\{f_n^{(k-1)}\}_{n=1}^{\infty}$ such that $\{f_n^{(k)}(p_j)\}_{n=1}^{\infty}$ converges for $j = 1, 2, \dots, k$. Take $g_n = f_n^{(n)}$. Then the sequence $\{g_n(p_j)\}_{n=1}^{\infty}$ converges for all $p_j \in S$.

Step 3: Uniform convergence in $C(K)$. Let $\epsilon > 0$. By equicontinuity, there exists $\delta > 0$ such that $d(x, y) < \delta \Rightarrow |g_n(x) - g_n(y)| < \frac{\epsilon}{3}$ for all n . Choose k such that $\frac{1}{k} < \delta$. For each $x \in K$, there exists $s \in S_k \subset S$ such that $d(x, s) < \frac{1}{k} < \delta$. Thus, $|g_n(x) - g_n(s)| < \frac{\epsilon}{3}$. Since S_k is finite, $\{g_n(s)\}$ converges uniformly for $s \in S_k$. Thus, there exists N such that for all $m, n \geq N$ and all $s \in S_k$, $|g_n(s) - g_m(s)| < \frac{\epsilon}{3}$. Then for any $x \in K$ and $m, n \geq N$, pick $s \in S_k$ such that $d(x, s) < \delta$. We have:

$$|g_n(x) - g_m(x)| \leq |g_n(x) - g_n(s)| + |g_n(s) - g_m(s)| + |g_m(s) - g_m(x)| \leq \frac{\epsilon}{3} + \frac{\epsilon}{3} + \frac{\epsilon}{3} = \epsilon$$

Therefore, $\{g_n\}$ is uniformly Cauchy in $C(K)$. Since $C(K)$ is complete, g_n converges uniformly to some $g \in C(K)$. ■

Theorem 4.5 (Arzelà-Ascoli II)

Let (K, d) be a compact metric space and $\mathcal{F}$ be a family of $C(K)$. $\mathcal{F}$ is relatively compact iff $\mathcal{F}$ is uniformly bounded and equicontinuous.

Remark. We say $\mathcal{F}$ is relatively compact if the closure of $\mathcal{F}$ is compact. Take any sequence in $\mathcal{F}$, if the sequence has a convergent subsequence, the limit of this subsequence, f , belongs to $\overline{\mathcal{F}}$. By definition, $\overline{\mathcal{F}}$ is sequentially compact and hence compact. Note that the first version of Arzelà-Ascoli tells us $f \in C(K)$, but f may not be in $\mathcal{F}$.

Proof.

($\Leftarrow$) Proved in Arzelà-Ascoli (I).

($\Rightarrow$) Suppose $\mathcal{F}$ is relatively compact, i.e., Take any sequence $\{f_n\}_{n \in \mathbb{N}} \subseteq \mathcal{F}$, $\exists$ a convergent subsequence $f_{n_k} \rightrightarrows f$ for some $f \in C(K)$.

Suppose, by contradiction, $\{f_n\}_{n \in \mathbb{N}}$ is not uniformly bounded.

For each $k \in \mathbb{N}$, $\exists x_k \in K$ and $\exists f_{n_k}$ s.t. $|f_{n_k}(x_k)| > k$.

Consider the sequence $\{f_{n_k}\}$. By assumption, $\exists$ convergent subsequence, denoted by $\{g_j\}_{j \in \mathbb{N}}$ s.t. $g_j \rightrightarrows g$, where $g \in C(K)$.

Since g is continuous on the compact set K , g is bounded.

Note that $\|g_j\|_\infty > j$ for each j (by our construction).

Now let $j \rightarrow \infty$, we have $\lim_{j \rightarrow \infty} \|g_j\|_\infty \rightarrow \infty$, which contradicts the fact that g_j converges uniformly to a bounded function g .

Suppose, by contradiction, $\{f_n\}_{n \in \mathbb{N}}$ is not equicontinuous.

$\exists \epsilon > 0$ s.t. $\forall \delta > 0$, $\exists x, y \in K$ s.t. $|x - y| < \delta \Rightarrow |f_n(x) - f_n(y)| > \epsilon$ for some n .

For each $k \in \mathbb{N}$, set $\delta = \frac{1}{k}$, $\exists x_k, y_k \in K$ and f_{n_k} s.t.

$$|x_k - y_k| < \frac{1}{k} \Rightarrow |f_{n_k}(x_k) - f_{n_k}(y_k)| > \epsilon$$

Consider the sequence $\{f_{n_k}\}$. By assumption, $\exists$ convergent subsequence, denoted by $\{h_j\}_{j \in \mathbb{N}}$ s.t. $h_j \rightrightarrows h$ for some $h \in C(K)$.

For this $\epsilon > 0$, $\exists N \in \mathbb{N}$ s.t. $\|h_j - h\|_\infty < \frac{\epsilon}{3} \quad \forall j \geq N$.

Moreover, since K is compact, h is uniformly continuous.

For this $\epsilon > 0$, $\exists \eta > 0$ s.t. $|x - y| < \eta \Rightarrow |h(x) - h(y)| < \frac{\epsilon}{3} \quad \forall x, y \in K$.

Note: $|x_j - y_j| < \frac{1}{j} \Rightarrow |h_j(x_j) - h_j(y_j)| > \epsilon \quad (*)$

Now take $j \rightarrow \infty$ large enough s.t. $|x_j - y_j| < \eta$. We apply the triangle inequality:

$$\begin{aligned} |h_j(x_j) - h_j(y_j)| &\leq |h_j(x_j) - h(x_j)| + |h(x_j) - h(y_j)| + |h(y_j) - h_j(y_j)| \\ &\leq \frac{\epsilon}{3} + \frac{\epsilon}{3} + \frac{\epsilon}{3} = \epsilon \end{aligned}$$

This directly contradicts our construction in $(*)$. Therefore, $\mathcal{F}$ must be equicontinuous. ■

Corollary

Let (K, d) be a compact metric space and $\mathcal{F}$ be a family of $C(K)$. $\mathcal{F}$ is compact iff $\mathcal{F}$ is closed, uniformly bounded and equicontinuous.

Proof.

$(\Leftarrow)$ Bounded and equicontinuous $\Rightarrow$ there exists a convergent subsequence. Since $\mathcal{F}$ is closed $\Rightarrow$ this subsequence converges to a function within $\mathcal{F}$. Thus, sequentially compact implies compact.

$(\Rightarrow)$ It suffices to show compact implies closed and bounded.

Proof of Boundedness:

Fix any $r > 0$ and let $\bigcup_{f \in \mathcal{F}} B(f, r)$ be an open cover of $\mathcal{F}$.

By compactness, there exists a finite subcover $\{f_1, f_2, \dots, f_n\}$ s.t. $\bigcup_{i=1}^n B(f_i, r) \supseteq \mathcal{F}$.

By definition, $\mathcal{F}$ is totally bounded and hence bounded.

To see this explicitly, fix $f \in \mathcal{F}$ and take any $g \in \mathcal{F}$.

WTS: $\exists M > 0$ s.t. $g \in B(f, M) \Leftrightarrow \|g - f\|_\infty \leq M$.

Since f, g belong to the union of the finite balls, they are near some centers f_i, f_j .

By the triangle inequality:

$$\begin{aligned} \|f - g\|_\infty &\leq \|f - f_i\|_\infty + \|f_i - f_j\|_\infty + \|f_j - g\|_\infty \\ &\leq r + \max_{i \neq j} \{\|f_i - f_j\|_\infty\} + r = M. \end{aligned}$$

This provides a uniform bound M for any pair in $\mathcal{F}$.

Proof of Closedness: Take any sequence $f_n \Rightarrow f$, where $f \in C(K)$.

By (sequential) compactness, $\exists f_{\phi(n)} \Rightarrow g$, where $g \in \mathcal{F}$.

By uniqueness of limit, we have $f = g \implies f \in \mathcal{F}$.

Therefore, $\mathcal{F}$ is closed. ■

Application: Compactness of Integral Operators

Let $K : [a, b] \times [a, b] \rightarrow \mathbb{R}$ be continuous. Define $T : C([a, b]) \rightarrow C([a, b])$ by

$$(Tf)(x) = \int_a^b K(x, y)f(y)dy$$

Then T is a compact operator. That is, for any bounded sequence $\{f_n\}_{n=1}^\infty \subseteq C([a, b])$ under the sup-norm, $\{Tf_n\}$ has a uniformly convergent subsequence in $C([a, b])$.

Proof.

Step 1: T is well-defined. We must show $Tf \in C([a, b])$. Given $f \in C([a, b])$, since f is continuous on a compact set, it is bounded, say $|f(y)| \leq M_f$. Also, $[a, b] \times [a, b]$ is compact, so K is uniformly continuous on it. For any fixed x , the integral is well-defined. By uniform continuity of K , for any $\epsilon > 0$, there exists $\delta > 0$ such that $|x - x'| < \delta \implies |K(x, y) - K(x', y)| < \frac{\epsilon}{M_f(b-a)}$. For $|x - x'| < \delta$:

$$\begin{aligned} |Tf(x) - Tf(x')| &= \left| \int_a^b (K(x, y) - K(x', y))f(y)dy \right| \\ &\leq \int_a^b |K(x, y) - K(x', y)||f(y)|dy < \int_a^b \frac{\epsilon}{M_f(b-a)} M_f dy = \epsilon. \end{aligned}$$

Thus Tf is continuous. T is well-defined.

Step 2: $\{Tf_n\}$ is uniformly bounded and equicontinuous. Assume $\{f_n\}$ is bounded by M , i.e., $\|f_n\|_\infty \leq M$. Since K is continuous on a compact set, there exists $C > 0$ such that $|K(x, y)| \leq C$. Then:

$$|(Tf_n)(x)| \leq \int_a^b |K(x, y)||f_n(y)|dy \leq \int_a^b CMdy = CM(b-a)$$

So $\{Tf_n\}$ is uniformly bounded. By uniform continuity of K , there exists $\delta > 0$ such that $|x - x'| < \delta \implies |K(x, y) - K(x', y)| < \frac{\epsilon}{M(b-a)}$. Then for $|x - x'| < \delta$:

$$|Tf_n(x) - Tf_n(x')| \leq \int_a^b |K(x, y) - K(x', y)||f_n(y)|dy < \int_a^b \frac{\epsilon}{M(b-a)} Mdy = \epsilon$$

This bound is independent of n . Hence $\{Tf_n\}$ is equicontinuous. By Arzelà-Ascoli, $\{Tf_n\}$ has a uniformly convergent subsequence in $C([a, b])$. ■

4.2 Finite-Dimensional Normed Spaces

Definition 4.6 (Operator Norm)

Let $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ be real normed vector spaces. A linear operator $T : X \rightarrow Y$ is called bounded if there exists a constant $c \geq 0$ such that $\|T\mathbf{x}\|_Y \leq c\|\mathbf{x}\|_X$ for all $\mathbf{x} \in X$. The smallest such constant is called the operator norm, defined as $\|T\|_{op} := \sup_{\mathbf{x} \neq \mathbf{0}} \frac{\|T\mathbf{x}\|_Y}{\|\mathbf{x}\|_X}$.

Let $\mathcal{L}(X, Y)$ be the space of all bounded linear operators from X to Y . One can verify that $(\mathcal{L}(X, Y), \|\cdot\|_{op})$ is a normed space. The topology on $\mathcal{L}(X, Y)$ induced by the operator norm $\|\cdot\|_{op}$ is called the **uniform operator topology**.

Theorem 4.6

Let X, Y be normed vector spaces, and let $T : X \rightarrow Y$ be a bounded linear operator. Then the operator norm $\|T\|_{op}$ satisfies the following equalities:

$$\|T\|_{op} = \inf\{c \geq 0 : \|T\mathbf{x}\| \leq c\|\mathbf{x}\| \text{ for all } \mathbf{x} \in X\} \quad (1)$$

$$= \sup\{\|T\mathbf{x}\| : \|\mathbf{x}\| \leq 1 \text{ and } \mathbf{x} \in X\} \quad (2)$$

$$= \sup\{\|T\mathbf{x}\| : \|\mathbf{x}\| < 1 \text{ and } \mathbf{x} \in X\} \quad (3)$$

$$= \sup\{\|T\mathbf{x}\| : \|\mathbf{x}\| \in \{0, 1\} \text{ and } \mathbf{x} \in X\} \quad (4)$$

Furthermore, if $X \neq \{\mathbf{0}\}$, the following equalities also hold:

$$\|T\|_{op} = \sup\{\|T\mathbf{x}\| : \|\mathbf{x}\| = 1 \text{ and } \mathbf{x} \in X\} \quad (5)$$

$$= \sup\left\{\frac{\|T\mathbf{x}\|}{\|\mathbf{x}\|} : \mathbf{x} \neq \mathbf{0} \text{ and } \mathbf{x} \in X\right\} \quad (6)$$

Proof.

We will prove that all six expressions are equal through a series of inequalities.

Step 1: Prove (5) = (6) (assuming $X \neq \{\mathbf{0}\}$)

For any $\mathbf{x} \neq \mathbf{0}$, we can define the unit vector $\mathbf{u} = \frac{\mathbf{x}}{\|\mathbf{x}\|}$, which satisfies $\|\mathbf{u}\| = 1$. By the linearity of T , we have:

$$\frac{\|T\mathbf{x}\|}{\|\mathbf{x}\|} = \left\|T\left(\frac{\mathbf{x}}{\|\mathbf{x}\|}\right)\right\| = \|T\mathbf{u}\|$$

This shows that the set of values for (6) is exactly the same as the set for (5).

Therefore, their supremums are identical, meaning (6) = (5).

Step 2: Prove (5) ≤ (2)

Let

$$A = \{\|T\mathbf{x}\| : \|\mathbf{x}\| = 1\}$$

$$B = \{\|T\mathbf{x}\| : \|\mathbf{x}\| \leq 1\},$$

it is an easy exercise $\sup A \leq \sup B$.

Step 3: Prove (2) = (3)

Clearly, $\{\mathbf{x} : \|\mathbf{x}\| < 1\} \subset \{\mathbf{x} : \|\mathbf{x}\| \leq 1\}$, which immediately gives (3) $\leq$ (2).

Conversely, given any $\mathbf{x}$ satisfying $\|\mathbf{x}\| \leq 1$ and any scalar $r \in (0, 1)$, the vector $r\mathbf{x}$ satisfies $\|r\mathbf{x}\| = r\|\mathbf{x}\| < 1$. Thus:

$$r\|T\mathbf{x}\| = \|T(r\mathbf{x})\| \leq \sup_{\|\mathbf{z}\| < 1} \|T\mathbf{z}\| = (3)$$

Taking the limit as $r \rightarrow 1^-$, we obtain $\|T\mathbf{x}\| \leq (3)$. Since this holds for all $\mathbf{x}$ with $\|\mathbf{x}\| \leq 1$, taking the supremum yields (2) $\leq$ (3). Therefore, (2) = (3).

Step 4: Prove (2) $\leq$ (1)

By the definition of (1), for any constant c in the set (meaning $c \geq 0$ and $\|T\mathbf{x}\| \leq c\|\mathbf{x}\|$ for all $\mathbf{x} \in X$), if we restrict our attention to $\|\mathbf{x}\| \leq 1$, we get:

$$\|T\mathbf{x}\| \leq c\|\mathbf{x}\| \leq c \cdot 1 = c$$

This implies that for all vectors in the closed unit ball, c is an upper bound for $\|T\mathbf{x}\|$. Consequently, the supremum (2) $\leq c$. Since (2) is less than or equal to every such bound c , it must be less than or equal to their infimum. Hence, (2) $\leq$ (1).

Step 5: Prove (1) $\leq$ (6)

Let $p = (6)$. By definition, for all $\mathbf{x} \neq \mathbf{0}$, we have $\frac{\|T\mathbf{x}\|}{\|\mathbf{x}\|} \leq p$. Since $\|\mathbf{x}\| > 0$ for $\mathbf{x} \neq \mathbf{0}$, we can safely multiply both sides by $\|\mathbf{x}\|$ to obtain:

$$\|T\mathbf{x}\| \leq p\|\mathbf{x}\| \quad \text{for all } \mathbf{x} \neq \mathbf{0}.$$

Now, we separately check the case where $\mathbf{x} = \mathbf{0}$. By the linearity of T , $\|T(\mathbf{0})\| = 0$, and $p\|\mathbf{0}\| = 0$. Thus, the inequality $\|T\mathbf{x}\| \leq p\|\mathbf{x}\|$ becomes $0 \leq 0$, which holds trivially.

Combining both cases, the inequality $\|T\mathbf{x}\| \leq p\|\mathbf{x}\|$ holds for *all* $\mathbf{x} \in X$. This means p satisfies the condition to be in the set defining (1). Since (1) is defined as the infimum of this set, (1) $\leq p = (6)$.

Summary of Steps 1-5: Combining the inequalities established above, we have:

$$(6) = (5) \leq (2) = (3) \leq (1) \leq (6)$$

This forces all the values to be strictly equal.

Step 6: Handle (4)

(4) is the supremum of the set $\{\|T\mathbf{x}\| : \|\mathbf{x}\| = 0 \text{ or } \|\mathbf{x}\| = 1\}$.

If $\mathbf{x} = \mathbf{0}$, then $\|T\mathbf{0}\| = 0$. If $\|\mathbf{x}\| = 1$, the supremum over these elements is precisely (5). Since the operator norm (5) ≥ 0 , including the $\mathbf{x} = \mathbf{0}$ case does not change the maximum possible value. Thus, (4) = max(0, (5)) = (5).

This completes the proof for all the equalities. ■

Definition 4.7

Let $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ be normed vector spaces, and let $\mathcal{L}(X, Y)$ denote the space of all bounded linear operators from X to Y . Let $(T_n)_{n \in \mathbb{N}}$ be a sequence of operators in $\mathcal{L}(X, Y)$ and another operator $T \in \mathcal{L}(X, Y)$. We say that T_n converges to T in the uniform operator topology (denoted $T_n \xrightarrow{\|\cdot\|_{\text{op}}} T$) if and only if:

$$\lim_{n \rightarrow \infty} \|T_n - T\|_{\text{op}} = 0 \iff \lim_{n \rightarrow \infty} \sup_{\|\mathbf{x}\|_X \leq 1} \|(T_n - T)(\mathbf{x})\|_Y = 0$$

This means that the error $\|T_n \mathbf{x} - T \mathbf{x}\|_Y$ converges to 0 uniformly for all vectors $\mathbf{x}$ within the unit ball.

Theorem 4.7

Let $T : X \rightarrow Y$ be a linear operator. The following are equivalent:

1. T is bounded.
2. T is continuous on X .
3. T is continuous at $\mathbf{0}$.

Proof.

(2 $\Rightarrow$ 3): Trivial.

(1 $\Rightarrow$ 2): If T is bounded, $\|T \mathbf{x} - T \mathbf{z}\|_Y = \|T(\mathbf{x} - \mathbf{z})\|_Y \leq \|T\|_{\text{op}} \|\mathbf{x} - \mathbf{z}\|_X$. Thus T is Lipschitz continuous, hence continuous.

(3 $\Rightarrow$ 1): If T is continuous at $\mathbf{0}$, for $\epsilon = 1$, there exists $\delta > 0$ such that $\|\mathbf{x}\| < \delta \Rightarrow \|T \mathbf{x}\| < 1$. For any $\mathbf{x} \neq \mathbf{0}$, consider $\mathbf{z} = \frac{\delta}{2} \frac{\mathbf{x}}{\|\mathbf{x}\|}$. Since $\|\mathbf{z}\| = \frac{\delta}{2} < \delta$, we have $\|T \mathbf{z}\| < 1$. By linearity, $\|T \left(\frac{\delta}{2} \frac{\mathbf{x}}{\|\mathbf{x}\|} \right)\| < 1$, which simplifies to $\|T \mathbf{x}\| < \frac{2}{\delta} \|\mathbf{x}\|$, meaning T is bounded. ■

Theorem 4.8

Let (X, d) be a metric space. If $\{x_n\}_{n=1}^{\infty}$ is a Cauchy sequence in X , then it is a bounded sequence.

Proof.

Since $\{x_n\}$ is a Cauchy sequence, we can choose a specific error bound, say $\epsilon = 1$. By the definition of a Cauchy sequence, there exists an integer $N \in \mathbb{N}$ such that for all $m, n \geq N$:

$$d(x_n, x_m) < 1$$

In particular, if we fix $m = N$, then for all $n \geq N$, we have:

$$d(x_n, x_N) < 1$$

This means that all terms from x_N onwards are contained within the open ball $B(x_N, 1)$ centered at x_N with radius 1.

Now, consider the terms before the N -th term. There are only finitely many such terms: $x_1, x_2, \dots, x_{N-1}$. We define a constant R to be the maximum distance from these finite terms to the center x_N , ensuring it is at least 1 to cover the tail:

$$R = \max\{d(x_1, x_N), d(x_2, x_N), \dots, d(x_{N-1}, x_N), 1\}$$

Since this is a maximum of a *finite* set of real numbers, R is a well-defined, finite positive constant.

Therefore, for all $n \in \mathbb{N}$, we have:

$$d(x_n, x_N) < R + 1$$

This shows that the entire sequence is contained within the closed ball $B(x_N, R + 1)$. By definition, the sequence $\{x_n\}$ is bounded in the metric space X . ■

Corollary

Let $(X, \|\cdot\|)$ be a normed vector space. If $\{x_n\}_{n=1}^{\infty}$ is a Cauchy sequence in X , then it is a bounded sequence.

Proof.

We may apply the previous theorem directly. Alternatively, we can use $\mathbf{0}$ as the center of the ball. Since $\{x_n\}$ is a Cauchy sequence, we can choose a specific error bound, say $\epsilon = 1$. By the definition of a Cauchy sequence, there exists an integer $N \in \mathbb{N}$ such that for all $m, n \geq N$:

$$\|x_n - x_m\| < 1$$

In particular, if we fix $m = N$, then for all $n \geq N$, we have:

$$\|x_n - x_N\| < 1$$

By the triangle inequality, for any $n \geq N$, the norm of x_n can be bounded by:

$$\|x_n\| = \|(x_n - x_N) + x_N\| \leq \|x_n - x_N\| + \|x_N\| < 1 + \|x_N\|$$

This shows that all terms from the N -th term onwards are bounded by the constant $1 + \|x_N\|$.

Now, consider the terms before the N -th term. There are only finitely many such terms: $x_1, x_2, \dots, x_{N-1}$. We can define a constant M to be the maximum of the

norms of all these terms, including our bound for the tail:

$$M = \max \{ \|x_1\|, \|x_2\|, \dots, \|x_{N-1}\|, 1 + \|x_N\| \}$$

Since this is a maximum of a *finite* set of real numbers, M is a well-defined, finite positive constant. Therefore, for all $n \in \mathbb{N}$, we have:

$$\|x_n\| \leq M$$

This proves that the Cauchy sequence $\{x_n\}$ is bounded. ■

Theorem 4.9

Let X be a normed vector space. If Y is a Banach space, then the space of all bounded linear operators $\mathcal{L}(X, Y)$ is a Banach space under the operator norm.

Proof.

Let $\{T_n\}_{n=1}^\infty$ be a Cauchy sequence in $\mathcal{L}(X, Y)$. By definition, for every $\epsilon > 0$, there exists an integer $N \in \mathbb{N}$ such that for all $m, n \geq N$,

$$\|T_n - T_m\|_{\text{op}} < \epsilon$$

Step 1: Pointwise convergence.

Fix any $\mathbf{x} \in X$. For $m, n \geq N$, we have:

$$\|(T_n - T_m)\mathbf{x}\|_Y \leq \|T_n - T_m\|_{\text{op}} \|\mathbf{x}\|_X < \epsilon \|\mathbf{x}\|_X$$

Since $\|\mathbf{x}\|_X$ is a fixed constant, this shows that $\{T_n\mathbf{x}\}_{n=1}^\infty$ is a Cauchy sequence in Y . Because Y is a complete space, this sequence must converge to some limit point in Y . We can therefore define a limit operator $T : X \rightarrow Y$ pointwise:

$$T\mathbf{x} := \lim_{n \rightarrow \infty} T_n\mathbf{x} \quad \text{for all } \mathbf{x} \in X.$$

Step 2: Uniform convergence ($T_n \rightarrow T$ in operator norm).

We need to show that $T_n \rightarrow T$ in the uniform operator topology. Let $\epsilon > 0$. Since $\{T_n\}$ is Cauchy, there exists $N \in \mathbb{N}$ such that for all $n, m \geq N$, $\|T_n - T_m\|_{\text{op}} < \frac{\epsilon}{2}$.

For any $\mathbf{x} \in X$ with $\|\mathbf{x}\|_X \leq 1$, and for a fixed $n \geq N$, we apply the triangle inequality for any $m \geq N$:

$$\|(T_n - T)\mathbf{x}\|_Y \leq \|(T_n - T_m)\mathbf{x}\|_Y + \|(T_m - T)\mathbf{x}\|_Y$$

Since $T_m\mathbf{x} \rightarrow T\mathbf{x}$ as $m \rightarrow \infty$, we can choose m large enough (with $m \geq N$) such that the second term $\|(T_m - T)\mathbf{x}\|_Y < \frac{\epsilon}{2}$. For the first term, since $n, m \geq N$ and

$\|\mathbf{x}\|_X \leq 1$, we have $\|(T_n - T_m)\mathbf{x}\|_Y \leq \|T_n - T_m\|_{\text{op}} \|\mathbf{x}\|_X < \frac{\epsilon}{2} \cdot 1 = \frac{\epsilon}{2}$.

Thus,

$$\|(T_n - T)\mathbf{x}\|_Y < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

Since this holds for all $\mathbf{x}$ with $\|\mathbf{x}\|_X \leq 1$, taking the supremum yields $\|T_n - T\|_{\text{op}} \leq \epsilon$ for all $n \geq N$. Hence, $T_n \rightarrow T$ uniformly.

Step 3: T is linear.

Let $\alpha, \beta \in \mathbb{R}$ and $\mathbf{x}, \mathbf{z} \in X$. Since each operator T_n is linear, we can use the algebraic properties of limits:

$$\begin{aligned} T(\alpha\mathbf{x} + \beta\mathbf{z}) &= \lim_{n \rightarrow \infty} T_n(\alpha\mathbf{x} + \beta\mathbf{z}) \\ &= \lim_{n \rightarrow \infty} (\alpha T_n\mathbf{x} + \beta T_n\mathbf{z}) \\ &= \alpha \lim_{n \rightarrow \infty} T_n\mathbf{x} + \beta \lim_{n \rightarrow \infty} T_n\mathbf{z} \\ &= \alpha T\mathbf{x} + \beta T\mathbf{z} \end{aligned}$$

Therefore, T is a linear operator.

Step 4: T is bounded.

Since $\{T_n\}$ is a Cauchy sequence in a normed space, it is uniformly bounded. This means there exists a constant $C > 0$ such that $\|T_n\|_{\text{op}} \leq C$ for all n .

For any $\mathbf{x} \in X$:

$$\|T\mathbf{x}\|_Y = \lim_{n \rightarrow \infty} \|T_n\mathbf{x}\|_Y \leq \lim_{n \rightarrow \infty} (\|T_n\|_{\text{op}} \|\mathbf{x}\|_X) \leq C \|\mathbf{x}\|_X$$

This shows that T is bounded, and its operator norm satisfies $\|T\|_{\text{op}} \leq C$.

Conclusion:

We have found a bounded linear operator $T \in \mathcal{L}(X, Y)$ such that $T_n \rightarrow T$ in the uniform operator topology. Thus, every Cauchy sequence in $\mathcal{L}(X, Y)$ converges and $T \in \mathcal{L}(X, Y)$, proving that $\mathcal{L}(X, Y)$ is a Banach space. ■

Proposition 11

Let X and Y be normed vector spaces. If $T : X \rightarrow Y$ is a bounded linear operator, then $\ker(T)$ is a closed linear subspace.

Proof.

First, we show that the singleton set $\{\mathbf{0}\}$ is closed in Y . By the properties of the norm (non-degeneracy and non-negativity), $\|\mathbf{y}\| > 0$ for any $\mathbf{y} \neq \mathbf{0}$. We can construct an open ball $B\left(\mathbf{y}, \frac{\|\mathbf{y}\|}{2}\right) \subset Y \setminus \{\mathbf{0}\}$. This implies $Y \setminus \{\mathbf{0}\}$ is open, and thus $\{\mathbf{0}\}$ is closed. By definition, $\ker(T) = T^{-1}(\{\mathbf{0}\})$. Since T is a bounded linear

operator, T is continuous. Because the preimage of a closed set under a continuous mapping is closed, $\ker(T)$ is a closed set in X . To show linear subspace, we left as an exercise. ■

Remark. Unlike the kernel, the image of a bounded linear operator, $\text{Im}(T)$, is not necessarily closed. Consider the space of square-summable sequences $X = Y = \ell^2$. We define a linear operator $T : \ell^2 \rightarrow \ell^2$ by:

$$T(x_1, x_2, x_3, \dots) = \left(x_1, \frac{1}{2}x_2, \frac{1}{3}x_3, \dots, \frac{1}{n}x_n, \dots \right)$$

1. T is bounded:

Clearly, $\|T\mathbf{x}\|_{\ell^2} \leq \|\mathbf{x}\|_{\ell^2}$, so T is bounded (with operator norm $\|T\|_{\text{op}} = 1$) and therefore continuous.

2. $\text{Im}(T)$ is not closed:

Consider a sequence of points $\{\mathbf{y}^{(k)}\}_{k=1}^{\infty}$ in Y , defined as:

$$\mathbf{y}^{(k)} = \left(1, \frac{1}{2}, \frac{1}{3}, \dots, \frac{1}{k}, 0, 0, \dots \right)$$

Notice that every $\mathbf{y}^{(k)} \in \text{Im}(T)$ because it is the image of $\mathbf{x}^{(k)} = (1, 1, 1, \dots, 1, 0, 0, \dots)$, which is a sequence with k ones and is perfectly well-defined in ℓ^2 .

As $k \rightarrow \infty$, the sequence $\mathbf{y}^{(k)}$ converges in ℓ^2 to the limit point:

$$\mathbf{y} = \left(1, \frac{1}{2}, \frac{1}{3}, \dots, \frac{1}{n}, \dots \right)$$

Since $\sum \frac{1}{n^2}$ converges, this limit point $\mathbf{y}$ is indeed in ℓ^2 . Thus, $\mathbf{y}$ belongs to the closure of the image, $\mathbf{y} \in \overline{\text{Im}(T)}$.

However, does $\mathbf{y}$ belong to $\text{Im}(T)$? If it did, there must exist some $\mathbf{x} \in \ell^2$ such that $T\mathbf{x} = \mathbf{y}$. Comparing the coordinates, we would need $\frac{1}{n}x_n = \frac{1}{n}$, which forces $x_n = 1$ for all n . This means $\mathbf{x} = (1, 1, 1, \dots)$. But this sequence has infinite length and is **not** in ℓ^2 . Therefore, no such $\mathbf{x} \in \ell^2$ exists, meaning $\mathbf{y} \notin \text{Im}(T)$.

Since the image contains a convergent sequence whose limit is strictly outside the image, $\text{Im}(T)$ is not a closed set.

Definition 4.8 (Topological Equivalence)

Let X be a set. Two metrics d_1 and d_2 on X are called **topologically equivalent** if they generate the same topology on X . That is, any set is d_1 -open iff it is d_2 -open.

Remark. Equivalently, topological equivalence implies that a sequence converges under d_1 if and only if it converges to the same limit under d_2 .

Theorem 4.10

If there exist positive constants $c, C > 0$ such that for any $\mathbf{x}, \mathbf{y} \in X$

$$cd_1(\mathbf{x}, \mathbf{y}) \leq d_2(\mathbf{x}, \mathbf{y}) \leq Cd_1(\mathbf{x}, \mathbf{y}), \quad (7)$$

then d_1 and d_2 are topologically equivalent.

Proof.

Let U be a d_1 -open set. To show that U is also d_2 -open, we take any $\mathbf{x} \in U$. Since U is d_1 -open, there exists $r > 0$ such that the open ball $B_{d_1}(\mathbf{x}, r) \subseteq U$.

Consider the d_2 -open ball $B_{d_2}(\mathbf{x}, cr)$. We want to show $B_{d_2}(\mathbf{x}, cr) \subseteq B_{d_1}(\mathbf{x}, r)$. Let $\mathbf{z} \in B_{d_2}(\mathbf{x}, cr)$, which means $d_2(\mathbf{x}, \mathbf{z}) < cr$. Using the first inequality in (7), we have:

$$d_1(\mathbf{x}, \mathbf{z}) \leq \frac{1}{c}d_2(\mathbf{x}, \mathbf{z}) < \frac{1}{c}(cr) = r$$

This implies $\mathbf{z} \in B_{d_1}(\mathbf{x}, r)$. Thus, we have established the nesting of open balls:

$$B_{d_2}(\mathbf{x}, cr) \subseteq B_{d_1}(\mathbf{x}, r) \subseteq U$$

Since for every $\mathbf{x} \in U$ we can find a d_2 -open ball fully contained within U , the set U is also d_2 -open.

The converse argument (showing any d_2 -open set is d_1 -open) can be proved analogously using the right side of the inequality $d_2(\mathbf{x}, \mathbf{y}) \leq Cd_1(\mathbf{x}, \mathbf{y})$, which is left as an exercise. ■

Remark. Metrics satisfying this condition are often called **Lipschitz equivalent** or **strongly equivalent**.

Definition 4.9 (Equivalent Norms)

Two norms $\|\cdot\|_1$ and $\|\cdot\|_2$ on a vector space X are said to be equivalent if there exists a constant $\alpha \geq 1$ such that:

$$\frac{1}{\alpha}\|\mathbf{x}\|_1 \leq \|\mathbf{x}\|_2 \leq \alpha\|\mathbf{x}\|_1 \quad \text{for all } \mathbf{x} \in X.$$

Remark (Symmetrization of the Equivalence Constant). In some literature, norm equivalence is initially defined using two distinct positive constants $c, C > 0$ such that $c\|\mathbf{x}\|_1 \leq \|\mathbf{x}\|_2 \leq C\|\mathbf{x}\|_1$. We can always elegantly symmetrize these constants into a single constant $\alpha \geq 1$ by defining $\alpha := \max\left(C, \frac{1}{c}, 1\right)$.

- Since $\alpha \geq C$, the upper bound naturally holds: $\|\mathbf{x}\|_2 \leq C\|\mathbf{x}\|_1 \leq \alpha\|\mathbf{x}\|_1$.
- Since $\alpha \geq \frac{1}{c}$, taking the reciprocal gives $\frac{1}{\alpha} \leq c$, which satisfies the lower bound: $\frac{1}{\alpha}\|\mathbf{x}\|_1 \leq c\|\mathbf{x}\|_1 \leq \|\mathbf{x}\|_2$.

This construction strictly guarantees $\alpha \geq 1$, which perfectly captures the symmetric nature of the equivalence relation.

Remark (Connection to Lipschitz Equivalence). Norm equivalence is essentially the vector space counterpart of Lipschitz equivalence. Using our symmetric definition, if $\|\cdot\|_1$ and $\|\cdot\|_2$ are equivalent with constant $\alpha \geq 1$:

$$\frac{1}{\alpha} \|\mathbf{x}\|_1 \leq \|\mathbf{x}\|_2 \leq \alpha \|\mathbf{x}\|_1 \quad \text{for all } \mathbf{x} \in X.$$

By substituting $\mathbf{x}$ with $\mathbf{u} - \mathbf{v}$, we immediately obtain the relationship between their induced metrics d_1 and d_2 :

$$\frac{1}{\alpha} d_1(\mathbf{u}, \mathbf{v}) \leq d_2(\mathbf{u}, \mathbf{v}) \leq \alpha d_1(\mathbf{u}, \mathbf{v})$$

This exactly matches the condition for Lipschitz equivalence (with constants $c = \frac{1}{\alpha}$ and $C = \alpha$). Consequently, equivalent norms always generate the exact same topology.

Theorem 4.11

Let X be a vector space, and let $\|\cdot\|_1$ and $\|\cdot\|_2$ be two norms on X . Let d_1 and d_2 be their respectively induced metrics, defined by $d_i(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|_i$. TFAE:

1. $\|\cdot\|_1$ and $\|\cdot\|_2$ are equivalent norms.
2. d_1 and d_2 are Lipschitz equivalent.
3. d_1 and d_2 are topologically equivalent.

Proof.

We will prove the most non-trivial implication: (3) $\Rightarrow$ (1). (The implications (1) $\Rightarrow$ (2) and (2) $\Rightarrow$ (3) follow directly from previous definitions and theorems regarding metric spaces).

Assume d_1 and d_2 are topologically equivalent. Consider the identity operator $I : (X, \|\cdot\|_1) \rightarrow (X, \|\cdot\|_2)$ defined by $I(\mathbf{x}) = \mathbf{x}$. Clearly, I is a linear operator.

Because d_1 and d_2 generate the same topology, any sequence converging to the origin $\mathbf{0}$ in $(X, \|\cdot\|_1)$ also converges to $\mathbf{0}$ in $(X, \|\cdot\|_2)$. Therefore, the operator I is continuous at the origin $\mathbf{0}$.

A fundamental property of bounded linear operators states that a linear operator is continuous at $\mathbf{0}$ if and only if it is bounded. Since I is bounded, there exists a constant $C_1 > 0$ such that for all $\mathbf{x} \in X$:

$$\|I(\mathbf{x})\|_2 \leq C_1 \|\mathbf{x}\|_1 \Rightarrow \|\mathbf{x}\|_2 \leq C_1 \|\mathbf{x}\|_1$$

Next, consider the inverse identity operator $I^{-1} : (X, \|\cdot\|_2) \rightarrow (X, \|\cdot\|_1)$, which is

also defined as $I^{-1}(\mathbf{x}) = \mathbf{x}$. By the exact same topological argument (since topological equivalence is symmetric), I^{-1} is continuous at $\mathbf{0}$, and therefore bounded. Thus, there exists a constant $C_2 > 0$ such that for all $\mathbf{x} \in X$:

$$\|I^{-1}(\mathbf{x})\|_1 \leq C_2 \|\mathbf{x}\|_2 \Rightarrow \|\mathbf{x}\|_1 \leq C_2 \|\mathbf{x}\|_2 \Rightarrow \frac{1}{C_2} \|\mathbf{x}\|_1 \leq \|\mathbf{x}\|_2$$

To symmetrize these bounds, let $\alpha = \max(C_1, C_2, 1)$. We then have:

$$\frac{1}{\alpha} \|\mathbf{x}\|_1 \leq \frac{1}{C_2} \|\mathbf{x}\|_1 \leq \|\mathbf{x}\|_2 \leq C_1 \|\mathbf{x}\|_1 \leq \alpha \|\mathbf{x}\|_1$$

This establishes that there exists a constant $\alpha \geq 1$ such that:

$$\frac{1}{\alpha} \|\mathbf{x}\|_1 \leq \|\mathbf{x}\|_2 \leq \alpha \|\mathbf{x}\|_1 \quad \text{for all } \mathbf{x} \in X.$$

Hence, $\|\cdot\|_1$ and $\|\cdot\|_2$ are equivalent norms. ■

Theorem 4.12

Let X be a finite-dimensional real vector space. Then any two norms on X are equivalent.

Proof.

Let $\{\mathbf{v}_i\}_{i=1}^n$ be a basis for X . For $\mathbf{x} = \sum x_i \mathbf{v}_i$, define the standard Euclidean norm $\|\mathbf{x}\|_2 = \left(\sum x_i^2\right)^{1/2}$. We just need to show any norm $\|\cdot\|$ is equivalent to $\|\cdot\|_2$. First,

$$\begin{aligned} \|\mathbf{x}\| &\leq \sum |x_i| \|\mathbf{v}_i\| \quad (\text{Triangle inequality}) \\ &\leq \left(\sum x_i^2\right)^{1/2} \left(\sum \|\mathbf{v}_i\|^2\right)^{1/2} \quad (\text{Cauchy-Schwarz for } (\mathbb{R}^n, \langle -, - \rangle_{\text{dot}})). \end{aligned}$$

So we can take $C = \left(\sum \|\mathbf{v}_i\|^2\right)^{1/2}$ to get $\|\mathbf{x}\| \leq C \|\mathbf{x}\|_2$.

Now, consider the unit sphere $S = \{\mathbf{x} \in X : \|\mathbf{x}\|_2 = 1\}$. Under $\|\cdot\|_2$, S is closed and bounded in $\mathbb{R}^n$, hence compact. The function $f(\mathbf{x}) = \|\mathbf{x}\|$ is continuous on S (since $|\|\mathbf{x}\| - \|\mathbf{y}\|| \leq \|\mathbf{x} - \mathbf{y}\| \leq C \|\mathbf{x} - \mathbf{y}\|_2$). Since S is compact, f attains an absolute minimum δ at some $\mathbf{z} \in S$. Since $\mathbf{z} \neq \mathbf{0}$, $\delta = \|\mathbf{z}\| > 0$. Thus $\|\mathbf{x}\| \geq \delta$ for all $\mathbf{x} \in S$. For any non-zero $\mathbf{w} \in X$, $\frac{\mathbf{w}}{\|\mathbf{w}\|_2} \in S$, so $\left\|\frac{\mathbf{w}}{\|\mathbf{w}\|_2}\right\| \geq \delta \Rightarrow \|\mathbf{w}\| \geq \delta \|\mathbf{w}\|_2$. This establishes equivalence. ■

Remark. This equivalence fails in infinite-dimensional spaces. For instance, on $(C[0, 1], \mathbb{R})$, we can define two norms:

$$\|f\|_\infty = \sup_{t \in [0, 1]} |f(t)| \quad \text{and} \quad \|f\|_2 = \left(\int_0^1 f^2(t) dt\right)^{\frac{1}{2}}$$

These two norms are not equivalent since there exists a sequence f_n that is Cauchy in L^2 but not in L^∞ , which directly implies $\dim(C[0, 1]) = \infty$.

Theorem 4.13

Every finite-dimensional normed space X is complete.

Proof.

By the previous theorem regarding equivalent norms, we know that $(X, \|\cdot\|_2) \cong (\mathbb{R}^n, \|\cdot\|_2)$, and we know $(\mathbb{R}^n, \|\cdot\|_2)$ is complete.

Therefore, X is complete with respect to the l_2 -metric induced by $\|\cdot\|_2$.

For any arbitrary norm $\|\cdot\|$ on X , it induces a metric $d : X \times X \rightarrow \mathbb{R}$.

Since all norms are equivalent, the metric spaces (X, d) and (X, l_2) generate the exact same topology.

Because (X, l_2) is complete and the topologies are equivalent via Lipschitz bounds, (X, d) is also complete. ■

Theorem 4.14

Let X be a normed space. Every finite-dimensional subspace of X is closed.

Proof.

Let $Y \subset X$ be a finite-dimensional subspace.

Take any sequence $\{\mathbf{y}_n\}_{n \in \mathbb{N}}$ in Y such that $\lim_{n \rightarrow \infty} \mathbf{y}_n = \mathbf{x}$ for some $\mathbf{x} \in X$.

Since convergent sequences are Cauchy, $\{\mathbf{y}_n\}_{n \in \mathbb{N}}$ is also a Cauchy sequence in Y .

Because $\dim(Y) < \infty$, the subspace Y itself is a complete normed space (by the previous theorem).

Therefore, this Cauchy sequence must converge to a limit within Y , meaning

$\lim_{n \rightarrow \infty} \mathbf{y}_n = \mathbf{y}$ for some $\mathbf{y} \in Y$.

By the uniqueness of limits in metric spaces, we must have $\mathbf{y} = \mathbf{x}$.

This implies $\mathbf{x} \in Y$. Since Y contains all its limit points, Y is closed. ■

Example 4.1 (Dense proper subspace may not be closed)

1. Let $C^1[0, 1] \subsetneq C[0, 1]$. Consider the function $f(t) = \left|t - \frac{1}{2}\right|$.
2. By the Stone-Weierstrass theorem, $\forall f \in C[0, 1], \exists \{f_n\}_{n \in \mathbb{N}} \subseteq C^1[0, 1]$ such that $f_n \Rightarrow f$. But $f \notin C^1[0, 1]$ (since it is not differentiable at $t = \frac{1}{2}$), so $C^1[0, 1]$ is not closed.

Theorem 4.15

Let X be a finite-dimensional normed space and $K \subseteq X$. Then K is compact if and only if K is closed and bounded.

Proof.

($\Rightarrow$) The forward direction (compact implies closed and bounded) is always true in any metric space.

($\Leftarrow$) For the reverse direction, since X is finite-dimensional, the topology induced by $(X, \|\cdot\|)$ and $(\mathbb{R}^n, \|\cdot\|_2)$ are equivalent.

The condition that K is closed and bounded in X is topologically equivalent to an associated set S being closed and bounded in $\mathbb{R}^n$.

By the standard Heine-Borel theorem in $\mathbb{R}^n$, S is compact in $\mathbb{R}^n$.

Due to the equivalence of topologies, this immediately implies K is compact in X . ■

Example 4.2 (Closed and bounded may not be compact)

Consider the Banach space $(C([0, 1]), \|\cdot\|_\infty)$. Let $B = \{f \in C([0, 1]) \mid \|f\|_\infty \leq 1\}$. This set is clearly closed and bounded.

To verify closedness, take any $\{f_n\}_{n \in \mathbb{N}} \subset B \subseteq C([0, 1])$ s.t. $f_n \Rightarrow f, f \in C([0, 1])$.

$$\|f\|_\infty \leq \|f - f_n\|_\infty + \|f_n\|_\infty < \epsilon + 1 \quad \forall \epsilon > 0 \implies \|f\|_\infty \leq 1 \implies f \in B.$$

Let $f_n(x) = x^n$ for $x \in [0, 1]$. The sequence $\{f_n\}_{n \in \mathbb{N}}$ is not equicontinuous. By the Arzelà-Ascoli theorem, B is not relatively compact, and hence not compact.

Theorem 4.16

Let $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ be normed vector spaces. If $\dim X < \infty$, then every linear operator $A : X \rightarrow Y$ is bounded.

Proof.

Let $A : X \rightarrow Y$ be a linear operator. We define a new norm on X by:

$$\|\mathbf{x}\|_A := \|\mathbf{x}\|_X + \|A\mathbf{x}\|_Y$$

(It is a straightforward exercise to verify that $\|\cdot\|_A$ indeed satisfies the axioms of a norm, utilizing the linearity of A and the triangle inequality of the norms).

Since X is finite-dimensional ($\dim X < \infty$), any two norms on X are equivalent.

Therefore, the newly defined norm $\|\cdot\|_A$ is equivalent to the original norm $\|\cdot\|_X$.

By the definition of equivalent norms, there exists a constant $C > 0$ such that for all $\mathbf{x} \in X$:

$$\frac{1}{C} \|\mathbf{x}\|_X \leq \|\mathbf{x}\|_A \leq C \|\mathbf{x}\|_X$$

Substituting the definition of $\|\mathbf{x}\|_A$ into the right side of the inequality, we obtain:

$$\|\mathbf{x}\|_X + \|A\mathbf{x}\|_Y \leq C \|\mathbf{x}\|_X$$

$$\implies \|A\mathbf{x}\|_Y \leq (C - 1) \|\mathbf{x}\|_X$$

This shows that there exists a constant $M = C - 1$ such that $\|A\mathbf{x}\|_Y \leq M\|\mathbf{x}\|_X$ for all $\mathbf{x} \in X$. Thus, A is a bounded linear operator. ■

Example 4.3 (Linear operator may not be bounded)

Consider the space $(C([- \pi, \pi], \mathbb{R}), \|\cdot\|_2)$, where $f \in C([- \pi, \pi], \mathbb{R})$ and $\|f\|_2 = \left(\int_{-\pi}^{\pi} |f(t)|^2 dt \right)^{\frac{1}{2}}$.

Let $\mathbf{e}_n^c = \frac{\cos(nt)}{\sqrt{\pi}}$ with $\|\mathbf{e}_n^c\| = 1$. Let $X = \text{Span} \left[\frac{\cos(nt)}{\sqrt{\pi}} \right]_{n \in \mathbb{N}}$, which implies $\dim X = \infty$.

Define a linear operator T to be such that $T(\mathbf{e}_n^c) = n$. i.e.,

for $\mathbf{x} \in X$, $\mathbf{x} = \sum_{j=1}^m \alpha_j \mathbf{e}_j^c \implies T(\mathbf{x}) = \sum_{j=1}^m \alpha_j j$.

Note that the operator norm is bounded below by its action on basis vectors:

$$\|T\| = \sup_{\mathbf{x} \neq \mathbf{0}} \frac{\|T\mathbf{x}\|_Y}{\|\mathbf{x}\|_X} \geq \sup_{\|\mathbf{x}\|=1} \|T\mathbf{x}\| \geq \|T\mathbf{e}_n^c\| = n \quad \forall n \in \mathbb{N} \quad (\because T\mathbf{e}_n^c = n, \|\mathbf{e}_n^c\| = 1)$$

Hence, $\|T\| = \infty$.

Definition 4.10 (Dual Space)

Let X be a normed vector space. The dual space of X , denoted as X^* , is defined as the space of all continuous linear functionals on X :

$$X^* = \mathcal{L}(X, \mathbb{R}) = \{\Lambda : X \rightarrow \mathbb{R} \mid \Lambda \text{ is linear and bounded}\}$$

Remark. Because $(\mathbb{R}, |\cdot|)$ is a complete normed space (a Banach space), the operator space X^* is also a Banach space, regardless of whether X itself is complete.

Example 4.4 (Dual of L^p Spaces)

Let $(M, \mathcal{A}, \mu)$ be a measurable space (Think of $\mathbb{R}^n$, measurable sets on $\mathbb{R}^n$, Lebesgue measure).

Let $1 < p < \infty$, and let q be the conjugate exponent of p such that:

$$\frac{1}{p} + \frac{1}{q} = 1 \quad \left(\text{i.e., } \frac{1}{q} = \frac{p-1}{p} \implies q(p-1) = p \right)$$

Recall the Hölder inequality: if $f \in L^p(\mu)$ and $g \in L^q(\mu)$, then $fg \in L^1(\mu)$, and:

$$\int_M fg \, d\mu \leq \int_M |f||g| \, d\mu \leq \|f\|_p \|g\|_q \quad \text{where } \|f\|_p = \left(\int_M |f|^p \, d\mu \right)^{\frac{1}{p}}$$

Note that $(L^p(\mu), \|\cdot\|_p)$ is a Banach space.

Now, given $g \in L^q(\mu)$, we can define a linear functional $\Lambda_g : L^p(\mu) \rightarrow \mathbb{R}$ by:

$$\Lambda_g(f) = \int_M fg \, d\mu \quad \text{for all } f \in L^p(\mu)$$

This implies $\Lambda_g \in (L^p(\mu))^*$. We can verify its properties:

- **Linearity:** Λ_g is linear on $L^p(\mu)$ because integrals are linear:

$$\Lambda_g(\alpha f_1 + \beta f_2) = \alpha \Lambda_g(f_1) + \beta \Lambda_g(f_2) \quad \forall \alpha, \beta \in \mathbb{R}, f_1, f_2 \in L^p(\mu)$$

- **Boundedness:** By Hölder's inequality, we have:

$$|\Lambda_g(f)| = \left| \int_M f g \, d\mu \right| \leq \|f\|_p \|g\|_q$$

Therefore, the operator norm is bounded by $\|g\|_q$:

$$\|\Lambda_g\| = \sup_{\substack{f \in L^p(\mu) \\ f \neq 0}} \frac{|\Lambda_g(f)|}{\|f\|_p} \leq \|g\|_q$$

This proves that Λ_g is a bounded linear operator.

Lemma (Riesz)

Let $(X, \|\cdot\|)$ be a normed vector space and $Y \subset X$ be a proper closed linear subspace. Then for any constant $0 < \delta < 1$, there exists $\mathbf{x} \in X$ such that $\|\mathbf{x}\| = 1$ and

$$\inf_{\mathbf{y} \in Y} \|\mathbf{x} - \mathbf{y}\| \geq 1 - \delta$$

Proof.

Suppose $Y \subsetneq X$ and Y is closed. Since $Y \neq X$, there exists some $\mathbf{x}_0 \in X \setminus Y$. Because $\mathbf{x}_0 \notin Y$ and Y is a closed set, the infimum distance from $\mathbf{x}_0$ to Y must be strictly positive. Let this distance be d :

$$d = \inf_{\mathbf{y} \in Y} \|\mathbf{x}_0 - \mathbf{y}\| > 0$$

Given $0 < \delta < 1$, we clearly have $d < \frac{d}{1 - \delta}$. By the definition of the infimum, there must exist some $\mathbf{y}_0 \in Y$ such that:

$$\|\mathbf{x}_0 - \mathbf{y}_0\| \leq \frac{d}{1 - \delta}$$

Now, we choose the normalized vector $\mathbf{x} = \frac{\mathbf{x}_0 - \mathbf{y}_0}{\|\mathbf{x}_0 - \mathbf{y}_0\|}$. It is obvious that $\|\mathbf{x}\| = 1$. For any arbitrary $\mathbf{y} \in Y$, we compute its distance to $\mathbf{x}$:

$$\|\mathbf{x} - \mathbf{y}\| = \left\| \frac{\mathbf{x}_0 - \mathbf{y}_0}{\|\mathbf{x}_0 - \mathbf{y}_0\|} - \mathbf{y} \right\| = \frac{1}{\|\mathbf{x}_0 - \mathbf{y}_0\|} \|\mathbf{x}_0 - (\mathbf{y}_0 + \|\mathbf{x}_0 - \mathbf{y}_0\| \mathbf{y})\|$$

Since Y is a linear subspace, and both $\mathbf{y}_0, \mathbf{y} \in Y$, the linear combination $\mathbf{y}_0 + \|\mathbf{x}_0 - \mathbf{y}_0\| \mathbf{y}$ is also an element of Y . By the definition of our infimum distance d , the distance from $\mathbf{x}_0$ to any element in Y is bounded below by d . Therefore:

$$\|\mathbf{x}_0 - (\mathbf{y}_0 + \|\mathbf{x}_0 - \mathbf{y}_0\| \mathbf{y})\| \geq d$$

Substituting this back into our equation, we obtain:

$$\|\mathbf{x} - \mathbf{y}\| \geq \frac{d}{\|\mathbf{x}_0 - \mathbf{y}_0\|} \geq \frac{d}{\frac{d}{1 - \delta}} = 1 - \delta$$

This completes the proof. ■

Remark. This lemma highlights a fundamental difference between general normed spaces and inner product spaces. In an inner product space, we can easily construct a vector that is perfectly orthogonal to a closed subspace, guaranteeing a distance of exactly 1.

However, in a general normed space, the notion of "orthogonality" does not exist. More importantly, the infimum distance $d = \inf_{\mathbf{y} \in Y} \|\mathbf{x}_0 - \mathbf{y}\|$ might not be attained by any actual vector in Y . Therefore, Riesz's Lemma acts as an analytic compromise: instead of demanding a perfectly perpendicular vector (which may not exist), we rely on the infimum property to find a point $\mathbf{y}_0$ that is "close enough" to the ideal shortest distance. By rescaling this imperfect vector, we ensure the distance is at least $1 - \delta$. We cannot always achieve 1, but we can get arbitrarily close.

Theorem 4.17 (Riesz's Lemma in Hilbert Spaces)

Let H be a Hilbert space and $Y \subset H$ be a proper closed subspace. Then there exists $\mathbf{x} \in H$ such that $\|\mathbf{x}\| = 1$ and

$$\inf_{\mathbf{y} \in Y} \|\mathbf{x} - \mathbf{y}\| = 1$$

Proof.

Since H is a Hilbert space and Y is a proper closed subspace, we can invoke the Orthogonal Decomposition Theorem, which states that $H = Y \oplus Y^\perp$.

Because $Y \subsetneq H$, its orthogonal complement $Y^\perp$ is non-trivial. Therefore, there exists some non-zero vector $\mathbf{z} \in Y^\perp$.

We define the normalized vector:

$$\mathbf{x} = \frac{\mathbf{z}}{\|\mathbf{z}\|}$$

It is immediate that $\|\mathbf{x}\| = 1$ and $\mathbf{x} \in Y^\perp$.

For any arbitrary $\mathbf{y} \in Y$, since $\mathbf{x} \in Y^\perp$, the vectors $\mathbf{x}$ and $\mathbf{y}$ are orthogonal ($\mathbf{x} \perp \mathbf{y}$).

We can strictly evaluate the distance using the Pythagorean theorem:

$$\|\mathbf{x} - \mathbf{y}\|^2 = \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 = 1 + \|\mathbf{y}\|^2$$

Since the norm is non-negative ($\|\mathbf{y}\|^2 \geq 0$), it directly follows that:

$$\|\mathbf{x} - \mathbf{y}\|^2 \geq 1 \implies \|\mathbf{x} - \mathbf{y}\| \geq 1$$

Furthermore, since the zero vector $\mathbf{0} \in Y$ and $\|\mathbf{x} - \mathbf{0}\| = \|\mathbf{x}\| = 1$, the infimum of the distance from $\mathbf{x}$ to Y is exactly attained at the origin (which corresponds to the orthogonal projection of $\mathbf{x}$ onto Y). Thus:

$$\inf_{\mathbf{y} \in Y} \|\mathbf{x} - \mathbf{y}\| = 1$$

This completes the proof. ■

Theorem 4.18 (Compactness of the Unit Ball)

Let $(X, \|\cdot\|)$ be a normed vector space. Let B be the closed unit ball and S be the unit sphere. The following statements are equivalent:

1. $\dim X < \infty$.
2. The closed unit ball B is compact.
3. The unit sphere S is compact.

Proof.

We will prove the implication $(2) \implies (1)$ (or equivalently $(3) \implies (1)$) by demonstrating its contrapositive: if $\dim X = \infty$, then S is not compact.

Suppose, by contradiction, that $\dim X = \infty$. Choose an arbitrary $\mathbf{x}_1 \in X$ with $\|\mathbf{x}_1\| = 1$. The subspace spanned by it, $Y_1 = \text{Span}\{\mathbf{x}_1\}$, is a finite-dimensional vector subspace and is therefore closed. Since $\dim X = \infty$, Y_1 is a proper subspace ($Y_1 \neq X$). By applying Riesz's Lemma with $\delta = \frac{1}{2}$, there exists $\mathbf{x}_2 \in X \setminus Y_1$ such that $\|\mathbf{x}_2\| = 1$ and $\|\mathbf{x}_2 - \mathbf{x}_1\| \geq \frac{1}{2}$.

We can proceed inductively. Suppose we have already chosen a set of linearly independent vectors $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\}$ on the unit sphere such that $\|\mathbf{x}_i - \mathbf{x}_j\| \geq \frac{1}{2}$ for all $i \neq j$. Let $Y_k = \text{Span}\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$. Because Y_k is finite-dimensional, it is a closed proper subspace of X . By Riesz's Lemma again with $\delta = \frac{1}{2}$, there exists $\mathbf{x}_{k+1} \in X$ such that $\|\mathbf{x}_{k+1}\| = 1$ and $\|\mathbf{x}_{k+1} - \mathbf{y}\| \geq \frac{1}{2}$ for all $\mathbf{y} \in Y_k$. In particular, $\|\mathbf{x}_{k+1} - \mathbf{x}_i\| \geq \frac{1}{2}$ for $i = 1, 2, \dots, k$.

This inductive process constructs an infinite sequence $\{\mathbf{x}_j\}_{j=1}^{\infty}$ situated entirely on the unit sphere S , possessing the property that the distance between any two distinct terms is at least $\frac{1}{2}$. Such a sequence cannot possess any Cauchy subse-

quence, and consequently, it cannot have any convergent subsequence. Therefore, S is not sequentially compact, which implies S (and hence B) is not compact. ■

Theorem 4.19 (Projection Theorem)

Let H be a Hilbert space and let $E \subset H$ be a nonempty closed convex subset. Then there exists a unique element $\mathbf{x}_0 \in E$ such that:

$$\|\mathbf{x}_0\| \leq \|\mathbf{x}\| \quad \text{for all } \mathbf{x} \in E$$

Proof.

Let $\delta := \inf\{\|\mathbf{x}\| \mid \mathbf{x} \in E\}$.

Uniqueness: First, we prove uniqueness. Suppose there exist two elements $\mathbf{x}_0, \mathbf{x}_1 \in E$ such that $\|\mathbf{x}_0\| = \|\mathbf{x}_1\| = \delta$. We want to show $\mathbf{x}_0 = \mathbf{x}_1$. Because E is a convex set, the midpoint $\frac{1}{2}\mathbf{x}_0 + \frac{1}{2}\mathbf{x}_1 \in E$. By the definition of the infimum δ , we must have $\left\|\frac{\mathbf{x}_0 + \mathbf{x}_1}{2}\right\| \geq \delta$, which implies $\|\mathbf{x}_0 + \mathbf{x}_1\|^2 \geq 4\delta^2$.

In a Hilbert space, the norm is induced by an inner product, so the parallelogram law holds. Applying it, we get:

$$\begin{aligned} 0 &\leq \|\mathbf{x}_0 - \mathbf{x}_1\|^2 = 2\|\mathbf{x}_0\|^2 + 2\|\mathbf{x}_1\|^2 - \|\mathbf{x}_0 + \mathbf{x}_1\|^2 \\ &\leq 2(\delta^2) + 2(\delta^2) - 4\delta^2 = 0 \end{aligned}$$

This forces $\|\mathbf{x}_0 - \mathbf{x}_1\|^2 = 0$, which means $\mathbf{x}_0 = \mathbf{x}_1$. The element is unique.

Existence: By the definition of the infimum, there must exist a sequence $\{\mathbf{x}_i\}_{i=1}^\infty$ in E such that $\lim_{i \rightarrow \infty} \|\mathbf{x}_i\| = \delta$. Given any $\epsilon > 0$, there exists an index i_0 such that if $i \geq i_0$, $\left|\|\mathbf{x}_i\|^2 - \delta^2\right| < \frac{\epsilon^2}{4}$, which implies $\|\mathbf{x}_i\|^2 < \delta^2 + \frac{\epsilon^2}{4}$.

For any $i, j \geq i_0$, convexity implies $\frac{\mathbf{x}_i + \mathbf{x}_j}{2} \in E$, so its norm is bounded below by δ : $\left\|\frac{\mathbf{x}_i + \mathbf{x}_j}{2}\right\| \geq \delta$. Applying the parallelogram law again to $\mathbf{x}_i$ and $\mathbf{x}_j$:

$$\begin{aligned} \|\mathbf{x}_i - \mathbf{x}_j\|^2 &= 2\|\mathbf{x}_i\|^2 + 2\|\mathbf{x}_j\|^2 - \|\mathbf{x}_i + \mathbf{x}_j\|^2 \\ &< 2\left(\delta^2 + \frac{\epsilon^2}{4}\right) + 2\left(\delta^2 + \frac{\epsilon^2}{4}\right) - 4\delta^2 = \epsilon^2 \end{aligned}$$

Taking the square root, we get $\|\mathbf{x}_i - \mathbf{x}_j\| < \epsilon$ for all $i, j \geq i_0$. This demonstrates that $\{\mathbf{x}_i\}_{i=1}^\infty$ is a Cauchy sequence. Since E is a closed subset residing within a complete Hilbert space H , E itself is complete. Therefore, the Cauchy sequence converges to some element $\mathbf{x}_0 \in E$. By the continuity of the norm function, we conclude:

$$\|\mathbf{x}_0\| = \lim_{i \rightarrow \infty} \|\mathbf{x}_i\| = \delta = \inf_{\mathbf{x} \in E} \|\mathbf{x}\|$$

This establishes the existence of the closest point. ■

Corollary

For any $p_1, p_2 \geq 1$, the ℓ_{p_1} and ℓ_{p_2} -metrics on $\mathbb{R}^n$ are topologically equivalent.

Corollary

If X is a finite-dimensional normed vector space, a subset $K \subseteq X$ is compact if and only if it is closed and bounded.

Corollary

If $\dim X < \infty$, then every linear operator $T : X \rightarrow Y$ is bounded.

Remark. On $C([0, 1], \mathbb{R})$, we can define $\|\cdot\|_\infty$ and $\|\cdot\|_2$. These two norms are not equivalent. Consequently, $\dim(C([0, 1])) = \infty$.